The Riemann–von Mangoldt clock and the zeros of the Riemann zeta function: an exact defect identity, prime cutoffs and a computable corrected clock

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Abstract

Let γ_k be the ordinates of the nontrivial zeros of the Riemann zeta function, counted with multiplicity, and let τ_k be the midpoint Riemann–von Mangoldt clock, $N_0(\tau_k) = k - \frac{1}{2}$, where N_0 is the smooth part of the Riemann–von Mangoldt formula. *Known:* $\delta_k = \tau_k - \gamma_k$ is bounded, $\limsup_k |\delta_k| \le 0.6331$ (Bellotti–Wong), its mean square over $T < \gamma_k \le 2T$ is $\sim 2\log\log T/\log^2 T$, and $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ is asymptotically normal with variance $\frac{1}{2\pi^2}\log\log T$ (Selberg, Korolev): the clock is close to the zeros in height but misplaces almost every zero by an unbounded number of positions. *New, unconditional:* (A) an exact identity turns sums of $g(k - \frac{1}{2} - F(\gamma_k))$ over consecutive ordinates into dF-integrals of $g(N - F)$, for every continuous g and every C^1 model F of the counting function; (B) for the clock equation corrected by the Euler phase S_X of the primes $p \le X$, the mean square of the ordinal defect $k - \frac{1}{2} - N_0(\gamma_k) - S_X(\gamma_k)$ is $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1 + \sqrt{\log\frac{\log T}{\log X}})$ uniformly for $2 \le X \le T^{1/2}$, also with a cutoff depending on the index, so the error in the position of the zeros becomes bounded in mean square exactly when $\log X \asymp \log T$; (C) an exact finite-height accounting of the computable one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$; (D) with $X = T^{1/\sqrt{\log\log T}}$ the computable clock satisfies $\langle(\tau_k^X - \gamma_k)^2\rangle = (\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$, smaller than the mean square of the bare clock by a factor $\sim \log\log\log T/(2\log\log T)$; (E) $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_t |S(t)|/\log t$ exactly, and $\limsup_k |\tau_k - \gamma_k| < 0.60832$. We also prove the Landau-type formula on the critical line needed for the prime responses of s_k, and that the cross term of the Euler phase at the clock points vanishes for $X \le T^{1/4-\varepsilon}$. A conditional law for primes $p = T^\theta$ and a determinantal model are kept separate. All observables are computed on the first 2 001 052 zeros of the Riemann zeta function.

Keywords: Riemann zeta function, zeros of the Riemann zeta function, Riemann–von Mangoldt formula, argument of the Riemann zeta function, truncated Euler product
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1. Introduction

1.1. The Riemann–von Mangoldt clock and the ordinates

Let $\zeta(s)$ be the Riemann zeta function, let $N(T)$ be the number of its nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, counted with multiplicity, and list their ordinates as $\gamma_1 \le \gamma_2 \le \cdots$, with multiplicity. Away from ordinates the Riemann–von Mangoldt formula reads [33, §§9.3–9.4]

$$N(T) = N_0(T) + \frac{1}{\pi}\arg\zeta\Big(\frac{1}{2}+iT\Big) + O\Big(\frac{1}{T}\Big), \qquad N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \tag{1.1}$$

the argument of the Riemann zeta function being obtained by continuous variation along $2 \to 2+iT \to \frac{1}{2}+iT$ from $\arg\zeta(2) = 0$. The *midpoint Riemann–von Mangoldt clock* (RvM clock) is the sequence τ_k defined by

$$N_0(\tau_k) = k - \frac{1}{2}, \qquad \tau_k = \frac{2\pi\,(k-\frac{11}{8})}{W_0\Big((k-\frac{11}{8})/e\Big)} \qquad (k \ge 1), \tag{1.2}$$

where W_0 is the principal branch of the Lambert function; the Lambert form appears in the work of França and LeClair [9, 10]. The clock is explicit and lies, in phase, half way between consecutive Gram points. This paper studies its relation to the ordinates γ_k of the zeros of the Riemann zeta function: in which sense the clock describes them, and how much prime information has to be added to it. The relation is measured in units of zeros by the ordinal displacement and by its prime-corrected version, the *ordinal defect*,

$$s_k = k - \frac{1}{2} - N_0(\gamma_k), \qquad u_k(X) = s_k - S_X(\gamma_k), \qquad S_X(t) = -\frac{1}{\pi}\sum_{p\le X}\arg\Big(1-p^{-1/2-it}\Big). \tag{1.3}$$

$u_k(X)$ measures how well the true ordinate satisfies the corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$; $|u_k(X)| < \frac{1}{2}$ means that the corrected count assigns to γ_k its correct index.

1.2. What is known and only used

The following facts are quoted with references in Section 3 and are not reproved.

- The displacement $\delta_k = \tau_k - \gamma_k$ is bounded, and $\limsup_k |\delta_k| \le 2\pi \cdot 0.10076 = 0.6331\ldots$; this follows at once from the explicit bound of Bellotti and Wong for $N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}$ [2], which improves those of Trudgian [34] and of Hasanalizade, Shen and Wong [15].

- The ordinal displacement s_k differs by less than 1 from minus the integer displacement of γ_k with respect to the Gram intervals, whose moments are those of a normal variable with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae as proved by Korolev [20, 19]. Hence $\langle s_k^2\rangle \sim \frac{1}{2\pi^2}\log\log T$ and $\langle \delta_k^2\rangle \sim 2\log\log T/\log^2 T$ over $T < \gamma_k \le 2T$: the bare clock is close to the zeros in height but misplaces almost all of them by an unbounded number of positions.

- In t-mean, $S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+it)$ is asymptotically normal (Selberg [28], Tsang [35]), and Selberg's approximation of $S(t)$ by prime sums up to a small power of T holds in $L^4(T, 2T)$ [28, 29], in the form proved by Fujii [12, §3]. These concern t-averages, not the ordinates.

- Landau's formula for $\sum_{0<\gamma\le t} n^\rho$ [21], in Gonek's uniform version [13], and Selberg's zero-density estimate [33, Thm. 9.19(C)].
- LeClair [22, Sect. 3] truncates the Euler product at a number of primes that tends to infinity with the height, and the truncated product is used in the equation of the individual zero with a fixed number of primes (10^6 for the 10^{100}-th zero, [11, §XXI, Note added 7]).

1.3. What this paper adds

Let $\mathcal{I}_T = \{k : T < \gamma_k \le 2T\}$ and let $\langle\cdot\rangle_{\mathcal{I}_T}$ be the mean over $\mathcal{I}_T$. Theorems A–E are unconditional. We have not found them in the literature; absence from a search is weak evidence and does not establish priority.

A. Exact defect identity (Theorems 4.1 and 4.2). For every real C^1 model F of the counting function, every continuous g with primitive $\mathcal{G}$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$, the sum of $\mathcal{G}(u_k + \frac{1}{2}) - \mathcal{G}(u_k - \frac{1}{2})$ over consecutive ordinates equals $\int g(N - F)\, dF$ up to two boundary terms. For $g(x) = x^{2j}$ this gives $\sum u_k^{2j}$ ($\frac{1}{12}$ per zero when $j = 1$); for $g(x) = e^{i\xi x}$ it identifies the characteristic function of the empirical law of u_k with a t-integral. No hypothesis on the zeros and no monotonicity of F is used.

B. How much prime information the clock needs (Theorems 5.1 and 5.11). Uniformly for $2 \le X \le T^{1/2}$, with $c = \log X/\log T$,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2}\log\frac{1}{c} + O\Big(1 + \sqrt{\log(1/c)}\Big), \tag{1.4}$$

and the same holds with the cutoff $X_k = \max(2, (\tau_k/2\pi)^c)$ varying with the index. A fixed set of primes leaves a mean-square defect $\sim (2\pi^2)^{-1}\log\log T$; primes up to T^c keep it bounded; boundedness holds if and only if $\liminf \log X/\log T > 0$ (Corollary 5.4). Thus the growth of the number of primes asked for in [22] is necessary but not sufficient.

C. Exact accounting for the computable clock (Theorem 6.1). For the one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$, the mean-square gain over any finite set of zeros equals a diagonal prime-response term minus a cross term, with no independence assumption; the residual of τ_k^X differs from $u_k(X)$ by an explicit transport term and by the exact nonnegative term $\chi_k = \frac{\gamma_k}{2\pi}(\frac{\tau_k}{\gamma_k} - 1 - \log\frac{\tau_k}{\gamma_k})$.

D. A computable clock that approaches the zeros by an order of magnitude (Theorem 7.3). For $2 \le X \le T^{1/2}$ the residual $r_k(X) = \mathcal{L}(\tau_k)(\tau_k^X - \gamma_k)$ satisfies $\langle r_k(X)^2\rangle_{\mathcal{I}_T} = \frac{1}{2\pi^2}\log\frac{1}{c} + O((1 + \sqrt{\log(1/c)})(1 + c^2\log\log T))$; the term $c^2\log\log T$ is the transport between τ_k and γ_k (Lemma 7.2). With $X = T^{1/\sqrt{\log\log T}}$, or with the cutoff $\tau_k^{1/\sqrt{\log\log\tau_k}}$ read off the clock itself,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\big(\tau_k^X - \gamma_k\big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}. \tag{1.5}$$

E. The pointwise envelope (Proposition 8.1 and Theorem 8.3). $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_{t\to\infty} |S(t)|/\log t$, with equality, also for the corrected clocks with $X \le (\log T)^{2-\varepsilon}$; and

$$\limsup_{k\to\infty} |\tau_k - \gamma_k| \le \frac{1}{\log 2} \int_0^1 \frac{\mathfrak{e}(\sigma)\, d\sigma}{\sqrt{\sigma(2-\sigma)}} < 0.60832 \tag{1.6}$$

for the exponent profile $\mathfrak{e}$ given by Bourgain's exponent pair and the functional equation, against 0.6331 before.

Main finding. The Euler phase of the primes up to a power of the height turns the clock from an approximation in height into an approximation in position. For the bare clock the error in units of zeros, $s_k = \mathcal{L}(\xi_k)\delta_k$, has mean square $\sim \frac{1}{2\pi^2}\log\log T$. With $X = T^c$ the error in units of zeros of the corrected clock equation has bounded mean square (Theorem B), so outside a proportion $\ll (1 + \log(1/c))/A^2$ of the zeros it is at most A; and the computable clock, which uses only τ_k and the primes, keeps the gain up to the transport between τ_k and γ_k (Theorem D). With $X = T^{1/\sqrt{\log\log T}}$,

$$\frac{\langle(\tau_k^X - \gamma_k)^2\rangle_{\mathcal{I}_T}}{\langle(\tau_k - \gamma_k)^2\rangle_{\mathcal{I}_T}} \sim \frac{\log\log\log T}{2\log\log T} \longrightarrow 0, \tag{1.7}$$

and in units of zeros the mean square drops from $\frac{1}{2\pi^2}\log\log T$ to $\frac{1}{4\pi^2}\log\log\log T$. The gain is a statement in mean square; for the worst zero, Proposition 8.1 shows that no cutoff $X \le (\log T)^{2-\varepsilon}$ changes $\limsup |\tau_k - \gamma_k|$, and the improvement (1.6) comes from the exponents of the Riemann zeta function, not from the primes.

Further unconditional results (Section 10). The symmetry $\rho \mapsto 1 - \bar\rho$ of the zeros and Selberg's density estimate give the Landau-type formula $\sum_{0<\gamma\le t} n^{i\gamma} = -\frac{t}{2\pi}\frac{\Lambda(n)}{\sqrt{n}} + O(\frac{t\log^2 n}{\log t} + \sqrt{n}\log^2 t)$ uniformly for $n \le t^{1/8}$ (Theorem 10.1); hence the prime responses of s_k tend to 1 uniformly for primes up to $(\log T)^{2-\varepsilon}$, and the mean-square gain of the full Euler phase at the clock points is $(2\lambda - \lambda^2)v_X + o(1)$ for $X \le (\log T)^{2-\varepsilon}$. The cross term at the clock points is $O(T^{-\varepsilon/2})$ for $X \le T^{1/4-\varepsilon}$; numerically it switches on when X^2 reaches the height in the scale $\tau/2\pi$, through prime pairs $pq \approx \tau_k/2\pi$. *Conditional results* (Section 10.4) are rigorous implications from two named hypotheses: the moving-prime law $1 - \Phi(\theta)$ for $p = T^\theta$, resting on an unconditional reduction to a pair correlation shifted by a prime frequency. *Heuristic support* (Section 11) is kept apart.

1.4. Organization

Section 2 lists the notation and Section 3 the external results. Sections 4–8 prove Theorems A–E. Section 9 computes the observables on the first 2 001 052 zeros of the Riemann zeta function. Section 10 contains the prime responses, unconditional and conditional, and Section 11 the heuristic support. Section 12 separates what is known from what is new and discusses the use of a finer knowledge of $\tau_k - \gamma_k$. All programs and outputs are in a public repository (Data availability).

2. Notation

2.1. List of symbols

Every symbol has one meaning throughout. The letter μ is not used (it is reserved for the Möbius function), π is only the number $3.14159\ldots$, and prime counts are written $\#\{p \le y\}$. Dummy variables of integration (t, y) are local.

symbol	meaning
$\zeta(s)$	the Riemann zeta function
$\rho_k = \beta_k + i\gamma_k$	nontrivial zeros of the Riemann zeta function, $0 < \gamma_1 \le \gamma_2 \le \cdots$ with multiplicity
$\nu(\gamma)$	number of zeros of the Riemann zeta function with ordinate γ (multiplicities included)
$N(t)$, $N(t^{\pm})$	number of zeros with $0 < \gamma \le t$; one-sided limits
ϑ, Θ_0, ϖ	Riemann–Siegel theta function; $\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}$; $\varpi = \vartheta - \Theta_0$
g_m	Gram points, $\vartheta(g_m) = m\pi$
N_0, $\mathcal{L}$	$N_0 = \Theta_0/\pi + 1$; $\mathcal{L} = N_0'$, $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$
S, S^*, G	$S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + it)$; midpoint value at ordinates; $G = N - N_0 = S + \varpi/\pi$
$\mathcal{I}_T$, $\langle\cdot\rangle_I$	$\{k : T < \gamma_k \le 2T\}$; mean over k in a finite index set I
τ_k, τ_x, W_0	RvM clock (2.5); the same for real $x \ge 1$; principal Lambert function
s_k, δ_k, ξ_k	$s_k = k - \frac{1}{2} - N_0(\gamma_k)$; $\delta_k = \tau_k - \gamma_k$; mean-value point in $s_k = \mathcal{L}(\xi_k)\delta_k$
Δ_k	integer displacement of γ_k with respect to the Gram intervals (Theorem 3.4)
$\mathfrak{d}_k$	$s_k + \Delta_k$, $\lvert\mathfrak{d}_k\rvert < 1$ (Remark 3.17)
$\mathfrak{s}_T$	$(\frac{1}{2\pi^2}\log\log T)^{1/2}$ (Remark 3.17)
Λ, a_n	von Mangoldt function; $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$
A, A_X, $\mathcal{P}$	set of prime powers with $\sum_{n\in A}\lvert a_n\rvert < \infty$; $A_X = \{p^m : p \le X\}$; finite set of primes
S_A, S_X, $S_{\mathcal{P}}$	$\sum_{n\in A} a_n \sin(t\log n)$; S_{A_X}; $S_{\{p^m : p\in\mathcal{P}\}}$
v_A, v_X, $v_{\mathcal{P}}$	$\frac{1}{2}\sum_{n\in A} a_n^2$; v_{A_X}; $v_{\{p^m : p\in\mathcal{P}\}}$
σ_X, M_X	bounds for $\lvert S_X\rvert$ and $\lvert S_X'\rvert$, (2.11)
F, F_X	a real C^1 model of N; $F_X = N_0 + S_X$
u_k, $u_k(X)$	$k - \frac{1}{2} - F(\gamma_k)$; $k - \frac{1}{2} - F_X(\gamma_k)$
Ψ_j, κ_j	sawtooth polynomial (4.2); $\kappa_j = \frac{(2j)!}{j!(2\pi)^{2j}}$
X, c, C, T_0	prime cutoff; $c = \log X/\log T$; absolute constants
Y, Y_0, S_Y^{pr}	polynomial length; $Y_0 = T^{1/160}$; $S_Y^{\mathrm{pr}}(t) = -\frac{1}{\pi}\sum_{p<Y} p^{-1/2}\sin(t\log p)$
$\mathcal{J}$, V, $\mathcal{W}$, $\mathcal{W}_1$	primes between X and Y_0; $V = \sum_{p\in\mathcal{J}} p^{-1}$; $\mathcal{W} = S_{Y_0}^{\mathrm{pr}} - S_X$; its part over $\mathcal{J}$ (5.4)
E_0, E_X, $\mathcal{E}_2$	$S - S_{Y_0}^{\mathrm{pr}}$; $S - S_X$; $\int_T^{2T} E_X^2\,dt$
z_n, $\mathcal{D}$, $\mathcal{D}_1$, $\mathcal{D}_2$, η_n, η_n'	coefficients and Dirichlet polynomials in Lemmas 5.8–5.9
τ_k^X, δ_k^X, $r_k(X)$, χ_k	one-step clock; $\tau_k^X - \gamma_k$; $\mathcal{L}(\tau_k)\delta_k^X$; second-order error (6.3)
$\mathcal{R}_n^{\gamma}(I)$, $\mathcal{R}_n^{\tau}(I)$	prime-power responses at the zeros and at the clock points (6.1)

symbol	meaning
$\mathrm{Cr}_A(I)$, $D_X(I)$	cross term (6.2); diagonal term (6.5)
λ	real multiplier of an Euler phase
ς_n, $\mathbf{Q}$, $\mathbf{b}$, $\boldsymbol{\lambda}$, $\Upsilon_{\boldsymbol{\lambda}}$	$\varsigma_n(k) = \sin(\tau_k \log n)$; Gram matrix; projection vector; coefficients; trigonometric correction (Remark 6.3)
$\mathrm{BW}(t)$	$0.10076 \log t + 0.24460 \log\log t + 8.08344$ (Bellotti–Wong)
t_k^X	root of $F_X(t) = k - \frac{1}{2}$ (Remark 5.6)
ℓ_n	$\log n$
ψ, t_1, t_2	test function and endpoints in Lemma 4.7
L_T, $\mathcal{N}_T$, θ	$\log(T/2\pi)$; $TL_T/2\pi$; $\log p / L_T$
$\Phi(\theta)$	$(1+\theta)\log(1+\theta) + (1-\theta)\log(1-\theta)$
$\mathcal{A}_{T,p}$, $\mathcal{S}_{T,p}$	one-particle term and shifted correlation (10.5)
$\mathcal{H}_n(t)$	$\int_{14}^{t} S(y) n^{iy}\, dy$ (proof of Theorem 10.3)
f_T, $\mathfrak{m}_T$, Z_T, $\mathcal{B}_T$	$S\mathbf{1}_{[T,2T]}$; its distributional derivative; Fourier transform of $\mathfrak{m}_T$; scaled bilinear integral (10.11)
ω, ω_0, α	frequency variables; $\alpha = \omega / L_T$
$\mathfrak{K}_\theta$, h	kernel (10.12); $h(q) = \max(1 - \lvert q\rvert, 0)$
R_0, $\mathfrak{n}$, $\Pi_{\mathfrak{n}}$, q, Ξ_q	model: circle length, number of modes, projection, lattice frequency, linear statistic
ι_1, ι_2, m_i	tilt parameters; multiplication operators (Section 11.1)
y_0, w_T	fixed threshold; threshold tending to infinity
K, B	$2\,001\,052$, the number of ordinates used; saturation level (Section 9.5)
$\mathfrak{g}$, $\mathfrak{G}$, ω	continuous test function and a primitive (Theorem 4.2); real frequency variable
$\mathfrak{t}_k(X)$, y_k	transport term $S_X(\gamma_k) - S_X(\tau_k)$; one-step correction $S_X(\tau_k)/\mathcal{L}(\tau_k)$
φ, $\mathfrak{h}$, $\mathcal{M}$, $\mathfrak{b}_T$, $\mathcal{U}_k$	cutoff function; $\varphi S_X'$; centred Hardy–Littlewood maximal operator; $2\pi/\log(3T)$; sampling intervals (Lemma 7.2)
X_k, $\tilde{u}_k$, $X_\pm$, $V_\pm$	variable cutoff; its defect; extreme cutoffs in a window; $\sum_{X_- < p \le X_+} p^{-1}$ (Theorem 5.11)
$\mathcal{Q}$, $\mathfrak{w}$	maximal partial prime sum; smooth window (Lemma 5.10)
$\mathcal{I}_T^\tau$, τ_k^{loc}	$\{k : T \le \tau_k < 2T\}$; one-step clock with cutoff $\tau_k^{1/\sqrt{\log\log \tau_k}}$ (Corollary 7.5)
$\mathfrak{a}$, BW_S	$\limsup_{t\to\infty} \lvert S(t)\rvert / \log t$; Bellotti–Wong bound for $\lvert S(t)\rvert$ (Theorem 3.2)
$\mathfrak{e}$, (k, l)	exponent profile of the Riemann zeta function; exponent pair (Section 8)
$\sigma_\circ$, ϱ, $\varkappa$, $\sigma^\sharp$, $\delta^\sharp$	centre and radius of the Jensen disc; power in Backlund's function; parameters of Backlund's trick (Theorem 3.16)
$N^+(\sigma, t)$, $\mathcal{K}_n(t)$	number of zeros with $\beta > \sigma$, $0 < \gamma \le t$; symmetric remainder (Theorem 10.1)

2.2. Counting functions

Let $\vartheta(t) = \Im \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ for $t > 0$. For $t > 0$ not an ordinate [33, Thms. 9.3–9.4],

$$N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t), \qquad S(t) = \frac{1}{\pi}\arg\zeta\Big(\tfrac{1}{2} + it\Big) = O(\log t), \tag{2.1}$$

with $\arg\zeta$ defined by continuous variation from 2 to $2 + it$ and then horizontally to $\frac{1}{2} + it$. At an ordinate we keep the right-continuous N and define S by (2.1); $S^*(\gamma) = \frac{1}{2}(S(\gamma^-) + S(\gamma^+))$. Put

$$\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}, \qquad N_0(t) = \frac{1}{\pi}\Theta_0(t) + 1, \qquad \varpi(t) = \vartheta(t) - \Theta_0(t). \tag{2.2}$$

By Stirling's formula $\varpi(t) = \frac{1}{48t} + O(t^{-3})$ and $\varpi'(t) = O(t^{-2})$; in particular $|\varpi(t)|/\pi < \frac{1}{100}$ for $t \geq 14$. For all $t > 0$,

$$N(t) = N_0(t) + S(t) + \frac{\varpi(t)}{\pi}, \qquad \mathcal{L}(t) = N_0'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}. \tag{2.3}$$

Since $N(2T) - N(T) = N_0(2T) - N_0(T) + O(\log T)$,

$$\#\mathcal{I}_T = T\mathcal{L}(T) + O(T). \tag{2.4}$$

2.3. The RvM clock and the displacements

N_0 is C^∞ and strictly increasing on $(2\pi, \infty)$, with $N_0(2\pi) = -\frac{1}{8}$ and $N_0(t) \to \infty$. Hence for every real $x \geq 1$ the equation $N_0(t) = x - \frac{1}{2}$ has exactly one solution $\tau_x > 2\pi$. Writing $t = 2\pi y$, the equation is $y\log(y/e) = x - \frac{11}{8} \geq -\frac{3}{8}$; its root $y > 1$ is $y = (x - \frac{11}{8})/W_0((x - \frac{11}{8})/e)$, with $y = e$ if $x = \frac{11}{8}$ [6]. Thus

$$\tau_x = \frac{2\pi\,(x - \frac{11}{8})}{W_0\Big((x - \frac{11}{8})/e\Big)}, \qquad \frac{d\tau_x}{dx} = \frac{1}{\mathcal{L}(\tau_x)} = \frac{2\pi}{\log(\tau_x/2\pi)}, \tag{2.5}$$

the derivative by implicit differentiation. Since $\Theta_0(\tau_k) = \pi(k - \frac{3}{2})$ and $\varpi = O(1/t)$, τ_k lies in phase half way between g_{k-2} and g_{k-1}, up to $O(\tau_k^{-1}(\log\tau_k)^{-1})$.

Definition 2.1. For $k \geq 1$, $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ and $\delta_k = \tau_k - \gamma_k$.

Since $\gamma_1 > 14 > 2\pi$ and $\tau_k > 2\pi$, the mean value theorem applied to $s_k = N_0(\tau_k) - N_0(\gamma_k)$ gives

$$s_k = \mathcal{L}(\xi_k)\,\delta_k \qquad \text{for some } \xi_k \text{ between } \gamma_k \text{ and } \tau_k. \tag{2.6}$$

Lemma 2.2. *For every $k \geq 1$,*

$$N(\gamma_k^-) - N_0(\gamma_k) + \frac{1}{2} \leq s_k \leq N(\gamma_k^+) - N_0(\gamma_k) - \frac{1}{2}, \tag{2.7}$$

and

$$\Big|s_k - \frac{\varpi(\gamma_k)}{\pi}\Big| \leq \max\Big(|S(\gamma_k^-)|, |S(\gamma_k^+)|\Big) - \frac{1}{2}. \tag{2.8}$$

Proof. The indices j with $\gamma_j = \gamma_k$ are exactly $N(\gamma_k^-) + 1, \dots, N(\gamma_k^+)$, so $N(\gamma_k^-) + 1 \leq k \leq N(\gamma_k^+)$; subtract $\frac{1}{2} + N_0(\gamma_k)$. By (2.3) at $\gamma_k^\pm$ and continuity of N_0 and ϖ, (2.7) reads $L + \frac{1}{2} \leq x \leq U - \frac{1}{2}$ with $x = s_k - \varpi(\gamma_k)/\pi$, $L = S(\gamma_k^-)$, $U = S(\gamma_k^+)$, and $U - L = \nu(\gamma_k) \geq 1$. Put $m = \max(|L|, |U|)$. Then $x \leq U - \frac{1}{2} \leq m - \frac{1}{2}$ and $x \geq L + \frac{1}{2} \geq -m + \frac{1}{2}$, which is (2.8). □

2.4. Euler phases and the corrected clock

For $n \geq 2$ let $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$. For a set A of prime powers with $\sum_{n\in A}|a_n| < \infty$,

$$S_A(t) = \sum_{n\in A} a_n \sin(t\log n), \qquad v_A = \frac{1}{2}\sum_{n\in A} a_n^2; \tag{2.9}$$

the series converges absolutely and uniformly in t, so S_A is continuous and bounded. For $X \geq 2$, $\sum_{n\in A_X}|a_n| \leq \frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{1}{mp^{m/2}} < \infty$ and, since $\arg(1-z) = -\Im\sum_{m\geq 1} z^m/m$ for $|z|<1$ with the principal argument,

$$S_X(t) = -\frac{1}{\pi}\sum_{p\leq X}\arg\bigl(1-p^{-1/2-it}\bigr) = -\frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{\sin(mt\log p)}{m\,p^{m/2}} = S_{A_X}(t). \tag{2.10}$$

The sum of principal arguments is a continuous argument of the finite Euler product, not necessarily its principal argument. The series of derivatives converges uniformly, because $\sum_{p\leq X}\log p\sum_{m\geq 1}p^{-m/2} < \infty$; hence $S_X \in C^\infty(\mathbb{R})$, its derivatives may be taken termwise, and

$$|S_X(t)| \leq \sigma_X := \frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{1}{m\,p^{m/2}}, \qquad |S_X'(t)| \leq M_X := \frac{1}{\pi}\sum_{p\leq X}\frac{\log p}{\sqrt{p}-1}, \tag{2.11}$$

with $\sigma_X \ll \sqrt{X}/\log X$ and $M_X \ll \sqrt{X}$ by the prime number theorem.

Definition 2.3. For $X \geq 2$ let $F_X = N_0 + S_X$ and $u_k(X) = k - \frac{1}{2} - F_X(\gamma_k) = s_k - S_X(\gamma_k)$. The one-step clock and its residual are

$$\tau_k^X = \tau_k - \frac{S_X(\tau_k)}{\mathcal{L}(\tau_k)}, \qquad \delta_k^X = \tau_k^X - \gamma_k, \qquad r_k(X) = \mathcal{L}(\tau_k)\,\delta_k^X. \tag{2.12}$$

Since $F_X(\tau_k) - (k-\frac{1}{2}) = S_X(\tau_k)$, τ_k^X is the first step of $t \mapsto t - (F_X(t)-k+\frac{1}{2})/\mathcal{L}(t)$ from $t = \tau_k$. At a simple ordinate, (2.3) gives $N_0(\gamma_k) + S^*(\gamma_k) + \varpi(\gamma_k)/\pi = k - \frac{1}{2}$, so $u_k(X) = S^*(\gamma_k) - S_X(\gamma_k) + \varpi(\gamma_k)/\pi$ is the error of the prime approximation of the argument of the Riemann zeta function at the zero.

3. External results

The results of this section are known; they are used, not reproved.

Theorem 3.1 ([33, Thms. 9.4, 9.9 and §9.12], [24])**.** *$S(t) = O(\log t)$; $\gamma_{k+1} - \gamma_k$ is bounded; and $S_1(t) = \int_0^t S(y)\,dy = O(\log t)$.*

Theorem 3.2 (Bellotti–Wong [2, Thm. 1.1, Cor. 1.5 and (1.2)])**.** *For $T \geq e$, $\bigl|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}\bigr| \leq \mathrm{BW}(T)$ and $|S(T)| \leq \mathrm{BW}_S(T)$, where*

$$\mathrm{BW}_S(t) = 0.10076\log t + \min\{0.24460\log\log t + 7.20844,\ 1.68845\log\log t + 1.50956\}. \tag{3.1}$$

Moreover $|S(T)| \leq 2.5167$ for $0 \leq T \leq 30\,610\,046\,000$.

Earlier explicit versions are due to Trudgian [34] and to Hasanalizade, Shen and Wong [15]. The last bound rests on Platt's rigorous isolation of the zeros up to that height [2, (1.2)].

Theorem 3.3 (Selberg [28]; see [12, p. 225]). *For every integer $j \geq 1$, $\int_T^{2T} S(t)^{2j}\,dt = \kappa_j\, T(\log\log T)^j + O_j\Big(T(\log\log T)^{j-1/2}\Big)$.*

Theorem 3.4 (Selberg; Korolev [20], [19, Thm. 4.2]). *Let Δ_k be the integer such that $g_{k+\Delta_k-2} < \gamma_k \leq g_{k+\Delta_k-1}$ (in [19] the Gram points are $t_m = g_{m-1}$ and $t_{m-1} < \gamma_k \leq t_m$ with $m = k + \Delta_k$), and let $0 < \varepsilon_0 < 10^{-3}$. For every fixed $j \geq 1$, as $n_0 \to \infty$, uniformly for $n_0^{27/82+0.9\varepsilon_0} \leq h_0 \leq n_0^{27/82+\varepsilon_0}$,*

$$\sum_{n_0<k\leq n_0+h_0} \Delta_k^{2j} = \kappa_j h_0(\log\log n_0)^j\Big(1+o(1)\Big), \qquad \Big|\sum_{n_0<k\leq n_0+h_0} \Delta_k^{2j-1}\Big| \ll_j h_0(\log\log n_0)^{j-1}. \tag{3.2}$$

Theorem 3.5 (Selberg's approximation in the form of Fujii [12, §3, pp. 232–233]). *$\int_T^{2T}\Big(S(t) - S_{Y_0}^{\mathrm{pr}}(t)\Big)^4 dt \ll T$, where $Y_0 = T^{1/160}$.*

In [12, §3] Selberg's difference between $\pi^{-1}\arg L(\frac12+it,\chi)$ and a prime sum is written as five explicit terms, each bounded on p. 233 in $L^{2k}(T,T+H)$ by $\ll k^2H^{1/2k}$ under the hypotheses of Fujii's Main Theorem. For the principal character (so that the L-function is the Riemann zeta function), $k=2$, $H=T$ and Fujii's parameter $a_2 = \frac14$, the prime sum runs over $p < T^{3/480} = Y_0$, and $\frac{1}{\pi}\Im p^{-1/2-it} = -\frac{1}{\pi}p^{-1/2}\sin(t\log p)$.

Theorem 3.6 (Montgomery–Vaughan [26, Cor. 3]). *For complex z_n, $\int_T^{2T}\Big|\sum_{n\leq Y} z_n n^{-it}\Big|^2 dt = \sum_{n\leq Y}|z_n|^2\Big(T+O(n)\Big)$ with an absolute implied constant.*

(Replace z_n by $z_n n^{-iT}$ to pass from $[0,T]$ to $[T,2T]$.)

Theorem 3.7 (Mertens [25]; see [32, Ch. I.1, Thms. 1.8 and 1.10]). *For $y \geq 2$, $\sum_{p\leq y}\frac{\log p}{p} = \log y + O(1)$ and $\sum_{p\leq y}\frac1p = \log\log y + O(1)$.*

Theorem 3.8 (Kusmin–Landau [14, Thm. 2.1]). *Let Ω be real with continuous monotone derivative on $[y_1,y_2]$ and $\|\Omega'\| \geq 1/H_{\mathrm{KL}} > 0$ there, where $\|\cdot\|$ is the distance to the nearest integer. Then $\sum_{y_1<n\leq y_2} e^{2\pi i\Omega(n)} \ll H_{\mathrm{KL}}$.*

Theorem 3.9 (Landau [21]; uniform version Gonek [13], as stated in [1, (1.1)]). *For integers $n \geq 2$ and real $t \geq 2$, $\sum_{0<\gamma\leq t} n^\rho = -\frac{t}{2\pi}\Lambda(n) + O\Big(n\log(2nt)\log\log(3n)\Big)$, zeros counted with multiplicity, with an absolute implied constant.*

(Gonek's sum runs over $0<\gamma<t$; the zeros with $\gamma = t$ contribute $O(n\log t)$.)

Theorem 3.10 (Selberg [33, Thm. 9.19(C)]). *Let $N^+(\sigma,t)$ be the number of zeros $\beta+i\gamma$ of the Riemann zeta function with $\beta > \sigma$ and $0 < \gamma \leq t$, counted with multiplicity. Uniformly for $\frac12 \leq \sigma \leq 1$ and $t \geq 2$, $N^+(\sigma,t) \ll t^{1-\frac14(\sigma-\frac12)}\log t$.*

Theorem 3.11 (Selberg [28]; Tsang [35]; see [7, (1.1)]). *For fixed real $y_1 < y_2$, $\frac1T\,\mathrm{meas}\{t\in[T,2T] : y_1 < S(t)/\mathfrak{s}_T \leq y_2\} \to \frac{1}{\sqrt{2\pi}}\int_{y_1}^{y_2} e^{-y^2/2}dy$, where $\mathfrak{s}_T = (\frac{1}{2\pi^2}\log\log T)^{1/2}$.*

In [7] the statement is that $\log\zeta(\frac12+it)/\sqrt{\log\log T}$ tends to a standard complex Gaussian for t uniform in $[T,2T]$; its imaginary part is $\pi S(t)/\sqrt{\log\log T}$, of variance $\frac12$.

Theorem 3.12 (Hardy–Littlewood [31, Thm. I.1], as quoted in [18, (1.2)]; Kinnunen [18, Thm. 1.4]). *Let $\mathcal{M}f(x)=\sup_{y>0}\frac{1}{2y}\int_{x-y}^{x+y}|f|$ be the centred maximal function of $f\in L^1_{\mathrm{loc}}(\mathbb{R})$. For $f\in L^2(\mathbb{R})$, resp. $L^4(\mathbb{R})$, $\|\mathcal{M}f\|_2\ll\|f\|_2$, resp. $\|\mathcal{M}f\|_4\ll\|f\|_4$. If f and f' belong to $L^4(\mathbb{R})$, then $\mathcal{M}f$ has a locally absolutely continuous representative, $(\mathcal{M}f)'\in L^4(\mathbb{R})$, and $|(\mathcal{M}f)'|\le\mathcal{M}(f')$ almost everywhere.*

Theorem 3.13 (Carleson [5], Hunt [16]; on $\mathbb{R}$ see [23]). *For $f\in L^2(\mathbb{R})\cap L^4(\mathbb{R})$ with Fourier transform $\mathcal{F}f(\omega)=\int_{\mathbb{R}}f(t)e^{-i\omega t}dt$, let $\mathcal{C}f(x)=\sup_{\omega_1\in\mathbb{R}}\bigl|\frac{1}{2\pi}\int_{-\infty}^{\omega_1}\mathcal{F}f(\omega)e^{i\omega x}d\omega\bigr|$. Then $\|\mathcal{C}f\|_4\ll\|f\|_4$.*

In [23] the statement is that $f\mapsto\sup_{\xi\in\mathbb{R}}|\mathcal{H}(e^{i\xi\cdot}f)|$, $\mathcal{H}$ the Hilbert transform, is bounded on $L^p(\mathbb{R})$ for $1<p<\infty$; since the partial Fourier integral up to ω_1 is $e^{i\omega_1 x}\cdot\frac12(\mathrm{Id}\pm i\mathcal{H})(e^{-i\omega_1\cdot}f)(x)$, the operator $\mathcal{C}f$ is at most $\frac12(|f|+\sup_\xi|\mathcal{H}(e^{i\xi\cdot}f)|)$.

Theorem 3.14 ([17, Lemma 2.8 and (7.57)]; Bourgain [4, §5, Thm. 6]). *(a) If (k,l) is an exponent pair, so is $A(\mathsf{k},\mathsf{l})=\bigl(\frac{\mathsf{k}}{2\mathsf{k}+2},\frac12+\frac{\mathsf{l}}{2\mathsf{k}+2}\bigr)$. (b) If (k,l) is an exponent pair with $\mathsf{l}-\mathsf{k}\ge\frac12$, then $\zeta(\mathsf{l}-\mathsf{k}+it)\ll|t|^{\mathsf{k}}\log|t|$ for $|t|\ge2$. (c) For every $\varepsilon>0$, $(\frac{13}{84}+\varepsilon,\frac{55}{84}+\varepsilon)$ is an exponent pair.*

For (b): [17, (7.57)] gives $\zeta(\sigma+it)\ll|t|^{(\mathsf{k}+\mathsf{l}-\sigma)/2}\log|t|$ for $\frac12\le\sigma\le\mathsf{l}-\mathsf{k}$; take $\sigma=\mathsf{l}-\mathsf{k}$.

Theorem 3.15 (Phragmén–Lindelöf principle in a strip [2, Lemma 2.5], [34, Lemma 3]). *Let $\sigma_1<\sigma_2$ and $e_1\ge e_2\ge0$. If $|\zeta(\sigma_i+it)|\le C_i|t|^{e_i}\log|t|$ for $|t|\ge2$ and $i=1,2$, then $|\zeta(\sigma+it)|\ll|t|^{e(\sigma)}\log|t|$ uniformly for $\sigma_1\le\sigma\le\sigma_2$ and $|t|\ge2$, where e is the linear function with $e(\sigma_i)=e_i$.*

(Apply [2, Lemma 2.5] to the entire function $(s-1)\zeta(s)$ of order 1, with exponents $1+e_i$ and one power of the logarithm on the two lines.)

Theorem 3.16 (Backlund's method [2, Lemmas 2.1, 2.3 and 2.4], after [34, 15]). *Let T be large, $\varkappa\in\mathbb{N}$, $\mathfrak{f}_\varkappa(s)=\frac12\bigl[((s+iT-1)\zeta(s+iT))^\varkappa+((s-iT-1)\zeta(s-iT))^\varkappa\bigr]$, and for a disc $D(\sigma_\circ,\varrho)$ let $\mathcal{Z}_\varkappa(\sigma_\circ,\varrho)=\frac{1}{\varkappa}\sum_z\log\frac{\varrho}{|z-\sigma_\circ|}$ over the zeros z of $\mathfrak{f}_\varkappa$ in $D(\sigma_\circ,\varrho)$. Let $1<\sigma_\circ<\varrho$, $\sigma^\sharp=\sigma_\circ+(\sigma_\circ-\frac12)^2/\varrho$, $\delta^\sharp=2\sigma_\circ-\sigma^\sharp-\frac12$, and assume $0<\delta^\sharp<\frac12$.*

(a) If $F_\circ:[-\pi,\pi]\to\mathbb{R}$ is even and $F_\circ(\theta)\ge\frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(\sigma_\circ+\varrho e^{i\theta})|$ for all $\varkappa$ and θ, then there are $\varkappa_m\to\infty$ with $\limsup_m\mathcal{Z}_{\varkappa_m}(\sigma_\circ,\varrho)\le\log\frac{\zeta(\sigma_\circ)}{\sqrt{(\sigma_\circ-1)^2+T^2}\,\zeta(2\sigma_\circ)}+\frac{1}{\pi}\int_0^\pi F_\circ(\theta)\,d\theta$.

(b) For every $\varkappa$, $\bigl|\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg\bigl((\sigma+iT-1)\zeta(\sigma+iT)\bigr)\bigr|\le\frac{\pi\mathcal{Z}_\varkappa(\sigma_\circ,\varrho)}{2\log(\varrho/(\sigma_\circ-\frac12))}+\frac{\pi}{100}+\frac{3\pi}{2\varkappa}+\frac{\pi}{4}$.

In (b) the term $\frac{\pi}{100}$ bounds $\frac12E(T,\delta^\sharp)$ of [2, Lemma 2.3], by the monotonicity in $\delta^\sharp$ and the bound for $E(T,\frac12)$ in [2, Lemma 2.4]; (a) is [2, Lemma 2.1], which also requires $\sigma_\circ-\varrho<\frac12<1<\sigma_\circ<\sigma^\sharp<\sigma_\circ+\varrho$, a consequence of $1<\sigma_\circ<\varrho$.

Remark 3.17 (Known consequences for the clock). (a) *Moments and law of s_k.* Since $\vartheta(\gamma_k)/\pi \in (k+\Delta_k-2, k+\Delta_k-1]$ and $\vartheta/\pi = N_0 - 1 + \varpi/\pi$, one has $s_k + \Delta_k = k + \Delta_k - \frac{1}{2} - N_0(\gamma_k) \in [-\frac{1}{2} + \frac{\varpi(\gamma_k)}{\pi}, \frac{1}{2} + \frac{\varpi(\gamma_k)}{\pi})$, so $s_k = -\Delta_k + \mathfrak{d}_k$ with $|\mathfrak{d}_k| < 1$. The index set $\mathcal{I}_T$ is the integer interval $(N(T), N(2T)]$, of length $\asymp T\log T$. Cover it by consecutive windows $(n_0, n_0+h_0]$ with $h_0 = \lfloor n_0^{27/82+0.95\varepsilon_0} \rfloor$, the last one shifted to end at $N(2T)$, so that it overlaps the previous one in at most $h_0 = O(T^{-1/2}\#\mathcal{I}_T)$ indices; in each window $\log\log n_0 = \log\log T + o(1)$. Summing (3.2) over the windows, and bounding the doubly counted indices by one window sum (for odd powers, by Hölder's inequality and the even moments), gives $\langle \Delta_k^{2j}\rangle_{\mathcal{I}_T} = \kappa_j(\log\log T)^j(1+o(1))$ and $\langle \Delta_k^{2j-1}\rangle_{\mathcal{I}_T} = O_j((\log\log T)^{j-1})$. Expanding $s_k^r = (-\Delta_k + \mathfrak{d}_k)^r$ and bounding $\langle |\Delta_k|^{r-i}\rangle_{\mathcal{I}_T} \le \langle \Delta_k^{2\lceil r/2\rceil}\rangle_{\mathcal{I}_T}^{(r-i)/2\lceil r/2\rceil} \ll (\log\log T)^{(r-i)/2}$ for $1 \le i \le r$, one obtains for every fixed $j \ge 1$

$$\langle s_k^{2j}\rangle_{\mathcal{I}_T} = \kappa_j(\log\log T)^j\Big(1+o(1)\Big), \qquad \langle s_k^{2j-1}\rangle_{\mathcal{I}_T} = O_j\Big((\log\log T)^{j-1}\Big). \tag{3.3}$$

With $\mathfrak{s}_T^2 = \frac{\log\log T}{2\pi^2}$, so that $\kappa_j(\log\log T)^j = \frac{(2j)!}{j!2^j}\mathfrak{s}_T^{2j}$, the normalized moments $\langle (s_k/\mathfrak{s}_T)^r\rangle_{\mathcal{I}_T}$ therefore tend to the moments of a standard normal variable Z for every $r \ge 1$: the even ones by the first relation, the odd ones to 0 because $(\log\log T)^{j-1}/\mathfrak{s}_T^{2j-1} \ll_j (\log\log T)^{-1/2}$. The even moments alone determine only the law of $|s_k|$; it is the odd-moment bound in (3.2) that gives the sign. The normal law is determined by its moments, so the method of moments [3, Thm. 30.2] shows that the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to that of Z. Only consequences for $|s_k|$ are used in this paper: $\langle s_k^2\rangle_{\mathcal{I}_T} \sim \frac{1}{2\pi^2}\log\log T$, $\limsup_k |s_k| = \infty$, and $\#\{k \in \mathcal{I}_T : |s_k| \le y_0\} = o(\#\mathcal{I}_T)$ for every fixed $y_0 > 0$; for these the even moments suffice.

(b) *Bound for δ_k.* Theorem 3.2, applied as $T \to \gamma_k^{\pm}$ (both sides have one-sided limits), gives $|N(\gamma_k^{\pm}) - N_0(\gamma_k) + \frac{7}{8}| \le \mathrm{BW}(\gamma_k)$, and (2.7) gives $-\mathrm{BW}(\gamma_k) - \frac{3}{8} \le s_k \le \mathrm{BW}(\gamma_k) - \frac{11}{8}$. As $\mathcal{L}$ is positive and increasing on $(2\pi,\infty)$ and $s_k = \int_{\gamma_k}^{\tau_k}\mathcal{L}$,

$$-\frac{\mathrm{BW}(\gamma_k)+3/8}{\mathcal{L}(\tau_k)} \le \delta_k \le \frac{\mathrm{BW}(\gamma_k)-11/8}{\mathcal{L}(\gamma_k)} \qquad (k \ge 1). \tag{3.4}$$

Hence $\delta_k = O(1)$, $\mathcal{L}(\tau_k)/\mathcal{L}(\gamma_k) \to 1$, and $\limsup_k |\delta_k| \le 2\pi \cdot 0.10076 = 0.6331\ldots$, an immediate consequence of [2]. A limit superior is not a supremum: $|\delta_{34}| = 0.993711$.

4. Theorem A: an exact defect identity at the zeros

Theorem 4.1. *Let $1 \le k_1 < k_2$ be indices, F a real C^1 function on an open interval containing $[\gamma_{k_1}, \gamma_{k_2}]$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$ for $k_1 \le k \le k_2$. For every integer $j \ge 1$,*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^{2j} F'(t)\,dt = \sum_{k=k_1+1}^{k_2-1} \Psi_j(u_k) + \frac{(\frac{1}{2}+u_{k_1})^{2j+1} + (\frac{1}{2}-u_{k_2})^{2j+1}}{2j+1}, \tag{4.1}$$

where

$$\Psi_j(u) = \frac{(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}}{2j+1} = \sum_{l=0}^{j}\binom{2j+1}{2l}\frac{4^{l-j}}{2j+1}\,u^{2l} = u^{2j} + \sum_{l<j}\binom{2j+1}{2l}\frac{4^{l-j}}{2j+1}u^{2l}. \tag{4.2}$$

In particular, since $\Psi_1(u) = u^2 + \frac{1}{12}$,

$$\sum_{k=k_1+1}^{k_2-1} u_k^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^2 F'(t)\,dt - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2} + u_{k_1})^3 + (\frac{1}{2} - u_{k_2})^3}{3}. \tag{4.3}$$

Proof. Existence. On $[\gamma_{k_1}, \gamma_{k_2}]$ the function N is a step function with finitely many jumps (Theorem 3.1) and F, F' are continuous; the integrand is bounded and piecewise continuous, hence Riemann integrable.

One gap. Fix k with $k_1 \le k < k_2$. If $\gamma_k < \gamma_{k+1}$, then $N(t) = k$ for $t \in (\gamma_k, \gamma_{k+1})$, because exactly k ordinates counted with multiplicity are $\le t$. The function $t \mapsto -(k - F(t))^{2j+1}/(2j+1)$ is C^1 with derivative $(k - F(t))^{2j} F'(t)$ (chain rule; no monotonicity of F is needed), so

$$\begin{aligned} \int_{\gamma_k}^{\gamma_{k+1}} \Big(k - F(t)\Big)^{2j} F'(t)\,dt &= \frac{(k - F(\gamma_k))^{2j+1} - (k - F(\gamma_{k+1}))^{2j+1}}{2j+1} \\ &= \frac{(\frac{1}{2} + u_k)^{2j+1} + (\frac{1}{2} - u_{k+1})^{2j+1}}{2j+1}, \end{aligned} \tag{4.4}$$

because $k - F(\gamma_k) = \frac{1}{2} + u_k$ and $k - F(\gamma_{k+1}) = -\frac{1}{2} + u_{k+1}$. If $\gamma_k = \gamma_{k+1}$, the left side of (4.4) is 0; then $u_{k+1} = u_k + 1$, and the right side equals $((\frac{1}{2} + u_k)^{2j+1} + (-\frac{1}{2} - u_k)^{2j+1})/(2j+1) = 0$. So (4.4) holds for every k, and the values of N at the finitely many ordinates do not affect the integrals.

Summation and boundary terms. Summing (4.4) over $k_1 \le k < k_2$ (a finite sum), each k with $k_1 < k < k_2$ contributes $(\frac{1}{2} + u_k)^{2j+1}$ from the gap starting at γ_k and $(\frac{1}{2} - u_k)^{2j+1}$ from the gap ending at γ_k, that is $\Psi_j(u_k)$; the unmatched terms are $(\frac{1}{2} + u_{k_1})^{2j+1}$ and $(\frac{1}{2} - u_{k_2})^{2j+1}$. In the binomial expansion of $(\frac{1}{2} + u)^{2j+1} + (\frac{1}{2} - u)^{2j+1}$ the odd powers of u cancel and u^{2l} has coefficient $2\binom{2j+1}{2l} 2^{2l-2j-1}$, which gives (4.2); the leading coefficient is $\binom{2j+1}{2j}/(2j+1) = 1$. □

Theorem 4.2. *Let $k_1 < k_2$, F and u_k be as in Theorem 4.1, and let $\mathfrak{g} : \mathbb{R} \to \mathbb{C}$ be continuous with a primitive $\mathfrak{G}$. Then*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \mathfrak{g}\Big(N(t) - F(t)\Big) F'(t)\,dt = \sum_{k=k_1+1}^{k_2-1} \Big[\mathfrak{G}\Big(u_k + \tfrac{1}{2}\Big) - \mathfrak{G}\Big(u_k - \tfrac{1}{2}\Big)\Big] + \mathfrak{G}\Big(u_{k_1} + \tfrac{1}{2}\Big) - \mathfrak{G}\Big(u_{k_2} - \tfrac{1}{2}\Big). \tag{4.5}$$

Proof. The integrand is bounded and piecewise continuous, as in the proof of Theorem 4.1. If $\gamma_k < \gamma_{k+1}$, then $N = k$ on (γ_k, γ_{k+1}) and $t \mapsto -\mathfrak{G}(k - F(t))$ is C^1 with derivative $\mathfrak{g}(k - F(t))F'(t)$, so the integral over $[\gamma_k, \gamma_{k+1}]$ is $\mathfrak{G}(k - F(\gamma_k)) - \mathfrak{G}(k - F(\gamma_{k+1})) = \mathfrak{G}(u_k + \frac{1}{2}) - \mathfrak{G}(u_{k+1} - \frac{1}{2})$. If $\gamma_k = \gamma_{k+1}$, the integral is 0 and $u_{k+1} = u_k + 1$, so the same formula holds. Summing over $k_1 \le k < k_2$, each k with $k_1 < k < k_2$ contributes $\mathfrak{G}(u_k + \frac{1}{2})$ from the gap that starts at γ_k and $-\mathfrak{G}(u_k - \frac{1}{2})$ from the gap that ends there. □

For $\mathfrak{g}(x) = x^{2j}$ this is (4.1), since $\mathfrak{G}(u + \frac{1}{2}) - \mathfrak{G}(u - \frac{1}{2}) = \Psi_j(u)$.

Corollary 4.3. *For real $\omega \neq 0$,*

$$\frac{2\sin(\omega/2)}{\omega} \sum_{k=k_1+1}^{k_2-1} e^{i\omega u_k} = \int_{\gamma_{k_1}}^{\gamma_{k_2}} e^{i\omega(N(t) - F(t))} F'(t)\,dt - \frac{e^{i\omega(u_{k_1} + 1/2)} - e^{i\omega(u_{k_2} - 1/2)}}{i\omega}. \tag{4.6}$$

Proof. Theorem 4.2 with $\mathfrak{g}(x) = e^{i\omega x}$ and $\mathfrak{G}(x) = e^{i\omega x}/(i\omega)$, for which $\mathfrak{G}(u+\frac{1}{2}) - \mathfrak{G}(u-\frac{1}{2}) = e^{i\omega u} \cdot 2\sin(\omega/2)/\omega$. □

Corollary 4.4 (Signed law of s_k from the t-law of S)**.** *For fixed real η, $\langle e^{i\eta s_k/\mathfrak{s}_T}\rangle_{\mathcal{I}_T} \to e^{-\eta^2/2}$. Hence the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to the standard normal law.*

Proof. Let $\eta \neq 0$, $\omega = \eta/\mathfrak{s}_T$, $k_1 = N(T)$ and $k_2 = N(2T)+1$, so that $\{k_1+1, \dots, k_2-1\} = \mathcal{I}_T$. Apply (4.6) with $F = N_0$, so $u_k = s_k$ and $N - F = G$. By Theorem 3.1 the intervals $[\gamma_{k_1}, T]$ and $[2T, \gamma_{k_2}]$ have bounded length, so the integral equals $\int_T^{2T} e^{i\omega G}\mathcal{L}\,dt + O(\log T)$; the boundary term is $O(1/|\omega|) = O(\mathfrak{s}_T)$; and $2\sin(\omega/2)/\omega = 1 + O(\omega^2)$. Divide by $\#\mathcal{I}_T = \int_T^{2T} \mathcal{L}\,dt + O(\log T)$. On $[T, 2T]$, $|e^{i\omega G} - e^{i\omega S}| \le |\omega||\varpi|/\pi \ll (\mathfrak{s}_T T)^{-1}$ and $\mathcal{L}(t) = \mathcal{L}(T)(1 + O(1/\log T))$, whence

$$\langle e^{i\omega s_k}\rangle_{\mathcal{I}_T} = \frac{1}{T}\int_T^{2T} e^{i\eta S(t)/\mathfrak{s}_T} dt + O\Big(\omega^2 + \frac{1}{\log T}\Big).$$

By Theorem 3.11 and the definition of convergence in distribution, applied to the bounded continuous function $y \mapsto e^{i\eta y}$, the integral tends to $e^{-\eta^2/2}$, and $\omega \to 0$. The last statement is Lévy's continuity theorem [3, Thm. 26.3]. □

Remark 4.5. $\Psi_j(u) = \int_{u-1/2}^{u+1/2} x^{2j} dx$ is the mean of x^{2j} over the unit interval centred at u; in particular $\Psi_1(u) = u^2 + \frac{1}{12}$ and $\frac{1}{12} = \int_{-1/2}^{1/2} x^2 dx$. The constant comes from the unit jump of $N - F$ at each zero, through the primitive in Theorem 4.2; when F is not monotone, $N - F$ need not run through an interval of length one between consecutive zeros, and the identities remain exact with the integral counted with sign. With $F = N_0$, Theorem 3.3 and induction on j through the lower-degree terms of Ψ_j recover the even moments in (3.3) without Gram intervals; Corollary 4.4 gives the sign without Korolev's odd moments. The main purpose of Theorem 4.1 is Theorem 5.1, where $F = F_X$ need not be monotone and no Gram-interval structure is available.

Corollary 4.6. *For $X \ge 2$ and $1 \le k_1 < k_2$,*

$$\begin{aligned}\sum_{k=k_1+1}^{k_2-1} u_k(X)^2 = {} & \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(S - S_X + \frac{\varpi}{\pi}\Big)^2 \big(\mathcal{L} + S_X'\big)dt \\ & - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2} + u_{k_1}(X))^3 + (\frac{1}{2} - u_{k_2}(X))^3}{3}.\end{aligned} \tag{4.7}$$

Proof. F_X is C^∞ on $(0, \infty)$ (Section 2.4), $F_X' = \mathcal{L} + S_X'$, and $N - F_X = S - S_X + \varpi/\pi$ by (2.3); apply Theorem 4.1 with $j = 1$. □

Lemma 4.7 (First-moment identity)**.** *Let $2\pi < t_1 < t_2$ be points which are not ordinates and $\psi \in C^1[t_1, t_2]$, real or complex. Then*

$$\sum_{t_1 < \gamma_k \le t_2} s_k \psi(\gamma_k) = \int_{t_1}^{t_2} G(t)\psi(t)\mathcal{L}(t)\,dt + \frac{1}{2}\Big[\psi\, G^2\Big]_{t_1}^{t_2} - \frac{1}{2}\int_{t_1}^{t_2} G(t)^2\psi'(t)\,dt. \tag{4.8}$$

Proof. On $[t_1, t_2]$ there are finitely many ordinates, none at t_1 or t_2. The function $G = N - N_0$ is right-continuous and of bounded variation, C^∞ with $G' = -\mathcal{L}$ between ordinates, and it jumps by $\nu(\gamma)$ at an ordinate γ. Let $G^*(\gamma) = \frac{1}{2}(G(\gamma^-) + G(\gamma^+))$. The indices with $\gamma_k = \gamma$

are $N(\gamma^-)+1,\dots,N(\gamma^-)+\nu(\gamma)$, so $\sum_{\gamma_k=\gamma}(k-\frac12)=\nu(\gamma)N(\gamma^-)+\frac{\nu(\gamma)^2}{2}=\nu(\gamma)\frac{N(\gamma^-)+N(\gamma^+)}{2}$ and, N_0 being continuous, $\sum_{\gamma_k=\gamma}s_k=\nu(\gamma)G^*(\gamma)$. The left side of (4.8) therefore equals the finite sum $\sum_{t_1<\gamma<t_2}\nu(\gamma)G^*(\gamma)\psi(\gamma)$ over distinct ordinates. As Lebesgue–Stieltjes measures on $(t_1,t_2]$,

$$d(G^2)=-2G\mathcal{L}\,dt+\sum_{t_1<\gamma<t_2}2G^*(\gamma)\nu(\gamma)\,\delta_\gamma, \tag{4.9}$$

because $(G^2)'=2GG'=-2G\mathcal{L}$ between ordinates and $G(\gamma^+)^2-G(\gamma^-)^2=(G(\gamma^+)+G(\gamma^-))\nu(\gamma)=2G^*(\gamma)\nu(\gamma)$. Integrating the bounded Borel function ψ against (4.9) gives $\sum_{t_1<\gamma<t_2}\nu(\gamma)G^*(\gamma)\psi(\gamma)=\frac12\int_{(t_1,t_2]}\psi\,d(G^2)+\int_{t_1}^{t_2}G\psi\mathcal{L}\,dt$. Since ψ is C^1 and G^2 is right-continuous of bounded variation and continuous at t_1 and t_2, integration by parts for Lebesgue–Stieltjes integrals [8, §3.5] gives $\int_{(t_1,t_2]}\psi\,d(G^2)=[\psi G^2]_{t_1}^{t_2}-\int_{t_1}^{t_2}G^2\psi'\,dt$. All integrands are bounded and measurable, so all integrals exist. □

5. Theorem B: how much prime information the clock needs

Theorem 5.1. *There are absolute constants T_0 and C such that for $T\ge T_0$ and every X with $2\le X\le T^{1/2}$, writing $c=\log X/\log T$,*

$$\Big|\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}u_k(X)^2-\frac{1}{2\pi^2}\log\frac1c\Big|\le C\Big(1+\sqrt{\log(1/c)}\Big). \tag{5.1}$$

Here $\log(1/c)=\log\log T-\log\log X\in[\log 2,\log\log T-\log\log 2]$. For fixed c both the main term and the error are bounded; the theorem does not identify a limit at fixed c. It concerns the defect at the true ordinate and does not select a root of $F_X(t)=k-\frac12$.

Corollary 5.2 (fixed cutoff)**.** *For fixed $X\ge 2$, $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}u_k(X)^2=\frac{1}{2\pi^2}\log\log T+O_X(\sqrt{\log\log T})$, and $\#\{k\in\mathcal{I}_T:|u_k(X)|\le y_0\}=o(\#\mathcal{I}_T)$ for every fixed $y_0>0$.*

Proof. The first statement is (5.1). Since $|u_k(X)-s_k|\le\sigma_X$, $|u_k(X)|\le y_0$ implies $|s_k|\le y_0+\sigma_X$; apply Remark 3.17(a). □

Corollary 5.3 (power cutoff)**.** *For fixed $0<c\le\frac12$ and $X=T^c$,*

$$\limsup_{T\to\infty}\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}u_k(T^c)^2\le\frac{1}{2\pi^2}\log\frac1c+C\Big(1+\sqrt{\log(1/c)}\Big), \tag{5.2}$$

and $\#\{k\in\mathcal{I}_T:|u_k(T^c)|>w_T\}=o(\#\mathcal{I}_T)$ whenever $w_T\to\infty$. The same holds for $X=N(T)^c$ with fixed $0<c<\frac12$, since then $X=T^{c'}$ with $c'=c+O(\log\log T/\log T)$.

Proof. (5.1) and Chebyshev's inequality. □

Corollary 5.4 (dichotomy)**.** *Let $X=X(T)$ with $2\le X\le T^{1/2}$. The mean square of $u_k(X)$ over $\mathcal{I}_T$ is bounded as $T\to\infty$ if and only if $\liminf_{T\to\infty}\log X/\log T>0$. For fixed $X_0\ge 2$ and $c\in(0,\frac12]$,*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(u_k(X_0)^2-u_k(T^c)^2\Big)=\frac{1}{2\pi^2}\log\log T+O\Big(\sqrt{\log\log T}\Big). \tag{5.3}$$

Proof. Both directions follow from (5.1), because $y \mapsto \frac{y}{2\pi^2} \pm C(1+\sqrt{y})$ is bounded on bounded sets and tends to infinity; (5.3) is the difference of the two cases of (5.1). □

Remark 5.5. Since $\log \#\{p \le X\} = \log X + O(\log\log X)$, the condition of Corollary 5.4 says that the logarithm of the number of primes must grow proportionally to $\log T$. A fixed number of primes, as in the computations reported in [11, §XXI, Note added 7], leaves almost all zeros violating the corrected equation by more than any fixed amount; the requirement that the number of primes tend to infinity [22, Sect. 3] is necessary but not sufficient. The range $c > \frac{1}{2}$ is not covered. In Theorem 5.1 the cutoff is constant on each window $\mathcal{I}_T$; Theorem 5.11 below allows the cutoff $X_k = (\tau_k/2\pi)^c$, which varies at every ordinate.

Remark 5.6 (Localization of the ordinates). If $M_X \le \frac{1}{2}\mathcal{L}(T)$, then $F_X' \ge \mathcal{L} - M_X \ge \frac{1}{2}\mathcal{L}(T)$ on $[T, 2T]$, so F_X is strictly increasing there. For $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac{1}{2} \le F_X(2T)$ the equation $F_X(t) = k - \frac{1}{2}$ has exactly one root $t_k^X \in [T, 2T]$ (intermediate value theorem), and $u_k(X) = F_X(t_k^X) - F_X(\gamma_k)$ gives $|\gamma_k - t_k^X| \le 2|u_k(X)|/\mathcal{L}(T)$ (mean value theorem). As $M_X \asymp \sqrt{X}$, this applies only for $X \ll \log^2 T$, and in that range the localization gains nothing in order (Proposition 7.6). For $X = T^c$, F_X need not be monotone, and Theorem 5.1 controls the equation at the true ordinates without selecting a root; Theorem 7.3 replaces the root by one Newton step.

5.1. Lemmas on prime polynomials

Fix $T \ge T_0$ and $2 \le X \le T^{1/2}$, let $Y_0 = T^{1/160}$, and split $S_X = S_X^{(1)} + S_X^{(2)} + S_X^{(\ge 3)}$ according to $m = 1$, $m = 2$, $m \ge 3$ in (2.10); the three series converge absolutely and uniformly. Let $\mathcal{J}$ be the set of primes in (X, Y_0) if $X < Y_0$ and in $[Y_0, X]$ if $X \ge Y_0$, and put

$$\begin{aligned} \mathcal{W}_1(t) &= -\frac{1}{\pi}\sum_{p\in\mathcal{J}} \frac{\sin(t\log p)}{\sqrt{p}}, \qquad V = \sum_{p\in\mathcal{J}} \frac{1}{p}, \\ \mathcal{W} &= S_{Y_0}^{\mathrm{pr}} - S_X = \pm\mathcal{W}_1 - S_X^{(2)} - S_X^{(\ge 3)}, \end{aligned} \tag{5.4}$$

with $+$ if $X < Y_0$ and $-$ if $X \ge Y_0$. The last identity holds because $S_X^{(1)}$ is the sum of $-\frac{1}{\pi}p^{-1/2}\sin(t\log p)$ over the primes $p \le X$, while $S_{Y_0}^{\mathrm{pr}}$ is the same sum over $p < Y_0$.

Lemma 5.7. $V = \log(1/c) + O(1)$ *uniformly for* $\log 2/\log T \le c \le \frac{1}{2}$.

Proof. By Theorem 3.7, $V = |\log\log Y_0 - \log\log X| + O(1) = |\log\frac{1}{160c}| + O(1)$. For $c \le \frac{1}{160}$ this is $\log\frac{1}{c} - \log 160 + O(1)$; for $\frac{1}{160} < c \le \frac{1}{2}$ both V and $\log\frac{1}{c}$ are $O(1)$. □

Lemma 5.8. *For real* z_n, $2 \le n \le Y$,

$$\int_T^{2T} \Big(\sum_{n\le Y} z_n \sin(t\log n)\Big)^2 dt = \frac{T}{2}\sum_{n\le Y} z_n^2 + O\Big(\sum_{n\le Y} n z_n^2 + \Big(\sum_{n\le Y}|z_n|\Big)^2\Big), \tag{5.5}$$

and the same with cos *in place of* sin.

Proof. With the finite sum $\mathcal{D}(t) = \sum z_n n^{-it}$, $\sum z_n \sin(t\log n) = -\Im\mathcal{D}$ and $(\Im\mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 - \Re\mathcal{D}^2)$. Theorem 3.6 evaluates $\int|\mathcal{D}|^2$, and $|\int_T^{2T}(n_1n_2)^{-it}dt| \le 2/\log(n_1n_2) \le 2/\log 4$ gives $|\int\mathcal{D}^2| \le (2/\log 4)(\sum|z_n|)^2$. For cos use $\Re\mathcal{D}$ and $(\Re\mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 + \Re\mathcal{D}^2)$. □

Lemma 5.9. *With absolute implied constants,*

$$\int_T^{2T} \mathcal{W}^2\,dt = \frac{T}{2\pi^2}V + O\Big(T(1+\sqrt{V})\Big), \tag{5.6}$$

$$\int_T^{2T} \mathcal{W}^4\,dt \ll T(1+V)^2, \tag{5.7}$$

$$\int_T^{2T} S_X'(t)^2\,dt \ll T(1+\log X)^2. \tag{5.8}$$

Proof. All primes involved are $\le \max(X, Y_0) \le T^{1/2}$. Lemma 5.8 with $z_p = \frac{1}{\pi\sqrt{p}}$ ($p \in \mathcal{J}$) gives $\int \mathcal{W}_1^2 = \frac{T}{2\pi^2}V + O(T^{1/2} + (\sum_{p\le T^{1/2}} p^{-1/2})^2) = \frac{T}{2\pi^2}V + O(T^{1/2})$. For $S_X^{(2)}(t) = -\frac{1}{2\pi}\sum_{p\le X} p^{-1}\sin(t\log p^2)$ the same lemma gives $\int (S_X^{(2)})^2 \ll T\sum_p p^{-2} + \#\{p \le X\} + (\sum_{p\le X} p^{-1})^2 \ll T$. Finally $|S_X^{(\ge 3)}| \le \frac{1}{\pi}\sum_p \sum_{m\ge 3} \frac{1}{mp^{m/2}} \le \frac{1}{\pi}\sum_p \frac{p^{-3/2}}{1-p^{-1/2}} < \infty$, uniformly in X. Expanding $\mathcal{W}^2$ and bounding the cross terms by Cauchy–Schwarz gives (5.6).

For (5.7) use $(y_1+y_2+y_3)^4 \le 27(y_1^4+y_2^4+y_3^4)$. Let $\mathcal{D}_1(t) = \sum_{p\in\mathcal{J}} p^{-1/2}p^{-it}$, so that $|\mathcal{W}_1| \le |\mathcal{D}_1|/\pi$ and $\mathcal{D}_1^2 = \sum_n \eta_n n^{-it}$ with $\eta_{p^2} = p^{-1}$, $\eta_{p_1p_2} = 2(p_1p_2)^{-1/2}$ for $p_1 < p_2$, all $n \le T$. Theorem 3.6 gives $\int |\mathcal{D}_1|^4 = \sum |\eta_n|^2(T + O(n)) \ll T\sum |\eta_n|^2 \le T(1+2V^2)$. For $S_X^{(2)}$ let $\mathcal{D}_2(t) = \sum_{p\le X} p^{-1}(p^2)^{-it}$, so that $|S_X^{(2)}| \le |\mathcal{D}_2|/2\pi$ and $\mathcal{D}_2^2 = \sum_n \eta_n' n^{-it}$ over $n = (p_1p_2)^2$ with $|\eta_n'| \le 2(p_1p_2)^{-1}$; then $\sum |\eta_n'|^2 \ll 1$ and $\sum n|\eta_n'|^2 \le 4\#\{p \le X\}^2 \le 4X^2 \le 4T$. $S_X^{(\ge 3)}$ is bounded.

For (5.8), $S_X'(t) = \sum_{n\in A_X} a_n \log n \cos(t\log n)$ (termwise differentiation, Section 2.4). The terms $m = 1$ have $|z_p| = \frac{\log p}{\pi\sqrt{p}}$; Lemma 5.8 and Theorem 3.7 give $\ll T\log X\sum_{p\le X}\frac{\log p}{p} + \sum_{p\le X}\log^2 p + (\sum_{p\le X}\frac{\log p}{\sqrt{p}})^2 \ll T(1+\log X)^2 + X\log X$, and $X \le T^{1/2}$. The terms $m = 2$ have $|z_{p^2}| = \frac{\log p}{\pi p}$ and contribute $\ll T + X\log X + \log^2 X$. The terms $m \ge 3$ are bounded by $\frac{1}{\pi}\sum_p \log p \sum_{m\ge 3} p^{-m/2} < \infty$. Combine with $(y_1+y_2+y_3)^2 \le 3(y_1^2+y_2^2+y_3^2)$. □

5.2. Proof of Theorem 5.1

Proof. Integrability. On $[T, 2T]$, S is bounded (Theorem 3.1) and piecewise continuous, and S_X, S_X', ϖ are continuous; all integrals below exist.

Second and fourth moments of E_X. Let $E_0 = S - S_{Y_0}^{\mathrm{pr}}$ and $E_X = S - S_X = E_0 + \mathcal{W}$. Theorem 3.5 and Hölder's inequality give

$$\int_T^{2T} E_0^4\,dt \ll T, \qquad \int_T^{2T} E_0^2\,dt \le T^{1/2}\Big(\int_T^{2T} E_0^4\,dt\Big)^{1/2} \ll T. \tag{5.9}$$

Let $\mathcal{E}_2 = \int_T^{2T} E_X^2\,dt$. From (5.6), (5.9) and Cauchy–Schwarz,

$$\mathcal{E}_2 = \frac{T}{2\pi^2}V + O\Big(T(1+\sqrt{V})\Big), \tag{5.10}$$

and from (5.7), (5.9) and $(y_1+y_2)^4 \le 8(y_1^4+y_2^4)$,

$$\int_T^{2T} E_X^4\,dt \ll T(1+V)^2. \tag{5.11}$$

Transfer to the ordinates. Let k_1, k_2 be the smallest and largest indices in $\mathcal{I}_T$ ($\mathcal{I}_T \neq \emptyset$ for $T \geq T_0$) and apply (4.7). By Lemma 2.2 and (2.11), $|u_k(X)| \ll \log T + \sqrt{X} \ll T^{1/4}$, so the boundary terms and $u_{k_1}(X)^2 + u_{k_2}(X)^2$ are $O(T^{3/4})$. The intervals $[T, \gamma_{k_1}]$ and $[\gamma_{k_2}, 2T]$ have bounded length (Theorem 3.1), and on them $E_X^2(\mathcal{L} + |S_X'|) \ll T^{1/2} \cdot T^{1/4}$. Since $|\varpi| \ll 1/T$, replacing $(E_X + \varpi/\pi)^2$ by E_X^2 costs $\ll T^{-1} \int_T^{2T} (|E_X| + 1)(\mathcal{L} + |S_X'|)\, dt \ll (1+V)^{1/2} \log T$ by Cauchy–Schwarz, (5.10) and (5.8). Hence

$$\sum_{k \in \mathcal{I}_T} u_k(X)^2 = \int_T^{2T} E_X^2 \mathcal{L}\, dt + \int_T^{2T} E_X^2 S_X'\, dt - \frac{\#\mathcal{I}_T}{12} + O(T^{3/4}). \tag{5.12}$$

Main term and derivative term. Since $0 \leq \mathcal{L}(t) - \mathcal{L}(T) \leq \frac{\log 2}{2\pi}$ on $[T, 2T]$, the first integral is $\mathcal{L}(T)\mathcal{E}_2 + O(\mathcal{E}_2)$. By Cauchy–Schwarz, (5.11) and (5.8),

$$\Big| \int_T^{2T} E_X^2 S_X'\, dt \Big| \leq \Big(\int_T^{2T} E_X^4 \Big)^{1/2} \Big(\int_T^{2T} S_X'^2 \Big)^{1/2} \ll T(1+V)(1+\log X). \tag{5.13}$$

Dividing by $\#\mathcal{I}_T = T\mathcal{L}(T) + O(T)$ and using (5.10),

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} u_k(X)^2 = \frac{V}{2\pi^2} + O\Big(1 + \sqrt{V} + \frac{1+V}{\log T} + (1+V)\, c \Big), \tag{5.14}$$

because $\log X = c \log T$. By Lemma 5.7, $(1+V)c \ll c(1 + \log \frac{1}{c}) \ll 1$ on $(0, \frac{1}{2}]$; $V \ll \log\log T$ since $c \geq \log 2 / \log T$, so $(1+V)/\log T = o(1)$. A second use of Lemma 5.7 replaces $\frac{V}{2\pi^2}$ by $\frac{1}{2\pi^2} \log \frac{1}{c} + O(1)$ and $\sqrt{V}$ by $\sqrt{\log(1/c)} + O(1)$, which proves (5.1). □

5.3. A cutoff varying with the index

Let $\mathcal{I}_T^\tau = \{k : T \leq \tau_k < 2T\}$; by (2.5), $\#\mathcal{I}_T^\tau = N_0(2T) - N_0(T) + O(1) \asymp T \log T$.

Lemma 5.10. *Let $T \geq T_0$ and $2 \leq X_- < X_+ \leq (T/3)^{1/2}$ with $X_+ \leq 4X_-$; let $V_\pm = \sum_{X_- < p \leq X_+} p^{-1}$ and*

$$\mathcal{Q}(t) = \max_{X_- \leq y \leq X_+} \Big| \sum_{X_- < p \leq y} p^{-1/2-it} \Big|.$$

Then $\mathcal{Q}$ is Lipschitz, $|\mathcal{Q}'(t)| \leq \max_{X_- \leq y \leq X_+} |\sum_{X_- < p \leq y} \log p\, p^{-1/2-it}|$ for almost every t, and

$$\int_{T-3}^{2T+3} \mathcal{Q}^4\, dt \ll T V_\pm^2, \qquad \int_{T-3}^{2T+3} |\mathcal{Q}'|^4\, dt \ll T(\log X_+)^4 V_\pm^2, \tag{5.15}$$

$$\langle \mathcal{Q}(\gamma_k)^2 \rangle_{\mathcal{I}_T} \ll V_\pm \Big(1 + \frac{\log X_+}{\log T} \sqrt{\log\log T} \Big) + \frac{X_+}{T}, \qquad \langle \mathcal{Q}(\tau_k)^2 \rangle_{\mathcal{I}_T^\tau} \ll V_\pm. \tag{5.16}$$

Proof. If $(X_-, X_+]$ contains no prime, $\mathcal{Q} = 0$. Otherwise let $p_1 < \cdots < p_J$ be its primes and $P_j(t) = \sum_{i \leq j} p_i^{-1/2-it}$ ($P_0 = 0$), so that $\mathcal{Q} = \max_{0 \leq j \leq J} |P_j|$, a maximum of finitely many Lipschitz functions; hence $\mathcal{Q}$ is Lipschitz and differentiable almost everywhere. At a point t where $\mathcal{Q}$ and all $|P_j|$ are differentiable and $\mathcal{Q}(t) = |P_j(t)|$, the function $\mathcal{Q} - |P_j| \geq 0$ has a minimum, so $\mathcal{Q}'(t) = |P_j|'(t)$, and $\big||P_j|'\big| \leq |P_j'|$ wherever $|P_j|$ is differentiable. This gives the bound for $\mathcal{Q}'$.

Window. Fix $\psi \in C_c^\infty(-\frac12,\frac12)$, real, even, nonnegative and not identically 0, let $\check\psi(x) = \int \psi(\omega) e^{i\omega x} d\omega$ and $W = |\check\psi|^2/\check\psi(0)^2$. Then $W \ge 0$ is a Schwartz function with $W(0) = 1$, and $\mathcal{F}W$ is supported in $[-1,1]$. Choose $0 < y_0 \le \frac14$ with $W \ge \frac12$ on $[-y_0, y_0]$ and put $\mathfrak{w}(t) = W(y_0(t - \frac32 T)/(\frac{T}{2} + 3))$. Then $\mathfrak{w} \ge \frac12$ on $[T-3, 2T+3]$, $\|\mathfrak{w}\|_4^4 \asymp T$, and $\mathcal{F}\mathfrak{w}$ and $\mathcal{F}(\mathfrak{w}^4)$ are supported in $|\omega| \le 2y_0/T$ and $|\omega| \le 8y_0/T$.

Proof of (5.15). Let $f_j = \mathfrak{w} P_j$. Then $\mathcal{F}f_j(\omega) = \sum_{i \le j} p_i^{-1/2} \mathcal{F}\mathfrak{w}(\omega + \log p_i)$, a sum of packets supported in $|\omega + \log p_i| \le 2y_0/T$. Since $\log p_{i+1} - \log p_i \ge \log(1 + 1/X_+) \ge (3/T)^{1/2}/2 > 4y_0/T$ for $T \ge T_0$, the packets are disjoint, and f_j is obtained from f_J by removing the packets of $p_{j+1}, \dots, p_J$, which lie below a point ω_j separating them from the others: $f_j(t) = f_J(t) - \frac{1}{2\pi}\int_{-\infty}^{\omega_j} \mathcal{F}f_J(\omega) e^{i\omega t} d\omega$. Hence $\max_j |f_j| \le |f_J| + \mathcal{C}f_J$, and on $[T-3, 2T+3]$, $\mathcal{Q} \le 2\max_j |f_j|$. By Theorem 3.13,

$$\int_{T-3}^{2T+3} \mathcal{Q}^4 \le 16 \int_{\mathbb{R}} \big(|f_J| + \mathcal{C}f_J\big)^4 \ll \int_{\mathbb{R}} \mathfrak{w}^4 |P_J|^4 \, dt.$$

Write $P_J^2 = \sum_n \eta_n n^{-it}$ with $\eta_{p^2} = p^{-1}$ and $\eta_{p_i p_{i'}} = 2(p_i p_{i'})^{-1/2}$ for $i < i'$, all $n \le X_+^2 \le T/3$ (unique factorization makes the representation unique). Expanding, $\int \mathfrak{w}^4 |P_J^2|^2 = \sum_{n,n'} \eta_n \eta_{n'} \mathcal{F}(\mathfrak{w}^4)(\log(n/n'))$, and $|\log(n/n')| \ge 1/\max(n,n') \ge 3/T > 8y_0/T$ for $n \ne n'$, so only the diagonal remains: $\int \mathfrak{w}^4 |P_J|^4 = \|\mathfrak{w}\|_4^4 \sum_n \eta_n^2 \ll T\big(\sum_p p^{-2} + 4V_\pm^2\big) \ll TV_\pm^2$, because $\sum_{X_- < p \le X_+} p^{-2} \le V_\pm/X_- \le 4V_\pm/X_+ \le 4V_\pm^2$. The same argument with the polynomials $\sum_{i \le j} \log p_i \, p_i^{-1/2-it}$, whose maximum dominates $|\mathcal{Q}'|$, gives the second bound, with η_n multiplied by $\log p_i \log p_{i'} \le (\log X_+)^2$.

At the ordinates. Let $T, 2T$ be non-ordinates (otherwise move them by less than the distance to the nearest ordinate, which changes neither side). As Lebesgue–Stieltjes measures on $(T, 2T]$, $dN = \mathcal{L}\,dt + dG$, and $\mathcal{Q}^2$ is Lipschitz, so integration by parts [8, §3.5] gives

$$\sum_{k \in \mathcal{I}_T} \mathcal{Q}(\gamma_k)^2 = \int_T^{2T} \mathcal{Q}^2 \mathcal{L}\, dt + \Big[\mathcal{Q}^2 G\Big]_T^{2T} - 2\int_T^{2T} G\, \mathcal{Q}\mathcal{Q}'\, dt, \tag{5.17}$$

multiplicities included. By Cauchy–Schwarz and (5.15) the first term is $\ll \log T \cdot T^{1/2}(TV_\pm^2)^{1/2} = TV_\pm \log T$. The boundary term is $\ll X_+ \log T$, since $\mathcal{Q} \le \sum_{p \le X_+} p^{-1/2} \ll X_+^{1/2}$ and $G \ll \log T$. By Hölder's inequality with exponents $2,4,4$, Theorem 3.3 ($\int_T^{2T} G^2 \ll T \log\log T$) and (5.15), the last term is $\ll (T\log\log T)^{1/2}(TV_\pm^2)^{1/4}(T(\log X_+)^4 V_\pm^2)^{1/4} = T\sqrt{\log\log T}\,(\log X_+)V_\pm$. Divide by $\#\mathcal{I}_T \gg T\log T$.

At the clock points. For consecutive indices, $\tau_{k+1} - \tau_k = \int_k^{k+1} dx/\mathcal{L}(\tau_x) \ge 2\pi/\log(3T) = \mathfrak{b}_T$ while $\tau_{k+1} \le 3T$, so the intervals $\mathcal{U}_k = [\tau_k, \tau_k + \mathfrak{b}_T]$, $k \in \mathcal{I}_T^\tau$, are disjoint and contained in $[T, 2T+1]$. For an absolutely continuous g on an interval $\mathcal{U}$ of length $\mathfrak{b}$ and $x \in \mathcal{U}$, $|g(x)| \le \frac{1}{\mathfrak{b}}\int_{\mathcal{U}} |g| + \int_{\mathcal{U}} |g'|$, hence $g(x)^2 \le \frac{2}{\mathfrak{b}}\int_{\mathcal{U}} g^2 + 2\mathfrak{b}\int_{\mathcal{U}} g'^2$. With $g = \mathcal{Q}$ and summing, $\sum_{k \in \mathcal{I}_T^\tau} \mathcal{Q}(\tau_k)^2 \le \frac{2}{\mathfrak{b}_T}\int_T^{2T+1} \mathcal{Q}^2 + 2\mathfrak{b}_T \int_T^{2T+1} \mathcal{Q}'^2 \ll TV_\pm \log T + \frac{T(\log X_+)^2 V_\pm}{\log T}$, by Cauchy–Schwarz and (5.15). Divide by $\#\mathcal{I}_T^\tau \gg T\log T$ and use $\log X_+ \le \log T$. □

Theorem 5.11. *For* $\frac{\log 2}{\log T} \le c \le \frac12$ *let* $X_k = \max\{2, (\tau_k/2\pi)^c\}$ *and* $\tilde u_k = s_k - S_{X_k}(\gamma_k)$. *Uniformly in* c,

$$\Big|\langle \tilde u_k^2 \rangle_{\mathcal{I}_T} - \frac{1}{2\pi^2}\log\frac{1}{c}\Big| \ll 1 + \sqrt{\log(1/c)}. \tag{5.18}$$

For fixed c, $\langle (\tilde u_k - u_k(X_-))^2 \rangle_{\mathcal{I}_T} \to 0$, *with* X_- *as in the proof.*

Proof. For $T \ge T_0$ and $k \in \mathcal{I}_T$, $|\delta_k| \le 1$ by (3.4), so $\tau_k \in [T-1, 2T+1]$ and $X_- \le X_k \le X_+$ with $X_\pm$ the values of $\max\{2, (y/2\pi)^c\}$ at $y = T-1$ and $y = 2T+1$. Then $X_+ \le ((2T+1)/2\pi)^{1/2} \le (T/3)^{1/2}$ and $X_+/X_- \le ((2T+1)/(T-1))^{1/2} \le 2$. The prime powers p^m with $X_- < p \le X_k$ are those in $S_{X_k} - S_{X_-}$, so by (2.10)

$$|S_{X_k}(\gamma_k) - S_{X_-}(\gamma_k)| \le \frac{1}{\pi}\mathcal{Q}(\gamma_k) + \frac{1}{\pi}\sum_{X_- < p \le X_+}\sum_{m\ge 2}\frac{1}{mp^{m/2}} \le \frac{1}{\pi}\mathcal{Q}(\gamma_k) + 1.1\,V_\pm,$$

since the terms $m = 1$ are $\frac{1}{\pi}\Im\sum_{X_- < p \le X_k} p^{-1/2-i\gamma_k}$ and $\sum_{m\ge 2} p^{-m/2}/m \le p^{-1}(1-2^{-1/2})^{-1}$. By Theorem 3.7, $V_\pm = \log\frac{\log X_+}{\log X_-} + O(\frac{1}{\log X_-}) \ll \frac{1}{\log X_-} \ll \frac{1}{1+c\log T}$ (and $V_\pm = 0$ if $X_- = X_+ = 2$). Lemma 5.10 gives

$$\langle(\tilde{u}_k - u_k(X_-))^2\rangle_{\mathcal{I}_T} \ll V_\pm + \frac{(1+c\log T)\sqrt{\log\log T}}{(1+c\log T)\log T} + T^{-1/2} + V_\pm^2 \ll 1,$$

which tends to 0 for fixed c. Theorem 5.1 with $X = X_-$, for which $\log(1/c_-) = \log(1/c) + O(1)$ with $c_- = \log X_-/\log T$, gives (5.1) for $u_k(X_-)$; (5.18) follows from $|\langle\tilde{u}^2\rangle - \langle u^2\rangle| \le 2\langle u^2\rangle^{1/2}\langle(\tilde{u}-u)^2\rangle^{1/2} + \langle(\tilde{u}-u)^2\rangle$. □

6. Theorem C: exact accounting for the one-step clock

For a finite nonempty set I of indices and a prime power n, define the response statistics at the zeros and at the clock points,

$$\mathcal{R}_n^\gamma(I) = \frac{2}{a_n}\langle s_k \sin(\gamma_k \log n)\rangle_I, \qquad \mathcal{R}_n^\tau(I) = \frac{2}{a_n}\langle s_k \sin(\tau_k \log n)\rangle_I, \tag{6.1}$$

and, for a set A of prime powers with $\sum_{n\in A}|a_n| < \infty$, the cross term

$$\mathrm{Cr}_A(I) = \langle S_A(\tau_k)^2\rangle_I - v_A. \tag{6.2}$$

For a prime p, $2/a_p = -2\pi\sqrt{p}$. These are finite-window statistics of the zeros of the Riemann zeta function, defined without any hypothesis; Cr_A contains the finite-window diagonal discrepancies and all off-diagonal products and has no fixed sign.

Theorem 6.1 (Finite-window accounting)**.** *Let I be a finite nonempty set of indices and A a set of prime powers with $\sum_{n\in A}|a_n| < \infty$.*

(a) For every k and $X \ge 2$,

$$\begin{aligned} &r_k(X) = u_k(X) + \big(S_X(\gamma_k) - S_X(\tau_k)\big) + \chi_k, \\ &\chi_k = \frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k} - 1 - \log\frac{\tau_k}{\gamma_k}\Big), \qquad \frac{\delta_k^2}{4\pi\max(\gamma_k,\tau_k)} \le \chi_k \le \frac{\delta_k^2}{4\pi\min(\gamma_k,\tau_k)}, \\ &|S_X(\gamma_k) - S_X(\tau_k)| \le M_X|\delta_k|. \end{aligned} \tag{6.3}$$

(b) For every real λ,

$$\Big\langle s_k^2 - (s_k - \lambda S_A(\tau_k))^2 \Big\rangle_I = \sum_{n\in A} a_n^2 \Big(\lambda\, \mathcal{R}_n^\tau(I) - \frac{\lambda^2}{2}\Big) - \lambda^2\, \mathrm{Cr}_A(I). \tag{6.4}$$

In particular, for the full local factors,

$$\begin{aligned} &\langle s_k^2 - (s_k - S_X(\tau_k))^2\rangle_I = D_X(I) - \mathrm{Cr}_X(I), \\ &D_X(I) = \sum_{n\in A_X} a_n^2 \Big(\mathcal{R}_n^\tau(I) - \frac{1}{2}\Big) = 2\langle s_k S_X(\tau_k)\rangle_I - v_X, \end{aligned} \tag{6.5}$$

with $\mathrm{Cr}_X = \mathrm{Cr}_{A_X}$.

(c) In units of zeros, the gain of the one-step clock over the bare clock is

$$\begin{aligned} &\Big\langle (\mathcal{L}(\tau_k)\delta_k)^2 - r_k(X)^2 \Big\rangle_I = D_X(I) - \mathrm{Cr}_X(I) + 2\langle \chi_k S_X(\tau_k)\rangle_I, \\ &|\langle \chi_k S_X(\tau_k)\rangle_I| \le \sigma_X \max_{k\in I} \chi_k. \end{aligned} \tag{6.6}$$

Proof. (a) By (2.12), $r_k(X) = \mathcal{L}(\tau_k)\delta_k - S_X(\tau_k)$. Put $\chi_k = \mathcal{L}(\tau_k)\delta_k - s_k$; since $s_k = N_0(\tau_k) - N_0(\gamma_k) = \int_{\gamma_k}^{\tau_k} \mathcal{L}$,

$$\chi_k = \int_{\gamma_k}^{\tau_k} \Big(\mathcal{L}(\tau_k) - \mathcal{L}(y)\Big) dy = \frac{1}{2\pi}\int_{\gamma_k}^{\tau_k} \log\frac{\tau_k}{y}\, dy = \frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k} - 1 - \log\frac{\tau_k}{\gamma_k}\Big),$$

the last step by $\int \log y\, dy = y\log y - y$. If $\gamma_k < \tau_k$, then $\frac{\tau_k - y}{\tau_k} \le \log\frac{\tau_k}{y} \le \frac{\tau_k - y}{y} \le \frac{\tau_k - y}{\gamma_k}$ for $y \in [\gamma_k, \tau_k]$, and integration gives $\frac{\delta_k^2}{4\pi\tau_k} \le \chi_k \le \frac{\delta_k^2}{4\pi\gamma_k}$; if $\tau_k < \gamma_k$, write $\chi_k = \frac{1}{2\pi}\int_{\tau_k}^{\gamma_k} \log\frac{y}{\tau_k}\, dy$ and use $\frac{y-\tau_k}{\gamma_k} \le \frac{y-\tau_k}{y} \le \log\frac{y}{\tau_k} \le \frac{y-\tau_k}{\tau_k}$; if $\tau_k = \gamma_k$, $\chi_k = 0$. Insert $s_k = u_k(X) + S_X(\gamma_k)$. The last bound is the mean value theorem with (2.11).

(b) The series $S_A(\tau_k) = \sum_n a_n \sin(\tau_k \log n)$ converges absolutely, and so does the double series for $S_A(\tau_k)^2$; since I is finite, the mean over I may be exchanged with these sums. Expanding, the left side is $2\lambda\langle s_k S_A(\tau_k)\rangle_I - \lambda^2\langle S_A(\tau_k)^2\rangle_I$, with $\langle s_k S_A(\tau_k)\rangle_I = \sum_n a_n \langle s_k \sin(\tau_k\log n)\rangle_I = \frac{1}{2}\sum_n a_n^2 \mathcal{R}_n^\tau(I)$ and $\langle S_A(\tau_k)^2\rangle_I = v_A + \mathrm{Cr}_A(I)$. Taking $\lambda = 1$ and $A = A_X$ gives (6.5).

(c) By (a), $\mathcal{L}(\tau_k)\delta_k = s_k + \chi_k$ and $r_k(X) = s_k + \chi_k - S_X(\tau_k)$, so $(\mathcal{L}(\tau_k)\delta_k)^2 - r_k(X)^2 = 2s_k S_X(\tau_k) - S_X(\tau_k)^2 + 2\chi_k S_X(\tau_k)$; average and use (6.5) and $|S_X| \le \sigma_X$. □

Remark 6.2. (6.5) is an exact sign criterion for the gain of the full Euler phase at the clock points: the gain is positive exactly when the diagonal responses outweigh the cross term. No independence of the phases $\tau_k \log p$ is assumed. A threshold $\mathcal{R}_p^\tau = \frac{1}{2}$ describes a single diagonal prime contribution and does not determine the optimal cutoff when Cr_X and the prime powers matter.

Remark 6.3 (Optimal trigonometric correction). For a finite set A of frequencies let $\varsigma_n(k) = \sin(\tau_k \log n)$, $\mathbf{Q} = (\langle \varsigma_{n_1}\varsigma_{n_2}\rangle_I)_{n_1,n_2\in A}$, $\mathbf{b} = (\langle s_k \varsigma_n(k)\rangle_I)_{n\in A}$ and $\Upsilon_{\boldsymbol{\lambda}} = \sum_n \lambda_n \varsigma_n$. Ordinary least squares gives $\langle (s_k - \Upsilon_{\boldsymbol{\lambda}}(k))^2\rangle_I = \langle s_k^2\rangle_I - 2\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{b} + \boldsymbol{\lambda}^{\mathsf{T}}\mathbf{Q}\boldsymbol{\lambda}$, minimized by $\boldsymbol{\lambda} = \mathbf{Q}^\dagger\mathbf{b}$ with gain $\mathbf{b}^{\mathsf{T}}\mathbf{Q}^\dagger\mathbf{b}$; for prime powers $b_n = \frac{1}{2}a_n\mathcal{R}_n^\tau(I)$. The Euler phase is the choice $\lambda_n = a_n$, and (6.4) measures what it loses through $\mathcal{R}_n^\tau \ne 1$ and $\mathrm{Cr}_A \ne 0$.

Remark 6.4 (Clock point against zero). For a prime power n, $\sin(\gamma_k\ell_n) - \sin(\tau_k\ell_n) = \int_0^1 \frac{d}{dy}\sin((\tau_k - y\delta_k)\ell_n)\,dy$, whence, exchanging the finite mean and the integral,

$$\mathcal{R}_n^{\gamma}(I) - \mathcal{R}_n^{\tau}(I) = -\frac{2\ell_n}{a_n}\int_0^1 \Big\langle s_k\delta_k \cos\Big((\tau_k - y\delta_k)\ell_n\Big)\Big\rangle_I\,dy. \tag{6.7}$$

For fixed n and $I = \mathcal{I}_T$ the right side is $O(\langle|s_k\delta_k|\rangle_{\mathcal{I}_T}) = O(\log\log T/\log T)$ by (2.6) and Remark 3.17; for $\ell_n = \theta L_T$ with fixed θ the phase change $\delta_k\ell_n$ has mean square $\theta^2 L_T^2\langle\delta_k^2\rangle_{\mathcal{I}_T} \sim 2\theta^2\log\log T$, so it is not small, the transport integral need not be small, and the computed responses do differ (Section 11); no asymptotic statement about this difference is proved.

Remark 6.5 (Fixed prime sets). For fixed X, $|\delta_k^X - \delta_k| \le \sigma_X/\mathcal{L}(\tau_k) \to 0$ and $|u_k(X) - s_k| \le \sigma_X$. A fixed set of primes changes neither $\limsup_k|\delta_k|$ nor the unboundedness of the defect.

7. Theorem D: a computable clock that approaches the zeros

The one-step clock $\tau_k^X = \tau_k - y_k$, $y_k = S_X(\tau_k)/\mathcal{L}(\tau_k)$, uses only τ_k and the primes $p \le X$. By Theorem 6.1(a),

$$r_k(X) = u_k(X) + \mathfrak{t}_k(X) + \chi_k, \qquad \mathfrak{t}_k(X) = S_X(\gamma_k) - S_X(\tau_k), \tag{7.1}$$

where Theorem 5.1 controls $u_k(X)$ and $0 \le \chi_k \ll 1/T$ on $\mathcal{I}_T$. This section controls the transport term $\mathfrak{t}_k(X)$ in mean square, with no assumption on the joint behaviour of δ_k and the phases $\tau_k\log p$.

Lemma 7.1. *For $T \ge T_0$ and $2 \le X \le T^{1/2}$,*

$$\int_{T-3}^{2T+3} S_X'(t)^4\,dt \ll T(1+\log X)^4, \qquad \int_{T-3}^{2T+3} S_X''(t)^4\,dt \ll T(1+\log X)^8.$$

Proof. By Section 2.4, the series

$$S_X'(t) = \sum_{n\in A_X} a_n\log n\cos(t\log n), \qquad S_X''(t) = -\sum_{n\in A_X} a_n\log^2 n\sin(t\log n)$$

converge absolutely and uniformly. Split according to $m = 1$, $m = 2$, $m \ge 3$ in $n = p^m$ and use $(y_1+y_2+y_3)^4 \le 27(y_1^4+y_2^4+y_3^4)$. For $\nu \in \{1,2\}$ the terms $m \ge 3$ of the ν-th derivative are bounded by $\frac{1}{\pi}\sum_p\sum_{m\ge3}\frac{(m\log p)^\nu}{mp^{m/2}} < \infty$, uniformly in X and t. The terms $m = 1$ are bounded by $\frac{1}{\pi}|\mathcal{D}_\nu(t)|$ with $\mathcal{D}_\nu(t) = \sum_{p\le X}\log^\nu p\; p^{-1/2-it}$; $\mathcal{D}_\nu^2 = \sum_n \eta_n n^{-it}$ over $n = pp' \le X^2 \le T$ with $\sum_n\eta_n^2 \le 2(\sum_{p\le X}\log^{2\nu}p/p)^2 \ll (1+\log X)^{4\nu}$ (Theorem 3.7 and partial summation), so Theorem 3.6 on an interval of length $T+6$ gives $\int|\mathcal{D}_\nu|^4 = \sum\eta_n^2(T+6+O(n)) \ll T(1+\log X)^{4\nu}$. The terms $m = 2$ are bounded by $\frac{1}{2\pi}|\mathcal{D}_\nu'(t)|$ with $\mathcal{D}_\nu'(t) = \sum_{p\le X}(2\log p)^\nu p^{-1-2it}$; here $\mathcal{D}_\nu'^2 = \sum_n\eta_n' n^{-it}$ over $n = (pp')^2 \le X^4$, with $\sum_n\eta_n'^2 \ll 1$ and $\sum_n n\eta_n'^2 \ll (\sum_{p\le X}\log^{2\nu}p)^2 \ll X^2(1+\log X)^{4\nu} \le T(1+\log X)^{4\nu}$, so again $\int|\mathcal{D}_\nu'|^4 \ll T(1+\log X)^{4\nu}$. □

Lemma 7.2 (Transport)**.** *For $T \ge T_0$ and $2 \le X \le T^{1/2}$,*

$$\big\langle \mathfrak{t}_k(X)^2\big\rangle_{\mathcal{I}_T} \ll \frac{\log\log T\,(1+\log X)^2}{\log^2 T}. \tag{7.2}$$

Proof. Fourth moment of δ_k. For $T \ge T_0$ and $k \in \mathcal{I}_T$, $|\delta_k| \le 1$ by (3.4); so the point ξ_k of (2.6) is $\ge T-1$ and $\mathcal{L}(\xi_k) \gg \log T$. By (3.3) with $j=2$,

$$\langle \delta_k^4 \rangle_{\mathcal{I}_T} = \langle s_k^4 \mathcal{L}(\xi_k)^{-4} \rangle_{\mathcal{I}_T} \ll \frac{(\log\log T)^2}{\log^4 T}. \tag{7.3}$$

Maximal function. Let $\varphi \in C^\infty(\mathbb{R})$ with $0 \le \varphi \le 1$, $\varphi = 1$ on $[T-2, 2T+2]$, $\varphi = 0$ outside $[T-3, 2T+3]$ and $|\varphi'| \le 2$, and let $\mathfrak{h} = \varphi S_X'$. For $k \in \mathcal{I}_T$ the segment between γ_k and τ_k lies in $[T-1, 2T+1]$, where $\mathfrak{h} = S_X'$, so

$$|\mathfrak{t}_k(X)| = \left| \int_{\tau_k}^{\gamma_k} \mathfrak{h}(y)\, dy \right| \le \int_{\tau_k - |\delta_k|}^{\tau_k + |\delta_k|} |\mathfrak{h}| \le 2|\delta_k|\, \mathcal{M}\mathfrak{h}(\tau_k) \tag{7.4}$$

(trivially if $\delta_k = 0$). The function $\mathfrak{h}$ is C^∞ with compact support. By Theorem 3.12, $g = \mathcal{M}\mathfrak{h}$ is locally absolutely continuous with $\|g\|_4 \ll \|\mathfrak{h}\|_4$ and $\|g'\|_4 \le \|\mathcal{M}(\mathfrak{h}')\|_4 \ll \|\mathfrak{h}'\|_4$; by Lemma 7.1, $\|\mathfrak{h}\|_4^4 \ll T(1+\log X)^4$ and, since $\mathfrak{h}' = \varphi' S_X' + \varphi S_X''$, $\|\mathfrak{h}'\|_4^4 \ll T(1+\log X)^8$.

Sampling at the clock points. As in the proof of Lemma 5.10, the intervals $\mathcal{U}_k = [\tau_k, \tau_k + \mathfrak{b}_T]$, $k \in \mathcal{I}_T$, are disjoint and contained in $[T-1, 2T+2]$. For $x \in \mathcal{U}$, an interval of length $\mathfrak{b}$, $|g(x)| \le \frac{1}{\mathfrak{b}} \int_{\mathcal{U}} |g| + \int_{\mathcal{U}} |g'|$; by Hölder's inequality and $(y_1+y_2)^4 \le 8(y_1^4+y_2^4)$, $g(x)^4 \le \frac{8}{\mathfrak{b}} \int_{\mathcal{U}} g^4 + 8\mathfrak{b}^3 \int_{\mathcal{U}} |g'|^4$. Summing over $k \in \mathcal{I}_T$ and dividing by $\#\mathcal{I}_T \gg T\log T$,

$$\langle g(\tau_k)^4 \rangle_{\mathcal{I}_T} \ll \frac{1}{T}\int_{\mathbb{R}} g^4 + \frac{1}{T\log^4 T}\int_{\mathbb{R}} |g'|^4 \ll (1+\log X)^4 + \frac{(1+\log X)^8}{\log^4 T} \ll (1+\log X)^4.$$

Conclusion. By (7.4) and Cauchy–Schwarz, $\langle \mathfrak{t}_k^2 \rangle \le 4\langle \delta_k^2 g(\tau_k)^2 \rangle \le 4\langle \delta_k^4 \rangle^{1/2} \langle g(\tau_k)^4 \rangle^{1/2}$; insert (7.3). □

Theorem 7.3 (Theorem D)**.** *There are absolute constants T_0 and C such that for $T \ge T_0$ and $2 \le X \le T^{1/2}$, with $c = \log X / \log T$,*

$$\left| \left\langle r_k(X)^2 \right\rangle_{\mathcal{I}_T} - \frac{1}{2\pi^2} \log \frac{1}{c} \right| \le C\Big(1 + \sqrt{\log(1/c)}\Big)\Big(1 + c^2 \log\log T\Big). \tag{7.5}$$

Proof. By Theorem 6.1(a), $0 \le \chi_k \le \delta_k^2/(4\pi(T-1)) \le 1/T$ on $\mathcal{I}_T$. Since $1 + \log X \le (1 + \frac{1}{\log 2}) c \log T$, Lemma 7.2 gives $\langle \mathfrak{t}_k^2 \rangle \ll c^2 \log\log T$, so $d_k = r_k(X) - u_k(X) = \mathfrak{t}_k + \chi_k$ has $\langle d_k^2 \rangle \ll c^2 \log\log T + T^{-2}$. Theorem 5.1 gives $\langle u_k(X)^2 \rangle \ll 1 + \log(1/c)$. Then $|\langle r_k^2 \rangle - \langle u_k^2 \rangle| \le 2\langle u_k^2 \rangle^{1/2} \langle d_k^2 \rangle^{1/2} + \langle d_k^2 \rangle \ll (1 + \sqrt{\log(1/c)})(c\sqrt{\log\log T} + T^{-1}) + c^2 \log\log T$, which is bounded by the right side of (7.5) because $y \le 1 + y^2$. Combine with (5.1). □

Corollary 7.4. *Let $X = T^{1/\sqrt{\log\log T}}$. Then*

$$\langle r_k(X)^2 \rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2} \log\log\log T + O\Big(\sqrt{\log\log\log T}\Big), \tag{7.6}$$

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} \Big(\tau_k^X - \gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}, \tag{7.7}$$

while $\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} (\tau_k - \gamma_k)^2 = (2 + o(1)) \log\log T / \log^2 T$; the ratio is asymptotic to $\log\log\log T / (2\log\log T)$.

Proof. Here $c = (\log\log T)^{-1/2}$, $2 \le X \le T^{1/2}$ for $T \ge T_0$, $\log(1/c) = \frac{1}{2}\log\log\log T$ and $c^2 \log\log T = 1$, so (7.5) gives (7.6). On $\mathcal{I}_T$, $\tau_k \in [T-1, 2T+1]$ and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1+O(\frac{1}{\log T}))$; divide r_k^2 by $\mathcal{L}(\tau_k)^2$. For the bare clock, $\delta_k = s_k/\mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log T}{2\pi}(1 + O(\frac{1}{\log T}))$, and $\langle s_k^2\rangle \sim \frac{1}{2\pi^2}\log\log T$ by (3.3). □

Corollary 7.5 (A cutoff read off the clock)**.** *For $\tau_k > e^e$ let $X_k^{\mathrm{loc}} = \tau_k^{1/\sqrt{\log\log\tau_k}}$ and $\tau_k^{\mathrm{loc}} = \tau_k - S_{X_k^{\mathrm{loc}}}(\tau_k)/\mathcal{L}(\tau_k)$. Then*

$$\frac{1}{\#\mathcal{I}_T^\tau}\sum_{k\in\mathcal{I}_T^\tau}\Big(\tau_k^{\mathrm{loc}} - \gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}.$$

Proof. Let $X_0 = T^{1/\sqrt{\log\log T}}$ and $X_0' = (2T)^{1/\sqrt{\log\log 2T}}$. The function $y \mapsto y/\sqrt{\log y}$ is increasing for $y > e^{1/2}$, so $X_0 \le X_k^{\mathrm{loc}} \le X_0'$ for $k \in \mathcal{I}_T^\tau$, and $\log X_0' - \log X_0 \le \log 2/\sqrt{\log\log T} = o(1)$; hence $X_0' \le 2X_0$ and $X_0' \le (T/3)^{1/2}$ for $T \ge T_0$. As in the proof of Theorem 5.11, $|S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k)| \le \frac{1}{\pi}\mathcal{Q}(\tau_k) + 1.1V_\pm$ with $X_\pm$ replaced by X_0, X_0', and Lemma 5.10 gives a mean square over $\mathcal{I}_T^\tau$ that is $\ll V_\pm \ll 1/\log X_0 = \sqrt{\log\log T}/\log T$. The sets $\mathcal{I}_T$ and $\mathcal{I}_T^\tau$ differ by $O(\log T)$ indices, those with γ_k or τ_k within distance 1 of T or $2T$, and for them $|r_k(X_0)| \le |s_k| + \chi_k + \sigma_{X_0} \ll \log T + X_0^{1/2}$ (Lemma 2.2); so the means of $r_k(X_0)^2$ over the two sets differ by $O(T^{-1/2})$, and (7.6) holds over $\mathcal{I}_T^\tau$. Finally $\mathcal{L}(\tau_k)(\tau_k^{\mathrm{loc}} - \gamma_k) = r_k(X_0) - (S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k))$; conclude with Cauchy–Schwarz and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1 + O(\frac{1}{\log T}))$. □

Proposition 7.6 (The root of the corrected equation in the monotone regime)**.** *Let $X = X(T) \ge 2$ with $M_X \le \frac{1}{2}\mathcal{L}(T)$, and let $\mathcal{I}_T'$ be the set of $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac{1}{2} \le F_X(2T)$, so that the root $t_k^X \in [T, 2T]$ of Remark 5.6 exists. Then $X \ll \log^2 T$, $\#(\mathcal{I}_T \setminus \mathcal{I}_T') \ll \log T$, and*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T'}\Big(t_k^X - \gamma_k\Big)^2 \ge \frac{4}{9\mathcal{L}(2T)^2}\Big(\frac{1}{2\pi^2}\log\log T - O\Big(\sqrt{\log\log T}\Big)\Big) \gg \frac{\log\log T}{\log^2 T}.$$

Proof. $M_X \ge \frac{1}{\pi}\sum_{p\le X}\log p/\sqrt{p} \gg \sqrt{X}$ by Chebyshev's estimate, so $X \ll \mathcal{L}(T)^2 \ll \log^2 T$ and $c = \log X/\log T \le (2\log\log T + O(1))/\log T$. On $[T, 2T]$, $0 < F_X' \le \mathcal{L} + M_X \le \frac{3}{2}\mathcal{L}(2T)$, so $|u_k(X)| = |F_X(t_k^X) - F_X(\gamma_k)| \le \frac{3}{2}\mathcal{L}(2T)|t_k^X - \gamma_k|$ for $k \in \mathcal{I}_T'$. An index $k \in \mathcal{I}_T$ is missing from $\mathcal{I}_T'$ only if $k - \frac{1}{2} < F_X(T)$ or $k - \frac{1}{2} > F_X(2T)$; since $N(T) < k \le N(2T)$ and $|F_X - N| \le |G| + \sigma_X \ll \log T$ at T and $2T$, there are $O(\log T)$ such k, each with $|u_k(X)| \ll \log T$ (Lemma 2.2). Hence $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T'} u_k(X)^2 = \langle u_k(X)^2\rangle_{\mathcal{I}_T} + O(\frac{\log^3 T}{T\log T})$, and Theorem 5.1 with $\log(1/c) \ge \log\log T - \log(2\log\log T + O(1))$ bounds it from below. □

Thus a root of the corrected equation can be selected only where it gains nothing in order: the error in height of t_k^X has mean square of the same order $\log\log T/\log^2 T$ as that of τ_k. Theorem 7.3 avoids roots and takes one Newton step instead.

Remark 7.7 (The Newton term and the restriction on c). Let ξ_k^X be a point between γ_k and τ_k with $\mathfrak{t}_k(X) = -S_X'(\xi_k^X)\delta_k$ (mean value theorem). Since $\delta_k = (r_k(X) + S_X(\tau_k))/\mathcal{L}(\tau_k)$ by (2.12), (7.1) becomes the exact relation

$$r_k(X)\Big(1 + \frac{S_X'(\xi_k^X)}{\mathcal{L}(\tau_k)}\Big) = u_k(X) + \chi_k - y_k\, S_X'(\xi_k^X). \tag{7.8}$$

The term $y_k S'_X(\xi_k^X)$ is the second-order term of Newton's method. Heuristically, $\langle S_X^2\rangle \approx v_X \approx \frac{1}{2\pi^2}\log\log X$ and $\langle S_X'^2\rangle/\mathcal{L}(T)^2 \approx \sum_{p\le X}\frac{\log^2 p}{2\pi^2 p}/\mathcal{L}(T)^2 \approx c^2$, so this term has mean square of order $c^2\log\log T$ if the two factors decorrelate: the order of Lemma 7.2 is the order of the Newton term, and the lemma cannot be improved by bounds that ignore the correlation between $u_k(X)$ and $\mathfrak{t}_k(X)$. At $T = 2^{19}$ (Table 5) the transport is smaller than this order and anticorrelated with $u_k(X)$, increasingly with c; a proof of this cancellation would allow a fixed c in Corollary 7.4 (Question 12.2). The denominator in (7.8) can vanish: $F'_X(\tau_k) < 0$ at 2.7% of the indices of that window for $c = \frac{1}{2}$.

8. Theorem E: the pointwise envelope

Let $\mathfrak{a} = \limsup_{t\to\infty} |S(t)|/\log t$. By Theorem 3.2, $\mathfrak{a} \le 0.10076$; whether $\mathfrak{a} = 0$ is not known.

Proposition 8.1 (The envelope of the clock is the envelope of S)**.** *(a)* $\limsup_{k\to\infty}|\tau_k - \gamma_k| = 2\pi\mathfrak{a}$. *(b) If* $X_k \ge 2$ *and* $\sigma_{X_k} = o(\log\tau_k)$*, in particular if* $X_k \le (\log\tau_k)^{2-\varepsilon}$*, then* $\limsup_{k\to\infty}|\tau_k^{X_k} - \gamma_k| = 2\pi\mathfrak{a}$.

Proof. (a) The values $S(\gamma_k^\pm)$ are limits of values of S at non-ordinates. For $\varepsilon > 0$, (2.8) gives $|s_k| \le (\mathfrak{a}+\varepsilon)\log\gamma_k + O_\varepsilon(1)$, and $|\delta_k| = |s_k|/\mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log\gamma_k}{2\pi}(1+o(1))$ since $\delta_k = O(1)$; so $\limsup|\delta_k| \le 2\pi\mathfrak{a}$. Conversely, let $t > \gamma_1$ be a non-ordinate, let $\gamma < t < \gamma'$ be the nearest ordinates and $k = N(t)$, so that $\gamma_k = \gamma$ and $\gamma_{k+1} = \gamma'$. On (γ,γ') the function G is continuous and decreasing ($G' = -\mathcal{L}$), so $G(\gamma'^-) \le G(t) \le G(\gamma^+)$, while $s_k = k - \frac12 - N_0(\gamma) = G(\gamma^+) - \frac12$ and $s_{k+1} = k + \frac12 - N_0(\gamma') = G(\gamma'^-) + \frac12$. Hence $s_k \ge G(t) - \frac12$ and $s_{k+1} \le G(t) + \frac12$, so $\max(|s_k|, |s_{k+1}|) \ge |G(t)| - \frac12$, and $|G(t) - S(t)| = |\varpi(t)|/\pi \to 0$. Both ordinates are within bounded distance of t (Theorem 3.1). Choosing non-ordinates $t\to\infty$ with $|S(t)| \ge (\mathfrak{a}-\varepsilon)\log t$ gives indices $k\to\infty$ with $|\delta_k| = |s_k|/\mathcal{L}(\xi_k) \ge 2\pi(\mathfrak{a}-\varepsilon) + o(1)$.

(b) $|\tau_k^{X_k} - \tau_k| = |S_{X_k}(\tau_k)|/\mathcal{L}(\tau_k) \le \sigma_{X_k}/\mathcal{L}(\tau_k) \to 0$, and $\sigma_X \ll \sqrt{X}/\log X$. □

So the corrected clocks with slowly growing cutoffs have exactly the pointwise envelope of the bare clock, and for every cutoff the envelope of τ_k is that of S. A smaller coefficient for the clock is a smaller coefficient for $S(t)$.

Proposition 8.2 (Finite-height envelope)**.** *For every* $k \ge 1$ *let* $E_k = \mathrm{BW}_S(\gamma_k) - \frac12 + |\varpi(\gamma_k)|/\pi$*, and* $E_k = 2.5167 - \frac12 + |\varpi(\gamma_k)|/\pi$ *if* $\gamma_k \le 30\,610\,046\,000$*. Then* $|s_k| \le E_k$ *and*

$$-\frac{E_k}{\mathcal{L}(\tau_k)} \le \delta_k \le \frac{E_k}{\mathcal{L}(\gamma_k)}. \tag{8.1}$$

Proof. Theorem 3.2 at $\gamma_k^\pm$ (one-sided limits; the bounds are continuous in t) and (2.8) give $|s_k| \le E_k$. Since $\mathcal{L}$ is positive and increasing on $(2\pi,\infty)$ and $s_k = \int_{\gamma_k}^{\tau_k}\mathcal{L}$, $s_k \ge \mathcal{L}(\gamma_k)\delta_k$ if $\delta_k \ge 0$ and $s_k \le \mathcal{L}(\tau_k)\delta_k$ if $\delta_k < 0$. □

Since $|\varpi|/\pi < 0.01$, the numerator of the lower bound in (8.1) is at most $0.10076\log\gamma_k + 0.24460\log\log\gamma_k + 6.71844$, against $\ldots + 8.45844$ in (3.4); the upper bound of (3.4) is smaller by 0.01 for large γ_k, and the minimum of the two may be used. In the range of Odlyzko's table both are superseded by the bound from 2.5167: $|\delta_k| \le 2.0267/\mathcal{L}(\cdot)$, which is 1.047 at $k = K$ (Section 9.2).

Theorem 8.3 (Theorem E)**.** *Let $\mathfrak{e}$ be continuous on $[\frac12,1]$ with $\mathfrak{e}(1)=0$, such that for every $\varepsilon>0$, $\zeta(\sigma+it)\ll_\varepsilon |t|^{\mathfrak{e}(\sigma)+\varepsilon}$ uniformly for $\frac12\le\sigma\le1$ and $|t|\ge2$, and put $\mathfrak{e}(\sigma)=\frac12-\sigma+\mathfrak{e}(1-\sigma)$ for $0\le\sigma<\frac12$. Then*

$$\mathfrak{a}\le\frac{1}{2\pi\log2}\int_0^1\frac{\mathfrak{e}(\sigma)}{\sqrt{\sigma(2-\sigma)}}\,d\sigma. \tag{8.2}$$

For the piecewise linear $\mathfrak{e}_{\mathrm B}$ through $(\frac12,\frac{13}{84})$, $(\frac{139}{194},\frac{13}{194})$, $(\frac{173}{207},\frac{13}{414})$, $(\frac{773}{854},\frac{13}{854})$, $(\frac{820}{867},\frac{13}{1734})$, $(\frac{3387}{3494},\frac{13}{3494})$, $(\frac{1149}{1169},\frac{13}{7014})$, $(\frac{13921}{14054},\frac{13}{14054})$, $(1,0)$, the right side is $0.0968160063\ldots$; hence

$$\limsup_{k\to\infty}|\tau_k-\gamma_k|=2\pi\mathfrak{a}\le0.6083129\ldots<0.60832. \tag{8.3}$$

Proof. Uniform bounds. Extend $\mathfrak{e}$ by 0 on $[1,\infty)$ and by $\frac12-\sigma$ on $(-\infty,0)$; the extension $\mathfrak{e}^*$ is continuous ($\mathfrak{e}(0)=\frac12+\mathfrak{e}(1)=\frac12$). For $\sigma\ge1$, $\zeta(\sigma+it)\ll\log|t|$ uniformly [33, Thm. 3.5]. For $-\frac{1}{50}\le\sigma<\frac12$, $\zeta(\sigma+it)=\chi(\sigma+it)\zeta(1-\sigma-it)$ with $|\chi(\sigma+it)|=(|t|/2\pi)^{1/2-\sigma}(1+O(1/|t|))$ uniformly [33, (4.12.3)], and $|\zeta(1-\sigma-it)|=|\zeta(1-\sigma+it)|$. Hence $\zeta(\sigma+it)\ll_\varepsilon|t|^{\mathfrak{e}^*(\sigma)+\varepsilon}$ uniformly for $-\frac{1}{50}\le\sigma\le4$, $|t|\ge2$.

Backlund's method. Let $0<\eta<\frac{1}{100}$, $\sigma_\circ=1+\eta$, $\varrho=1+2\eta$. Then $1<\sigma_\circ<\varrho$, $\sigma^\sharp=\frac54+\frac32\eta$, $\delta^\sharp=\frac14+\frac{\eta}{2}\in(0,\frac12)$, and $\varrho/(\sigma_\circ-\frac12)=2$. For $s=\sigma_\circ+\varrho e^{i\theta}$, $\Re s\in[-\eta,2+3\eta]$ and $|\Im s|\le2$, so for T large

$$\frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(s)|\le\log\max_{\pm}|(s\pm iT-1)\zeta(s\pm iT)|\le\log(T+4)+(\mathfrak{e}^*(\Re s)+\varepsilon)\log(T+2)+C_\varepsilon=:F_\circ(\theta),$$

an even continuous function. By Theorem 3.16(a), along $\varkappa_m\to\infty$,

$$\limsup_m\mathcal{Z}_{\varkappa_m}(\sigma_\circ,\varrho)\le\log T\cdot\frac{1}{\pi}\int_0^\pi\Big(\mathfrak{e}^*(\sigma_\circ+\varrho\cos\theta)+\varepsilon\Big)d\theta+O_{\varepsilon,\eta}(1),$$

the terms $\log(T+4)$ and $-\log\sqrt{\eta^2+T^2}$ cancelling up to $O(1/T)$. The argument defining $S(T)$ runs along $2\to2+iT\to\sigma^\sharp+iT\to\frac12+iT$, and on the first two segments $\Re s>1$, where $\log\zeta$ is holomorphic; so $\pi S(T)=\arg\zeta(\sigma^\sharp+iT)+\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg((\sigma+iT-1)\zeta(\sigma+iT))-\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg(\sigma+iT-1)$, with $|\arg\zeta(\sigma^\sharp+iT)|\le\log\zeta(\sigma^\sharp)$ and the last term bounded by π. Theorem 3.16(b) with $\varkappa=\varkappa_m\to\infty$ gives, for T large,

$$|S(T)|\le\frac{\log T}{2\log2}\cdot\frac{1}{\pi}\int_0^\pi\Big(\mathfrak{e}^*(1+\eta+(1+2\eta)\cos\theta)+\varepsilon\Big)d\theta+O_{\varepsilon,\eta}(1).$$

Limit. Divide by $\log T$, let $T\to\infty$, then $\varepsilon\to0$ and $\eta\to0$; $\mathfrak{e}^*$ is bounded and continuous, so by dominated convergence $\mathfrak{a}\le\frac{1}{2\pi\log2}\int_0^\pi\mathfrak{e}^*(1+\cos\theta)\,d\theta$. The integrand vanishes for $0\le\theta\le\frac{\pi}{2}$; on $[\frac{\pi}{2},\pi]$ substitute $\sigma=1+\cos\theta$, $d\sigma=-\sin\theta\,d\theta$, $\sin\theta=\sqrt{\sigma(2-\sigma)}$, which gives (8.2).

The profile. Let $\varepsilon'>0$, $(\mathsf{k}_0,\mathsf{l}_0)=(\frac{13}{84}+\varepsilon',\frac{55}{84}+\varepsilon')$ and $(\mathsf{k}_j,\mathsf{l}_j)=A^j(\mathsf{k}_0,\mathsf{l}_0)$ for $1\le j\le7$, exponent pairs by Theorem 3.14(a),(c), with $\mathsf{l}_j-\mathsf{k}_j\ge\frac12$; by (b), $|\zeta(\mathsf{l}_j-\mathsf{k}_j+it)|\ll|t|^{\mathsf{k}_j}\log|t|$, and $|\zeta(1+it)|\ll\log|t|$. The points $(\mathsf{l}_j-\mathsf{k}_j,\mathsf{k}_j)$ and $(1,0)$ have decreasing ordinates, and Theorem 3.15 between consecutive points gives the hypothesis of the theorem for the piecewise linear function through them. For $\varepsilon'=0$ these are the vertices listed in the statement; the right side of (8.2) depends continuously on ε', and its value at $\varepsilon'=0$, computed piece by piece with 40-digit quadrature by `opc_envelope_coefficient_v10.py`, is $0.0968160063238\ldots$. Multiply by 2π and use Proposition 8.1(a). □

Remark 8.4. (a) With the explicit exponents used by Bellotti and Wong ($\frac{27}{164}$ at $\sigma=\frac12$, Yang's lines $\sigma_j=1-j/(2^j-2)$ with exponents $1/(2^j-2)$, $4\le j\le 9$ [2]) and linear interpolation between $\sigma=0$ and $\sigma=\frac12$, as in [2, §3.1], the right side of (8.2) is 0.10074, the limit of their $C_1=0.10076$ as their parameters tend to $(1,1)$. The same right-hand profile reflected by the functional equation gives 0.09783, i.e. $2\pi\mathfrak{a}\le 0.6147$, with explicit inputs only; this uses that $\mathfrak{e}$ is convex, so that the reflected profile lies below the chord. (b) For the Lindelöf profile ($\mathfrak{e}=0$ on $[\frac12,1]$) the right side is 0.07863 ($2\pi\cdot$ it is 0.4940): Backlund's method with a disc cannot give less, whatever the exponents. (c) Theorem 8.3 is asymptotic: the explicit constants of the bounds for ζ do not enter, which is why non-explicit exponents may be used.

Proposition 8.5 (An explicit bound valid for all $T\ge e$)**.** *For $T\ge e$,*

$$\begin{aligned}&|S(T)|\le 0.09783\log T+0.722\log\log T+13,\\&\Big|N(T)-\frac{T}{2\pi}\log\frac{T}{2\pi e}\Big|\le 0.09783\log T+0.722\log\log T+14.\end{aligned}\tag{8.4}$$

Proof. Explicit profile. Let $\sigma_m=1-m/(2^m-2)$, $4\le m\le 9$. From the bounds collected in [2, (2.1), (2.3), (2.4)], for $|t|\ge 3$: $|\zeta(\frac12+it)|\le 2|t|^{27/164}\log|t|$ (below e^{105} from $0.618|t|^{1/6}\log|t|$, since $|t|^{1/492}\le 2/0.618$ there; above, $66.7|t|^{27/164}\le 2|t|^{27/164}\log|t|$), $|\zeta(\sigma_m+it)|\le 2|t|^{1/(2^m-2)}\log|t|$ and $|\zeta(1+it)|\le 2\log|t|$. Let $g_{\mathrm e}$ be the linear interpolation through $(\frac12,\frac{27}{164})$, $(\sigma_m,\frac{1}{2^m-2})$, $(1,0)$, extended by 0 on $[1,3]$. The Phragmén–Lindelöf principle [2, Lemma 2.5] for $(s-1)\zeta(s)$ with shift $Q=10$, the boundary values for $|t|\le 3$ being bounded by the elementary integral representation of ζ, gives $|\zeta(\sigma+it)|\le 10\,t^{g_{\mathrm e}(\sigma)}\log t$ for $\frac12\le\sigma\le 3$, $t\ge 10$. With $|\chi(\sigma+it)|\le 3t^{1/2-\sigma}$ for $-0.01\le\sigma\le\frac12$, $t\ge 10$ (Stirling's formula with explicit remainder), the reflected profile $h_{\mathrm e}$ ($h_{\mathrm e}=g_{\mathrm e}$ on $[\frac12,1]$, $\frac12-\sigma+g_{\mathrm e}(1-\sigma)$ on $[0,\frac12]$, $\frac12-\sigma$ below 0, 0 above 1) satisfies $|\zeta(\sigma+it)|\le 30\,t^{h_{\mathrm e}(\sigma)}\log t$ for $-0.01\le\sigma\le 3$, $t\ge 10$.

Fixed parameters. Take $\sigma_\circ=1.000002$, $\varrho=1.000005$ and, in the notation of [2, Thm. 1.2], $\eta=10^{-6}$; then $\sigma_\circ-\varrho<1-\sigma_\circ<-\eta$ and $\delta^\sharp\in(\frac14,\frac12)$. Put $D=\log(\varrho/(\sigma_\circ-\frac12))$ and $I=\int_0^\pi h_{\mathrm e}(\sigma_\circ+\varrho\cos\theta)\,d\theta$. On each linear piece $h_{\mathrm e}(\sigma)=\alpha\sigma+\beta$ a primitive is $(\alpha\sigma_\circ+\beta)\theta+\alpha\varrho\sin\theta$, the endpoints being arccosines of rationals, and interval arithmetic with directed rounding gives $0.097829231878517946338 76<I/(2\pi D)<0.097829231878517946338 77$ and $1/(2D)<0.721346479766175634597 98$.

Backlund's method. For $T\ge 100$ the bound for ζ gives $\frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(\sigma_\circ+\varrho e^{i\theta})|\le(1+h_{\mathrm e}(\sigma_\circ+\varrho\cos\theta))\log(T+3)+\log\log(T+3)+\log 30$, and $\zeta(\sigma_\circ)/\zeta(2\sigma_\circ)\le 1+1/(\sigma_\circ-1)=500001$. Theorem 3.16 and the decomposition of the argument in the proof of Theorem 8.3 give $|S(T)|\le\frac{1}{2D}\big(-\log T+\log 500001+(1+\frac{I}{\pi})\log(T+3)+\log\log(T+3)+\log 30\big)+R(T)$, where $R(T)<1$ collects $\frac14$, the variation of $\arg(s-1)$, $E(T,\delta^\sharp)/2\pi$ and $\pi^{-1}\log\zeta(\sigma^\sharp)$. Since $\log(30\cdot 500001)/(2D)<11.92$ and the replacement of $\log(T+3)$ by $\log T$ costs less than 0.05, the constant is <13 for $T\ge 100$; for $e\le T\le 100$, $|S(T)|\le 2.5167$ (Theorem 3.2). Finally $N(T)-\frac{T}{2\pi}\log\frac{T}{2\pi e}=\frac78+S(T)+\varpi(T)/\pi$ and $\frac78+|\varpi|/\pi<1$. □

The coefficient 0.09783 is the value of (8.2) for the explicit profile reflected by the functional equation (Remark 8.4(a)), now reached at fixed admissible parameters. The lower-order terms are larger than those of (3.1), and (8.4) is smaller than $\mathrm{BW}_S(T)$ only for $T\ge e^{4000}$; the minimum of the two bounds is valid for all $T\ge e$.

9. Computations

9.1. Data and methods

The ordinates are those of Odlyzko's table of the first $K = 2\,001\,052$ zeros of the Riemann zeta function, accurate to within $4 \cdot 10^{-9}$ [27], converted to IEEE double precision by correctly rounded decimal-to-binary conversion ($\gamma_K = 1\,132\,490.658714\ldots$). Spot checks against `mpmath.zetazero` at eight indices differ by at most $3.5 \cdot 10^{-10}$. The clock values τ_k are the double-precision numbers nearest to the exact roots of (2.5). They are computed by Newton's method in extended precision and certified for every k by evaluating $N_0 - (k - \frac{1}{2})$ at the rounded value and at both neighbouring floating-point numbers; the 20 679 indices at which the root lies within the evaluation error of a rounding midpoint are decided with `mpmath` at 40 significant digits, and an independent `mpmath` recomputation at 19 900 indices, including the first and last 100, agrees bit for bit. The Euler phase uses full local factors, evaluated either as $-\pi^{-1} \sum_{p \le X} \arg(1 - p^{-1/2-it})$ with the principal argument, or by (2.10) truncated where $p^{-m/2} \le 10^{-17}$. All means are over complete windows, without subsampling or fitted parameters. Windows of ordinals are $(0, 10^3]$, $(10^3, 10^4]$, $(10^4, 10^5]$, $(10^5, 5 \cdot 10^5]$, $(5 \cdot 10^5, 10^6]$, $(10^6, K]$; windows of height are $(T, 2T]$.

9.2. The clock and the ordinates

Figure 1 shows $\delta_k = \tau_k - \gamma_k$ for $k \le 2000$ together with the residual of the one-step clock with $\mathcal{P} = \{2, 3, 5, 7\}$. The root mean square of δ_k decreases from 0.3587 on $(0, 10^3]$ to 0.1789 on $(10^6, K]$, of the order of magnitude of $(2 \log\log T)^{1/2} / \log T$ (Remark 3.17). The maxima per window decrease only from 0.9937 to 0.7174 (Table 4); all of them exceed 0.6331, which bounds only the limit superior. The finite-height envelope (3.4) holds at every index and is used to at most 15% of its size. The envelope (8.1) with the bound 2.5167, valid for all ordinates of the table, gives $|s_k| \le 2.0267$ and is used to at most 66%: $\max_k |s_k| = 1.3325$, at $k = 1\,249\,694$ ($\gamma_k = 735\,839.17$, $\delta_k = -0.7174$).

9.3. Direct defects (Theorem B)

For $T \in \{2^{14}, 2^{16}, 2^{18}, 2^{19}\}$ and $c \in \{\frac{1}{10}, \frac{1}{5}, \frac{1}{3}, \frac{1}{2}\}$ we compute $u_k(X)$ at the true ordinates with $X = T^c$ constant on $(T, 2T]$; the windows contain 21 519, 100 536, 459 982 and 977 804 ordinates. Smaller exponents give $X < 2$ at these heights and are excluded, as in Theorem 5.1.

At fixed c the measured means are essentially independent of T, as Theorem 5.1 requires of a bounded quantity, and they decrease monotonically in c. Between $c = \frac{1}{2}$ and $c = \frac{1}{10}$ the increase of $\langle u_k^2 \rangle$ at $T = 2^{19}$ is 0.0534, against 0.0815 for the main term; at these heights the error term $O(1 + \sqrt{\log(1/c)})$ is not small compared with the main term, and the data do not determine a limit at fixed c. The one-step residual $r_k(X)$ agrees with $u_k(X)$ to within 3% for $c \le \frac{1}{5}$ and exceeds it at $c = \frac{1}{2}$ (0.0146 against 0.0122 at $T = 2^{19}$): this is the cost of evaluating S_X at τ_k instead of γ_k (Theorem 6.1(a)).

9.4. Exact accounting (Theorem C)

For $X \in \{7, 61, 401, 907, 1999\}$ and the five windows above $k = 1000$ we compute the gain $\langle s_k^2 - (s_k - S_X(\tau_k))^2 \rangle$, the diagonal term D_X and the cross term Cr_X of (6.5). Table 3 lists three windows; Figure 3 shows all five.

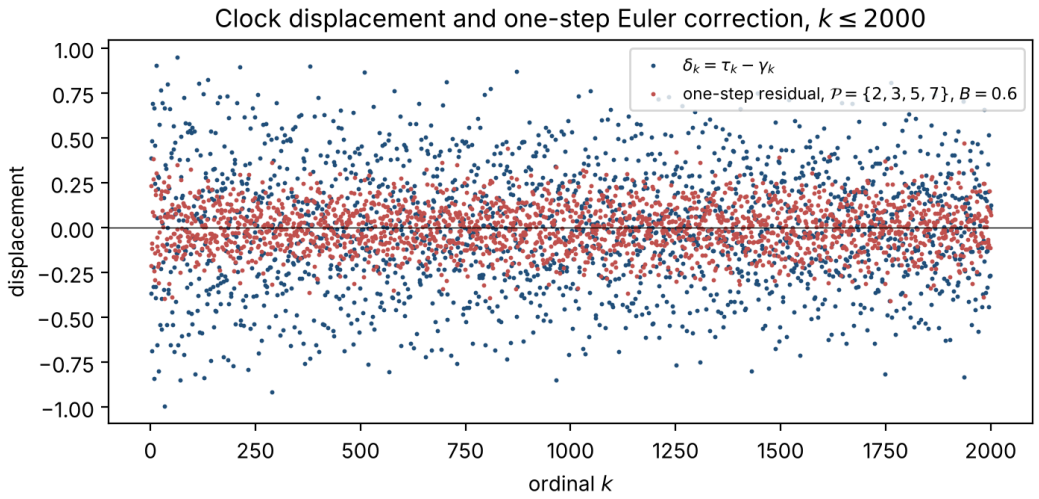


Figure 1: The displacement $\delta_k = \tau_k - \gamma_k$ of the RvM clock and the residual $\tau_k^{\mathcal{P}} - \gamma_k$ of the one-step clock with $\mathcal{P} = \{2, 3, 5, 7\}$, for $k \le 2000$; the correction $S_{\mathcal{P}}(\tau_k)/\mathcal{L}(\tau_k)$ is capped at $\pm B$ with $B = 0.6$ (saturation, Section 9.5).

Table 2: Mean squares of the defect $u_k(X)$ and of the one-step residual $r_k(X)$ over $T < \gamma_k \le 2T$, $X = T^c$.

T	c	X	$\#\{p \le X\}$	$\langle u_k(X)^2 \rangle$	$\langle r_k(X)^2 \rangle$	$\frac{1}{2\pi^2} \log \frac{1}{c}$
2^{14}	1/10	2.64	1	0.066738	0.066312	0.116650
2^{14}	1/5	6.96	3	0.039487	0.038425	0.081535
2^{14}	1/3	25.40	9	0.020174	0.018347	0.055656
2^{14}	1/2	128.00	31	0.010929	0.010701	0.035115
2^{16}	1/10	3.03	2	0.056337	0.055961	0.116650
2^{16}	1/5	9.19	4	0.039588	0.039045	0.081535
2^{16}	1/3	40.32	12	0.022282	0.021508	0.055656
2^{16}	1/2	256.00	54	0.011533	0.012316	0.035115
2^{18}	1/10	3.48	2	0.062696	0.062544	0.116650
2^{18}	1/5	12.13	5	0.041377	0.041265	0.081535
2^{18}	1/3	64.00	18	0.022737	0.022855	0.055656
2^{18}	1/2	512.00	97	0.011983	0.013848	0.035115
2^{19}	1/10	3.73	2	0.065599	0.065517	0.116650
2^{19}	1/5	13.93	6	0.040530	0.040623	0.081535
2^{19}	1/3	80.63	22	0.022911	0.023446	0.055656
2^{19}	1/2	724.08	128	0.012187	0.014567	0.035115

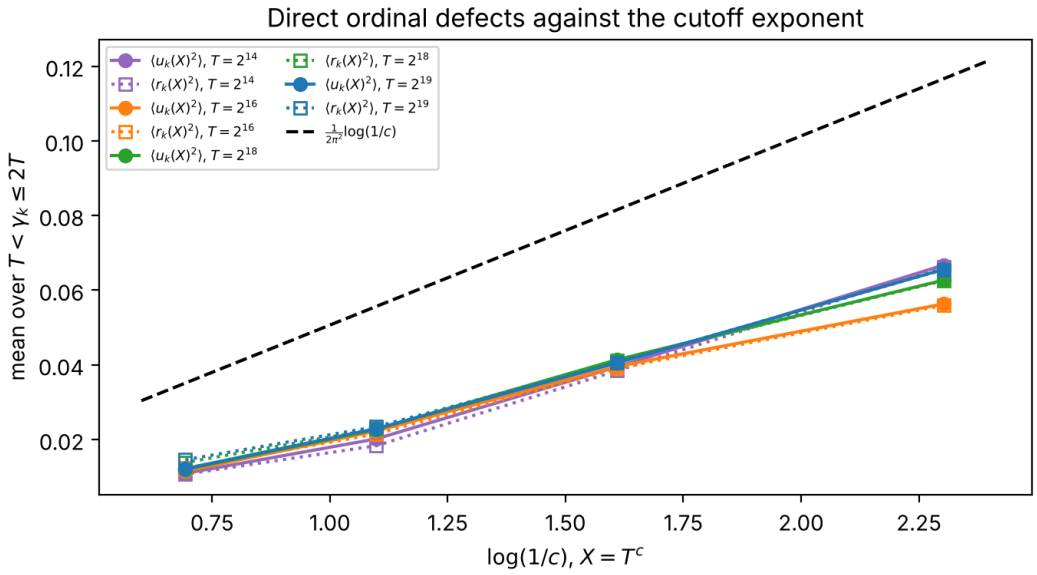


Figure 2: Mean square of $u_k(X)$ (filled markers) and of $r_k(X)$ (open markers) over $T < \gamma_k \le 2T$ against $\log(1/c)$, $X = T^c$, for the four heights of Table 2; dashed: $\frac{1}{2\pi^2}\log(1/c)$.

Table 3: Exact decomposition (6.5) of the one-step gain into the diagonal term D_X and the cross term Cr_X.

window of k	$\langle s_k^2\rangle$	X	gain	D_X	Cr_X	v_X
$(10^3, 10^4]$	0.085195	7	0.063224	0.063984	0.000760	0.066252
		61	0.075868	0.085258	0.009390	0.093853
		401	0.067945	0.097561	0.029616	0.111486
		907	0.062337	0.100483	0.038146	0.117730
		1999	0.055885	0.101829	0.045944	0.123202
$(10^5, 5\cdot 10^5]$	0.106325	7	0.064185	0.064201	0.000016	0.066252
		61	0.086367	0.086579	0.000212	0.093853
		401	0.093276	0.096932	0.003657	0.111486
		907	0.092779	0.099690	0.006911	0.117730
		1999	0.091379	0.101921	0.010542	0.123202
$(10^6, K]$	0.113224	7	0.064441	0.064444	0.000003	0.066252
		61	0.087434	0.087478	0.000044	0.093853
		401	0.098011	0.098495	0.000483	0.111486
		907	0.098607	0.101389	0.002782	0.117730
		1999	0.098325	0.103474	0.005149	0.123202

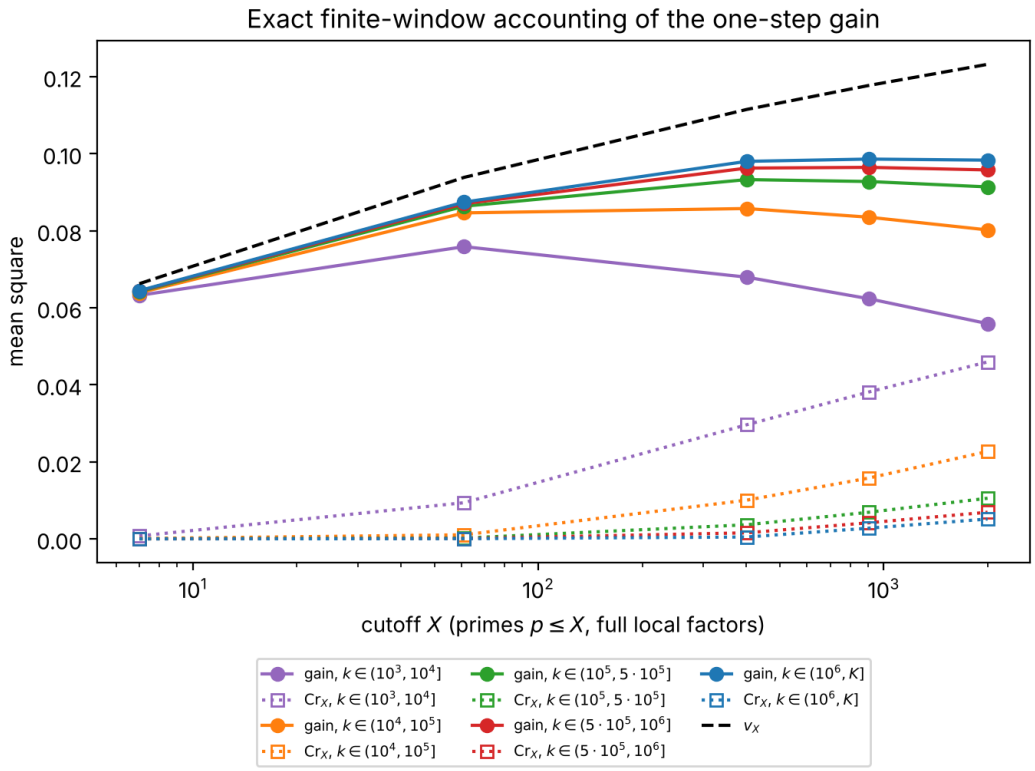


Figure 3: Gain of the one-step clock (filled markers) and cross term Cr_X (open markers) against the cutoff X, for five windows of ordinals; dashed: v_X.

The diagonal term D_X increases with X in every window and stays below v_X; the cross term Cr_X grows with X at fixed height and decreases with the height at fixed X. The gain is therefore maximal at a cutoff that increases with the height: $X = 61$ on $(10^3, 10^4]$, between 401 and 1999 on $(10^5, 5 \cdot 10^5]$, and near 907 on $(10^6, K]$. No parameter is fitted, and the identity gain $= D_X - \mathrm{Cr}_X$ is exact.

9.5. The one-step clock against the cutoff

For comparison with [9, 22] we also report the root mean square and the maximum of $\tau_k^X - \gamma_k$ for 40 cutoffs $2 \le X \le 1999$ (Figure 4) and for five rules, with saturation $\tau_k - \mathrm{sgn}(y_k)\min(|y_k|, B)$, $y_k = S_X(\tau_k)/\mathcal{L}(\tau_k)$, where indicated (Table 4, Figure 5). The minimizing cutoff increases from 7 on $(0, 10^3]$ to 907 on $(10^6, K]$, in agreement with the accounting above, and a least-squares fit $\log X_{\mathrm{opt}} = \theta_{\mathrm{fit}} \log(\tau/2\pi)$ over the five upper windows gives $\theta_{\mathrm{fit}} = 0.5744$. This exponent is exploratory: the curves are flat near the minimum (in the top window the root mean square is 0.06447 at $X = 701$, 0.06427 at $X = 907$ and 0.06428 at $X = 1009$), and it is not asserted to be asymptotically optimal. At these heights the minimizing cutoff lies beyond the onset $X \approx (\tau/2\pi)^{1/2}$ of the cross term of the primes (Section 9.7), where the gain of further primes is offset by the resonant pairs; θ_{fit} describes this finite-height balance. A large fixed cutoff is harmful at small heights: $\mathcal{P} = \{p \le 1999\}$ gives the residual 3.247 at $k = 1$. The largest residual for $\mathcal{P} = \{2, 3, 5, 7\}$, at $k = 258\,318$, follows the close pair $\gamma_{258317} = 175\,681.6491$, $\gamma_{258318} = 175\,681.7159$ (normalized gap 0.109), at which $S^*(\gamma_{258318}) = 1.039$, so that $S(\gamma^+_{258318}) = 1.539$, and $\delta_{258318} = 0.638$.

Table 4: Root mean square, maximum of $|\tau_k^X - \gamma_k|$ and number of residuals $\ge \frac{1}{2}$ over all K ordinates, and the same quantities in the top window $(10^6, K]$.

rule	RMS	max	$\# \ge \frac{1}{2}$	RMS, top	max, top	$\# \ge \frac{1}{2}$, top
bare clock τ_k	0.188066	0.993711	11589	0.178900	0.717359	2971
$\mathcal{P} = \{2, 3, 5, 7\}$, $B = 0.6$	0.120691	0.522347	1	0.117413	0.489293	0
$\mathcal{P} = \{p \le 61\}$, $B = 0.6$	0.085635	0.457259	0	0.085354	0.400144	0
$\mathcal{P} = \{p \le 1999\}$, no saturation	0.070757	3.247397	156	0.064897	0.528267	7
$X_k = \max(7, (\tau_k/2\pi)^{0.57})$, $B = 0.6$	0.066608	0.513886	2	0.064259	0.513886	1

9.6. Transport and the Newton term (Theorem D)

Table 5 decomposes the residual (7.1) on the window $T = 2^{19}$ of Table 2 (all 977 804 ordinates, full local factors). The identity $r_k = u_k + \mathfrak{t}_k + \chi_k$ holds to $2 \cdot 10^{-16}$, and $0 \le \chi_k \le 6.8 \cdot 10^{-8}$ up to rounding.

The transport is smaller than $c^2 \log\log T$ ($= 0.103$, 0.287, 0.644) by factors 21–35, and it is anticorrelated with $u_k(X)$, increasingly with c; this is why $\langle r_k^2 \rangle$ exceeds $\langle u_k^2 \rangle$ by only 0.0001–0.0024. The root mean square of $S'_X/\mathcal{L}$ is close to c, as in Remark 7.7, and the Newton term $y_k S'_X(\tau_k)$ has correlation 0.66–0.77 with $-\mathfrak{t}_k(X)$; at $c = \frac{1}{2}$ it overestimates the transport, and the denominator of (7.8) matters.

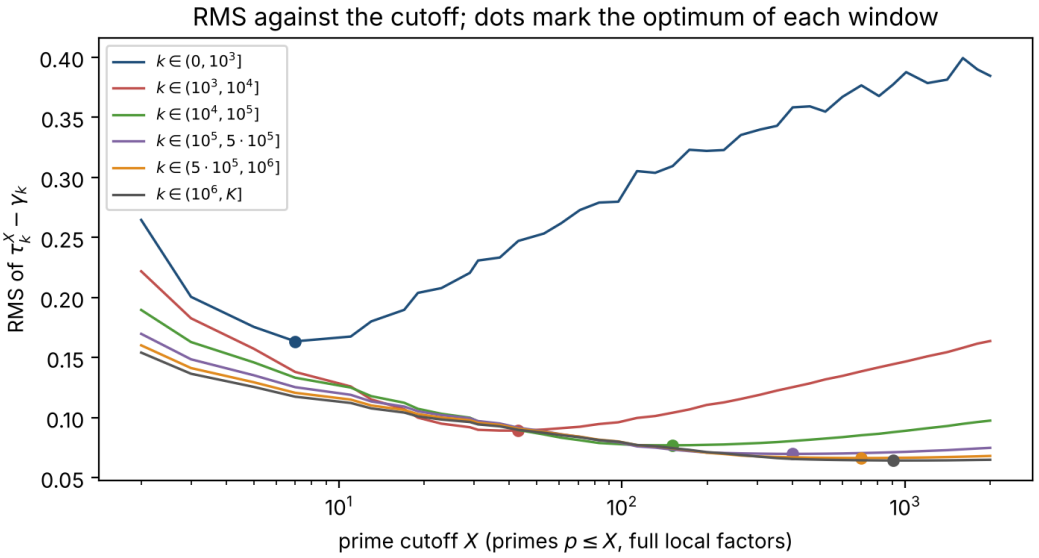


Figure 4: Root mean square of $\tau_k^X - \gamma_k$ (no saturation) against the cutoff X, by window of ordinals; dots mark the minimum in each window.

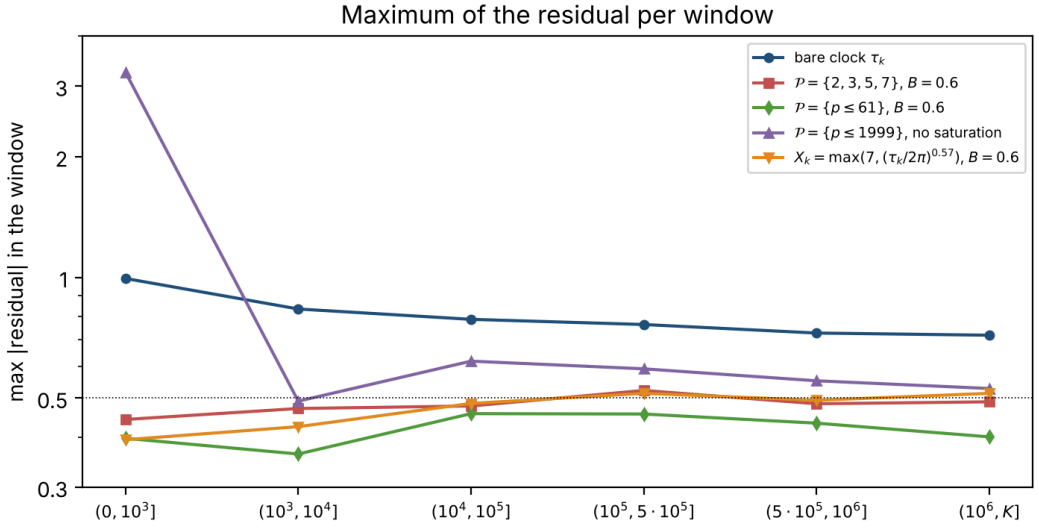


Figure 5: Maximum of $|\tau_k^X - \gamma_k|$ in each window of ordinals for the five rules of Table 4, logarithmic scale; dotted: $\frac{1}{2}$. The value 3.247 of the fixed set $\{p \le 1999\}$ in the first window occurs at $k = 1$.

Table 5: Transport term and Newton term of the one-step clock over $2^{19} < \gamma_k \le 2^{20}$, $X = T^c$: mean squares, the correlation of $u_k(X)$ with $\mathfrak{t}_k(X)$, the root mean square of $S'_X(\tau_k)/\mathcal{L}(\tau_k)$, and the proportion of indices with $F'_X(\tau_k) < 0$.

c	X	$\langle u_k^2 \rangle$	$\langle \mathfrak{t}_k^2 \rangle$	$\mathrm{corr}(u_k, \mathfrak{t}_k)$	$\langle r_k^2 \rangle$	$\langle (y_k S'_X(\tau_k))^2 \rangle$	RMS $\frac{S'_X(\tau_k)}{\mathcal{L}(\tau_k)}$	$F'_X(\tau_k) < 0$
1/5	13.93	0.040530	0.004905	−0.171	0.040623	0.003559	0.225	0.002%
1/3	80.63	0.022911	0.011391	−0.336	0.023446	0.014386	0.361	0.27%
1/2	724.08	0.012187	0.018283	−0.533	0.014567	0.046866	0.525	2.7%

Table 6: Cross term at the clock points on $(10^6, K]$; $(\tau_{\min}/2\pi)^{1/2} = 309.1$.

X	$\#\{p \le X\}$	Cr_X	$\mathrm{Cr}_X^{(1)}$
100	25	0.000083	0.000000
200	46	0.000149	0.000000
300	62	0.000186	0.000000
316	65	0.000195	0.000004
350	70	0.000248	0.000048
400	78	0.000449	0.000235
500	95	0.001107	0.000857
700	125	0.002056	0.001761
907	155	0.002782	0.002447
1200	196	0.003608	0.003210
1999	303	0.005149	0.004662

9.7. Onset of the cross term

Table 6 gives Cr_X on the top window $(10^6, K]$, where $\tau_k/2\pi$ runs over $[95\,536, 180\,241]$, for the full Euler phase and for its prime part $S_X^{(1)}(t) = -\frac{1}{\pi}\sum_{p\le X} p^{-1/2}\sin(t\log p)$, with $\mathrm{Cr}_X^{(1)} = \langle S_X^{(1)}(\tau_k)^2\rangle - \frac{1}{2}\sum_{p\le X} a_p^2$.

$\mathrm{Cr}_X^{(1)}$ vanishes to six decimals until X^2 reaches $\tau_{\min}/2\pi$ and then grows. For $X = 1999$ the 91 809 ordered pairs of primes split into the 7 807 pairs with $pq \in [\tau_{\min}/2\pi, \tau_{\max}/2\pi]$, which contribute 0.004657, and the others, which contribute 0.000005. The full Euler phase has earlier resonances, from products of prime powers in the same range, with smaller weights. This is the mechanism of Proposition 10.4: the phase $\omega\tau_x/2\pi$ with $\omega = \log n$ is stationary in the sense of the Poisson summation formula exactly at $\tau_x = 2\pi n$. A stationary-phase evaluation (heuristic: endpoint terms and the variation of the amplitude are neglected) at x_0 with $\tau_{x_0} = 2\pi n$, where $\omega\tau_{x_0}/2\pi - x_0 = n - \frac{11}{8}$ and the second derivative of the phase is $-2\pi/(\tau\log^2(\tau/2\pi))$, gives $\sum_{k\in I} e^{i\tau_k\log n} \approx -\sqrt{n}\log n$, hence a contribution $\log(pq)/(2\pi^2 \#I)$ of each resonant ordered pair to $\mathrm{Cr}_X^{(1)}$. Summed over the 7 807 pairs this is 0.004663, against 0.004657 measured.

10. Prime responses at the zeros and at the clock points

Subsections 10.1–10.3 are unconditional. Subsection 10.4 contains rigorous implications from two named hypotheses. Nothing heuristic is used; Theorems A–E do not depend on this section.

10.1. Fixed and slowly moving frequencies

Theorem 10.1 (Landau's formula on the critical line)**.** *Uniformly for $t \ge 2$ and integers $2 \le n \le t^{1/8}$,*

$$\sum_{0<\gamma\le t} n^{i\gamma} = -\frac{t}{2\pi}\frac{\Lambda(n)}{\sqrt{n}} + O\Big(\frac{t\log^2 n}{\log t} + \sqrt{n}\log^2 t\Big), \tag{10.1}$$

zeros counted with multiplicity. For fixed n the error is $O(t/\log t)$.

Proof. Let $\ell = \log n$. By the functional equation and $\zeta(\bar{s}) = \overline{\zeta(s)}$, $\rho \mapsto 1-\bar{\rho}$ maps the zeros with $0 < \gamma \le t$ onto themselves, preserving multiplicities and ordinates. Hence $\sum n^{\rho} = \sum n^{1-\bar{\rho}}$, and averaging the two expressions, $\sum_{0<\gamma\le t} n^{\rho} = \sqrt{n}\sum_{0<\gamma\le t} n^{i\gamma}\cosh((\beta-\frac12)\ell)$. Thus

$$\sum_{0<\gamma\le t} n^{i\gamma} = \frac{1}{\sqrt{n}}\sum_{0<\gamma\le t} n^{\rho} - \mathcal{K}_n(t), \qquad \mathcal{K}_n(t) = \sum_{0<\gamma\le t} n^{i\gamma}\Big(\cosh((\beta-\tfrac12)\ell)-1\Big).$$

The zeros with $\beta < \frac12$ correspond to those with $\beta > \frac12$, and those with $\beta = \frac12$ contribute 0, so $|\mathcal{K}_n(t)| \le 2\sum_{\beta>1/2}(\cosh((\beta-\frac12)\ell)-1)$. Writing $\cosh(x\ell)-1 = \int_0^x \ell\sinh(\ell y)\,dy$ and exchanging the finite sum with the integral,

$$\sum_{\beta>1/2,\,0<\gamma\le t}\Big(\cosh((\beta-\tfrac12)\ell)-1\Big) = \int_0^{1/2} \ell\sinh(\ell y)\,N^+(\tfrac12+y,t)\,dy \ll t\log t\cdot\ell\int_0^{\infty}\sinh(\ell y)e^{-ay}dy,$$

with $a = \frac14\log t$, by Theorem 3.10. For $\ell \le a/2$ the last integral is $\ell/(a^2-\ell^2) \le \frac43\ell/a^2$, so $\mathcal{K}_n(t) \ll t\ell^2/\log t$. Theorem 3.9 gives $n^{-1/2}\sum n^{\rho} = -\frac{t}{2\pi}\Lambda(n)n^{-1/2} + O(\sqrt{n}\log(2nt)\log\log(3n))$, and $\log(2nt)\log\log(3n) \ll \log^2 t$ for $n \le t^{1/8}$. □

The symmetry removes the term linear in $\beta-\frac12$; Selberg's estimate then bounds the quadratic remainder. Without the symmetry one needs $\sum_{0<\gamma\le t}|\beta-\frac12| = o(t)$, which is not known.

Lemma 10.2. *For fixed real $\omega \ne 0$, $\sum_{k\in\mathcal{I}_T} e^{i\omega\tau_k} \ll_\omega \log T$. Hence for fixed integers $n_1, n_2 \ge 2$, $\langle \sin(\tau_k\log n_1)\sin(\tau_k\log n_2)\rangle_{\mathcal{I}_T} \to \frac12\mathbf{1}_{n_1=n_2}$.*

Proof. Take $\omega > 0$ (conjugate otherwise) and $\Omega(x) = \omega\tau_x/2\pi$ for real $x \ge 1$. By (2.5), $\Omega'(x) = \omega/\log(\tau_x/2\pi)$ is continuous, positive and decreasing, and for $x \in [\min\mathcal{I}_T, \max\mathcal{I}_T]$ and $T \ge T_0(\omega)$ it lies in $[\omega/\log(T/\pi), \frac12]$, so that $\|\Omega'\| \ge \omega/\log(T/\pi)$. $\mathcal{I}_T$ is a set of consecutive integers, and Theorem 3.8 gives the bound. For the second claim use $2\sin y_1\sin y_2 = \cos(y_1-y_2)-\cos(y_1+y_2)$ with $\omega = \log(n_1/n_2)$ (if $n_1 \ne n_2$) and $\omega = \log(n_1n_2)$, and $\#\mathcal{I}_T \asymp T\log T$. □

Theorem 10.3. *Uniformly for $T \ge T_0$ and integers $2 \le n \le T^{1/8}$,*

$$\langle s_k \sin(\gamma_k\log n)\rangle_{\mathcal{I}_T} = \frac{a_n}{2} + O\Big(\frac{\log n\,\log\log T}{\log T} + \frac{\sqrt{n}\log^2 T}{T\log n}\Big), \tag{10.2}$$

and the same with τ_k in place of γ_k. Hence for primes $p \le T^{1/8}$, $\mathcal{R}_p^{\gamma}(\mathcal{I}_T)$ and $\mathcal{R}_p^{\tau}(\mathcal{I}_T)$ are $1 + O(\frac{\sqrt{p}\log p\log\log T}{\log T} + \frac{p\log^2 T}{T\log p})$; they tend to 1 uniformly for $p \le (\log T)^{2-\varepsilon}$, and for every fixed prime power n.

Proof. For $t > 14$ not an ordinate let $\mathcal{H}_n(t) = \int_{14}^t S(y)n^{iy}dy$; the integrand is bounded and piecewise continuous. As measures on $(14,t]$, $dS = dN - (\mathcal{L}+\varpi'/\pi)dy$, and $y \mapsto n^{iy}/(i\ell_n)$ is a C^1 primitive of n^{iy}; integration by parts for Lebesgue–Stieltjes integrals gives

$$\mathcal{H}_n(t) = \Big[\frac{S(y)n^{iy}}{i\ell_n}\Big]_{14}^{t} - \frac{1}{i\ell_n}\sum_{14<\gamma_k\le t} n^{i\gamma_k} + \frac{1}{i\ell_n}\int_{14}^{t} n^{iy}\Big(\mathcal{L}+\frac{\varpi'}{\pi}\Big)dy = \frac{i}{\ell_n}\sum_{0<\gamma_k\le t} n^{i\gamma_k} + O(\log t), \tag{10.3}$$

uniformly in n (since $\ell_n \ge \log 2$), because $\gamma_1 > 14$, $S = O(\log t)$, $\int_{14}^t n^{iy}\mathcal{L}\,dy = O(\log t/\ell_n)$ by a second integration by parts ($\mathcal{L}'(y) = 1/(2\pi y)$), and $\varpi' = O(y^{-2})$. By Theorem 10.1, for $T \le t \le 2T+1$, $\mathcal{H}_n(t) = -i\frac{\Lambda(n)}{2\pi\sqrt{n}\,\ell_n}t + O(\frac{T\ell_n}{\log T} + \frac{\sqrt{n}\log^2 T}{\ell_n} + \log T)$. Integrating by parts once more, with $\frac{d}{dt}(t\mathcal{L}(t)) = \mathcal{L}(t) + \frac{1}{2\pi}$,

$$\int_T^{2T} S(t)n^{it}\mathcal{L}(t)\,dt = \Big[\mathcal{H}_n\mathcal{L}\Big]_T^{2T} - \int_T^{2T}\frac{\mathcal{H}_n(t)}{2\pi t}\,dt = -i\frac{\Lambda(n)}{2\pi\sqrt{n}\,\ell_n}\int_T^{2T}\mathcal{L}\,dt + O\Big(T\ell_n + \frac{\sqrt{n}\log^3 T}{\ell_n}\Big). \tag{10.4}$$

Taking imaginary parts, $\int_T^{2T} S\sin(\ell_n t)\mathcal{L}\,dt = \frac{a_n}{2}\int_T^{2T}\mathcal{L}\,dt + O(T\ell_n + \sqrt{n}\log^3 T/\ell_n)$. Apply Lemma 4.7 with $\psi(t) = \sin(\ell_n t)$ on $[t_1, t_2]$, where $t_1 \in [T, T+1/T]$ and $t_2 \in [2T, 2T+1/T]$ are non-ordinates; the change of endpoints alters both sides by $O(\log^2 T)$. The ϖ-term is $O(\log T)$, the boundary term is $O(\log^2 T)$, and $\frac12\int G^2|\psi'| \le \frac{\ell_n}{2}\int_T^{2T} G^2 \ll \ell_n T\log\log T$ by Theorem 3.3. Divide by $\#\mathcal{I}_T = \int_T^{2T}\mathcal{L}\,dt + O(\log T) \gg T\log T$. For τ_k: $|\sin(\ell_n\tau_k) - \sin(\ell_n\gamma_k)| \le \ell_n|\delta_k|$, and by (2.6) and (3.3), $\langle|s_k\delta_k|\rangle_{\mathcal{I}_T} \ll \langle s_k^2\rangle_{\mathcal{I}_T}/\log T \ll \log\log T/\log T$. For primes multiply by $2/|a_p| = 2\pi\sqrt{p}$. □

The same computation without the term $\frac12\int G^2\psi'$ gives, for the one-particle term of Section 10.2, $\mathcal{A}_{T,p} = 1 + O(\frac{\sqrt{p}\log p}{\log T} + \frac{p\log^2 T}{T\log p})$ uniformly for $p \le T^{1/8}$; so Hypothesis U below holds for $p \le (\log T)^2/(\log\log T)^{2+\varepsilon}$, where $\theta \to 0$, and not for fixed $\theta > 0$.

Proposition 10.4 (No cross term below $T^{1/4}$)**.** *For $\varepsilon > 0$ there is $T_0(\varepsilon)$ such that for $T \ge T_0(\varepsilon)$ and $2 \le X \le T^{1/4-\varepsilon}$, $|\operatorname{Cr}_X(\mathcal{I}_T)| \ll T^{-\varepsilon/2}\sqrt{\log\log T}$.*

Proof. Let $Y = T^{1/2-\varepsilon}$, $A' = \{n \in A_X : n \le Y\}$, $A'' = A_X \setminus A'$, and $S_X = S' + S''$ accordingly. For each $p \le X$ the first m with $p^m > Y$ has $p^{-m/2} < Y^{-1/2}$, so $|S''| \le \sum_{A''}|a_n| \le \frac{\#\{p\le X\}}{\pi(1-2^{-1/2})}Y^{-1/2} \le 2T^{-\varepsilon/2}$. Next, $2\langle S'(\tau_k)^2\rangle = \sum_{A'} a_n^2 + \sum_{n_1,n_2\in A'} a_{n_1}a_{n_2}\Big(\mathbf{1}_{n_1\ne n_2}\langle\cos(\tau_k\log\frac{n_1}{n_2})\rangle - \langle\cos(\tau_k\log n_1n_2)\rangle\Big)$, and every frequency ω that occurs satisfies $1/Y \le |\omega| \le 2\log Y = (1-2\varepsilon)\log T$. For $k \in \mathcal{I}_T$, $\tau_k \in [T-1, 2T+1]$, and as in Lemma 10.2 the phase $\Omega(x) = |\omega|\tau_x/2\pi$ has $\Omega'(x) = |\omega|/\log(\tau_x/2\pi) \in [\frac{1}{2Y\log T}, 1-\varepsilon]$ for $T \ge T_0(\varepsilon)$; Theorem 3.8 gives $|\sum_{k\in\mathcal{I}_T} e^{i\omega\tau_k}| \ll Y\log T + \varepsilon^{-1}$. Since $(\sum_{A_X}|a_n|)^2 \le \sigma_X^2 \ll X$, the off-diagonal part is $\ll X(Y\log T + \varepsilon^{-1})/(T\log T) \ll T^{-3\varepsilon}$, and $\frac12\sum_{A'} a_n^2 = v_X + O(T^{-\varepsilon})$. Finally $\operatorname{Cr}_X = \langle S'^2\rangle - v_X + 2\langle S'S''\rangle + \langle S''^2\rangle$, and $|\langle S'S''\rangle| \le \langle S'^2\rangle^{1/2}\,2T^{-\varepsilon/2}$ with $\langle S'^2\rangle \ll 1 + v_X \ll \log\log T$. □

Beyond this range the cross term is produced by products n_1n_2 of prime powers near $\tau_k/2\pi$, where the clock phase is stationary; Section 9.7 shows its onset for the primes at $X \approx (\tau/2\pi)^{1/2}$ and its size.

Corollary 10.5 (Gain at the clock points)**.** *(a) For a finite set A of prime powers and $\lambda \in \mathbb{R}$, $\langle s_k^2 - (s_k - \lambda S_A(\tau_k))^2\rangle_{\mathcal{I}_T} \to (2\lambda - \lambda^2)v_A$. (b) For fixed $\varepsilon > 0$ and λ, uniformly for $2 \le X \le (\log T)^{2-\varepsilon}$, with full local factors,*

$$\Big\langle s_k^2 - (s_k - \lambda S_X(\tau_k))^2\Big\rangle_{\mathcal{I}_T} = (2\lambda - \lambda^2)v_X + O_\lambda\Big(\frac{\sqrt{X}\log\log T}{\log T} + T^{-1/40}\Big).$$

Proof. (a) In (6.4), $\mathcal{R}_n^{\tau}(\mathcal{I}_T) \to 1$ by Theorem 10.3 and $\mathrm{Cr}_A(\mathcal{I}_T) \to 0$ by Lemma 10.2. (b) The left side is $2\lambda\langle s_k S_X(\tau_k)\rangle - \lambda^2\langle S_X(\tau_k)^2\rangle$, and $\langle S_X(\tau_k)^2\rangle = v_X + O(T^{-1/40})$ by Proposition 10.4. Let $Y = T^{1/8}$. For $n \in A_X$ with $n \le Y$, (10.2) multiplied by a_n and summed gives $\frac{1}{2}\sum a_n^2$ plus an error $\ll \frac{\log\log T}{\log T}\sum_{A_X}|a_n|\log n + \frac{\log^2 T}{T}\sum_{n\le Y}|a_n|\frac{\sqrt{n}}{\log n} \ll \frac{\sqrt{X}\log\log T}{\log T} + \frac{X\log^2 T}{T}$, since $\sum_{A_X}|a_n|\log n = \frac{1}{\pi}\sum_{p\le X}\frac{\log p}{\sqrt{p}-1} \ll \sqrt{X}$ and $|a_{p^m}|\sqrt{p^m}/\log p^m \le 1/(m^2\log p)$. For $n > Y$, $|\sum a_n\langle s_k \sin(\tau_k \log n)\rangle| \le \langle s_k^2\rangle^{1/2}\sum_{A_X, n>Y}|a_n| \ll \sqrt{\log\log T}\, XY^{-1/2}$, and $\frac{1}{2}\sum_{n>Y} a_n^2 \ll X/Y$. □

For $X = T^c$ the gain of the one-step clock is controlled by Theorem 7.3 instead.

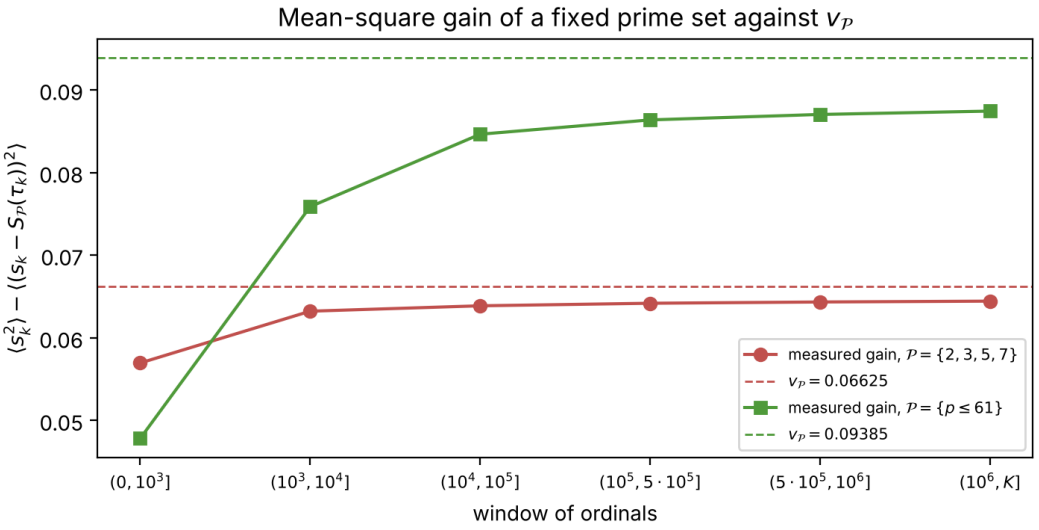


Figure 6: Mean-square gain $\langle s_k^2\rangle - \langle (s_k - S_{\mathcal{P}}(\tau_k))^2\rangle$ of two fixed prime sets $\mathcal{P}$ (full local factors) in each window of ordinals, and the constants $v_{\mathcal{P}}$ (dashed) of Corollary 10.5.

10.2. The response at the ordinates and its Fourier reduction

Let $T, 2T$ be non-ordinates, p a prime, and

$$\mathcal{A}_{T,p} = -\frac{2\pi\sqrt{p}}{\#\mathcal{I}_T}\int_T^{2T} G(t)\sin(t\log p)\mathcal{L}(t)\,dt, \qquad \mathcal{S}_{T,p} = \frac{1}{T}\int_T^{2T} S(t)^2 p^{it}\,dt. \tag{10.5}$$

Proposition 10.6 (Unconditional separation of one- and two-point terms)**.**

$$\begin{aligned}\mathcal{R}_p^{\gamma}(\mathcal{I}_T) = \mathcal{A}_{T,p} &+ \frac{\pi\sqrt{p}\,T\log p}{\#\mathcal{I}_T}\Re\mathcal{S}_{T,p} - \frac{\pi\sqrt{p}}{\#\mathcal{I}_T}\Big[G^2\sin(t\log p)\Big]_T^{2T} \\ &+ \frac{\pi\sqrt{p}\log p}{\#\mathcal{I}_T}\int_T^{2T}(G^2 - S^2)\cos(t\log p)\,dt.\end{aligned} \tag{10.6}$$

If $\theta = \log p/L_T$ stays in a compact subset of $(0,1)$, the last two terms are $o(1)$ and

$$\mathcal{R}_p^{\gamma}(\mathcal{I}_T) = \mathcal{A}_{T,p} + 2\pi^2\theta\sqrt{p}\,\big(1 + O(1/L_T)\big)\Re\mathcal{S}_{T,p} + o(1). \tag{10.7}$$

Proof. Lemma 4.7 with $\psi(t) = \sin(t\log p)$ on $[T,2T]$, multiplied by $-2\pi\sqrt{p}/\#\mathcal{I}_T$, gives (10.6), because $\int_T^{2T} G^2\cos(t\log p)\,dt = T\Re\mathcal{S}_{T,p} + \int_T^{2T}(G^2-S^2)\cos(t\log p)\,dt$. On $[T,2T]$, $|G^2-S^2| \le 2|S||\varpi|/\pi + \varpi^2/\pi^2 \ll \log T/T$, so the last term is $O(\sqrt{p}\log p/T)$, and the boundary term is $O(\sqrt{p}\log^2 T/(T\log T))$. Both are $o(1)$ for $p \le T^{\theta_1}$ with $\theta_1 < 1$. Finally $\#\mathcal{I}_T = \frac{TL_T}{2\pi}(1+O(1/L_T))$. □

Let $f_T = S\mathbf{1}_{[T,2T]}$, let $\mathfrak{m}_T$ be its distributional derivative (a finite signed measure, since f_T has compact support and bounded variation), and

$$Z_T(\omega) = \int_{\mathbb{R}} e^{i\omega t}\,d\mathfrak{m}_T(t), \qquad \widehat{f_T}(\omega) = \int_{\mathbb{R}} f_T(t)e^{i\omega t}dt. \tag{10.8}$$

Theorem 10.7 (Unconditional shifted two-point representation)**.** *For non-ordinates $T, 2T$,*

$$Z_T(\omega) = \sum_{T<\gamma<2T} \nu(\gamma)e^{i\omega\gamma} - \int_T^{2T}\Big(\mathcal{L}(t) + \frac{\varpi'(t)}{\pi}\Big)e^{i\omega t}\,dt + S(T)e^{i\omega T} - S(2T)e^{2i\omega T}, \tag{10.9}$$

$Z_T(\omega) = -i\omega\widehat{f_T}(\omega)$, so that $Z_T(0) = 0$ and $Z_T(\omega)/\omega$ extends continuously by $-i\widehat{f_T}$; and for every real ω_0

$$\int_T^{2T} S(t)^2 e^{i\omega_0 t}\,dt = -\frac{1}{2\pi}\int_{\mathbb{R}} \frac{Z_T(\omega)Z_T(\omega_0-\omega)}{\omega(\omega_0-\omega)}\,d\omega, \tag{10.10}$$

the integral converging absolutely.

Proof. On $(T,2T)$, $dS = dN - (\mathcal{L}+\varpi'/\pi)dt$ as measures, with atoms $\nu(\gamma)\delta_\gamma$; the cutoff adds $S(T)\delta_T$ and $-S(2T)\delta_{2T}$. This gives $\mathfrak{m}_T$ and (10.9). Since f_T is of bounded variation with compact support, integration by parts gives $\int e^{i\omega t}d\mathfrak{m}_T = -i\omega\int f_T e^{i\omega t}dt$, the boundary terms vanishing. As $f_T \in L^1\cap L^2$ is real, Plancherel's identity applied to f_T and $f_T e^{-i\omega_0 t}$ gives $\int f_T^2 e^{i\omega_0 t}dt = \frac{1}{2\pi}\int \widehat{f_T}(\omega)\widehat{f_T}(\omega_0-\omega)\,d\omega$, absolutely convergent since $\int|\widehat{f_T}(\omega)\widehat{f_T}(\omega_0-\omega)|d\omega \le \|\widehat{f_T}\|_2^2 = 2\pi\|f_T\|_2^2$. Substitute $\widehat{f_T} = iZ_T/\omega$. □

With $\omega_0 = \theta L_T = \log p$ and $\omega = \alpha L_T$ put

$$\mathcal{B}_T(\theta) = \int_{\mathbb{R}} \frac{Z_T(\alpha L_T)Z_T((\theta-\alpha)L_T)}{\alpha(\theta-\alpha)}\,d\alpha, \qquad \text{so that} \qquad \mathcal{S}_{T,p} = -\frac{\mathcal{B}_T(\theta)}{2\pi T L_T}. \tag{10.11}$$

Expanding the zero sums gives the weights $e^{i\alpha L_T\gamma}e^{i(\theta-\alpha)L_T\gamma'} = p^{i\gamma'}e^{i\alpha L_T(\gamma-\gamma')}$: $\mathcal{S}_{T,p}$ is a pair correlation of the zeros of the Riemann zeta function *shifted by a prime frequency*, which a stationary pair correlation (a function of $\gamma-\gamma'$ only) does not determine.

10.3. The kernel $\mathfrak{K}_\theta$

For $0<\theta<1$ define the continuous, piecewise linear function

$$\mathfrak{K}_\theta(\alpha) = \begin{cases} \theta, & \alpha \le -1, \\ \theta-1-\alpha, & -1 \le \alpha \le \theta-1, \\ 0, & \theta-1 \le \alpha \le 1, \\ \alpha-1, & 1 \le \alpha \le 1+\theta, \\ \theta, & \alpha \ge 1+\theta, \end{cases} \tag{10.12}$$

equivalently, with $h(q) = \max(1 - |q|, 0)$,

$$\mathfrak{K}_\theta(\alpha) = 1 - h(\alpha) - h(\theta - \alpha) - h(\theta) + 2\max\Big(1 - \max(|\alpha|, |\theta - \alpha|, \theta), 0\Big), \tag{10.13}$$

as one checks on each of the five intervals.

Proposition 10.8 (Integral and support of the kernel)**.** *For $0 < \theta < 1$,*

$$\int_{\mathbb{R}} \frac{\mathfrak{K}_\theta(\alpha)}{\alpha(\theta - \alpha)}\, d\alpha = -\frac{2\Phi(\theta)}{\theta}, \tag{10.14}$$

the integral being absolutely convergent, and $\mathfrak{K}_\theta(\alpha) = 0$ whenever $|\alpha| + |\theta - \alpha| + \theta < 2$.

Proof. $\mathfrak{K}_\theta$ is bounded and vanishes on $[\theta - 1, 1]$, which contains 0 and θ; the integrand is bounded, continuous outside $[\theta - 1, 1]$ and $O(\alpha^{-2})$ as $|\alpha| \to \infty$, hence absolutely integrable. It is invariant under $\alpha \mapsto \theta - \alpha$, which maps $[1, \infty)$ onto $(-\infty, \theta - 1]$, so the integral is twice that over $[1, \infty)$. On $[1 + \theta, \infty)$, $\frac{\theta}{\alpha(\theta-\alpha)} = \frac{1}{\alpha} - \frac{1}{\alpha-\theta}$, whose integral is $\lim_{y\to\infty}\Big[\log\frac{\alpha}{\alpha-\theta}\Big]_{1+\theta}^{y} = -\log(1+\theta)$. On $[1, 1+\theta]$, $\frac{\alpha-1}{\alpha(\theta-\alpha)} = -\frac{1}{\theta\alpha} + \frac{\theta-1}{\theta(\theta-\alpha)}$, whose integral is $-\frac{\log(1+\theta)}{\theta} + \frac{\theta-1}{\theta}\log(1-\theta)$. The sum is $-\Phi(\theta)/\theta$; doubling gives (10.14). If $\alpha > 1$ then $|\alpha|+|\theta-\alpha|+\theta \geq \alpha+(\alpha-\theta)+\theta = 2\alpha > 2$, and if $\alpha < \theta - 1$ the same follows by the symmetry; elsewhere $\mathfrak{K}_\theta = 0$. □

Proposition 10.9 (The square of the prime polynomial has no prime frequency)**.** *For $Y \geq 2$, $T \geq 2$ and every prime p, $\left|\frac{1}{T}\int_T^{2T} S_Y^{\mathrm{pr}}(t)^2 p^{it}\, dt\right| \ll Y^3/T$, with an absolute constant. In particular, with $E_0 = S - S_{Y_0}^{\mathrm{pr}}$, uniformly for primes $p \leq T^{\theta_1}$ with fixed $\theta_1 < 1$,*

$$\mathcal{S}_{T,p} = \frac{1}{T}\int_T^{2T} \Big(2S_{Y_0}^{\mathrm{pr}}(t)E_0(t) + E_0(t)^2\Big)p^{it}\, dt + o(p^{-1/2}).$$

Proof. Expanding the product of sines, $S_Y^{\mathrm{pr}}(t)^2 p^{it}$ is a combination, with coefficients of modulus at most $\frac{1}{4\pi^2}(qr)^{-1/2}$, of $e^{i\lambda t}$ with $\lambda = \log(pq^{\pm1}r^{\pm1})$, $q, r < Y$ primes. No λ vanishes: $p = qr$, $pq = r$, $pr = q$ and $pqr = 1$ are impossible for primes. Write $e^\lambda = A/B$ with coprime positive integers; then $|\lambda| \geq \log(1 + 1/\min(A, B)) \geq 1/(2\min(A, B))$, and $\min(A, B) \leq \max(r, qr) < Y^2$ in all cases, or $|\lambda| \geq \log 8$. Hence $|\int_T^{2T} e^{i\lambda t}dt| \leq 2/|\lambda| \leq 4Y^2$, and the sum of the coefficients is $\ll (\sum_{q<Y} q^{-1/2})^2 \ll Y$. For the second statement take $Y = Y_0 = T^{1/160}$: $Y_0^3/T = T^{-157/160} = o(T^{-\theta_1/2})$. □

Thus the moving-prime law below is a statement about the terms that contain Selberg's remainder E_0. The bound $\int_T^{2T} E_0^4 \ll T$ of Theorem 3.5 gives only $|\frac{1}{T}\int E_0^2 p^{it}| \ll 1$ and $|\frac{1}{T}\int S_{Y_0}^{\mathrm{pr}} E_0 p^{it}| \ll \sqrt{\log\log T}$, without the oscillation of size $p^{-1/2}$ that the law requires.

10.4. The moving-prime law under Hypotheses U and W

Hypothesis 10.10 (Hypothesis U: uniform one-particle input)**.** For every $0 < \theta_0 < \theta_1 < 1$, uniformly for primes with $\theta_0 \leq \log p/L_T \leq \theta_1$, $\mathcal{A}_{T,p} = 1 + o(1)$.

A sufficient condition is the uniform estimate $\sup_{T\leq t\leq 2T} |\sum_{T<\gamma_k\leq t} p^{i\gamma_k} + \frac{(t-T)\log p}{2\pi\sqrt{p}}| = o(T\log p/\sqrt{p})$ in that range, by two integrations by parts as in the proof of Theorem 10.3. Theorem 10.1 gives it only for $p \leq T^{o(1)}$, through the error term $t\log^2 p/\log t$.

Hypothesis 10.11 (Hypothesis W: weighted arithmetic bridge)**.** Uniformly for primes with θ in any compact subset of $(0,1)$,

$$\int_{\mathbb{R}} \frac{Z_T(\alpha L_T) Z_T((\theta-\alpha)L_T) + \mathcal{N}_T p^{-1/2} \mathfrak{K}_\theta(\alpha)}{\alpha(\theta-\alpha)}\, d\alpha = o\Big(\frac{TL_T}{\sqrt{p}}\Big), \tag{10.15}$$

with the continuous extensions of $Z_T(\omega)/\omega$ at $\omega = 0$.

Proposition 10.12 (Moving-prime law under Hypotheses U and W)**.** *Under Hypothesis W,*

$$\mathcal{S}_{T,p} = -\frac{1}{2\pi^2\sqrt{p}} \frac{\Phi(\theta)}{\theta} + o(p^{-1/2}). \tag{10.16}$$

Under Hypotheses U and W, or U and the real part of (10.16)*, uniformly for θ in compact subsets of* $(0,1)$*,*

$$\mathcal{R}_p^{\gamma}(\mathcal{I}_T) = 1 - \Phi(\theta) + o(1). \tag{10.17}$$

Conversely, under Hypothesis U, (10.17) *is equivalent to the real part of* (10.16)*. Nothing is asserted about $\mathcal{R}_p^{\tau}$.*

Proof. The integral of the kernel term in (10.15) converges absolutely (Proposition 10.8); hence so does that of the first term, which is $\mathcal{B}_T(\theta)$, and by (10.14) $\mathcal{B}_T(\theta) = \frac{\mathcal{N}_T}{\sqrt{p}} \frac{2\Phi(\theta)}{\theta} + o(TL_T/\sqrt{p}) = \frac{TL_T}{\pi\sqrt{p}} \frac{\Phi(\theta)}{\theta} + o(TL_T/\sqrt{p})$; (10.11) gives (10.16). Then $\sqrt{p}\,\Re\mathcal{S}_{T,p} = O(1)$, so in (10.7) the factor $1 + O(1/L_T)$ contributes $o(1)$ and $2\pi^2\theta\sqrt{p}\,\Re\mathcal{S}_{T,p} = -\Phi(\theta) + o(1)$; with $\mathcal{A}_{T,p} = 1 + o(1)$ this is (10.17). Conversely, solve (10.7) for $\Re\mathcal{S}_{T,p}$. □

Remark 10.13. $\Phi(\theta) = 2\sum_{j\ge1} \frac{\theta^{2j}}{(2j-1)2j} = \theta^2 + \frac{\theta^4}{6} + \frac{\theta^6}{15} + \cdots$, so $1 - \theta^2$ is only the quadratic truncation of (10.17), and $1 - \Phi(\theta)$ vanishes at $\theta = 0.9036\ldots$. Hypotheses U and W are not consequences of Theorem 10.1, which reaches only $\theta \to 0$. By Proposition 10.9, (10.16) is a statement about $\frac{1}{T}\int (2S_{Y_0}^{\mathrm{pr}} E_0 + E_0^2) p^{it}$, with required error $o(p^{-1/2}) = T^{-\theta/2+o(1)}$.

11. Heuristic support and numerical evidence

This section contains no statement about the Riemann zeta function that is used elsewhere. Subsection 11.1 proves a theorem about a random-matrix model; its relevance to the zeros of the Riemann zeta function is a heuristic.

11.1. A determinantal model whose connected kernel is $\mathfrak{K}_\theta$

On a circle of length R_0 let $\Pi_{\mathfrak{n}}$ be the orthogonal projection onto $\mathfrak{n}$ consecutive Fourier modes, and consider its determinantal point process, which has exactly $\mathfrak{n}$ points. For $q \in R_0^{-1}\mathbb{Z}$ let Ξ_q be the centred linear statistic of $y \mapsto e^{2\pi i q y}$, so that $\Xi_0 = 0$ and $\overline{\Xi_q} = \Xi_{-q}$. Let $R_0, \mathfrak{n} \to \infty$ with $\mathfrak{n}/R_0 \to 1$, so that the local kernel tends to the sine kernel. For lattice $\theta > 0$ and real ι_1, ι_2 the tilt

$$d\mathbb{P}_{\iota_1,\iota_2} = \frac{\exp\{-\iota_1 \Re\Xi_{-\theta} - \iota_2 \Im\Xi_{-\theta}\}}{\mathbb{E}\exp\{-\iota_1 \Re\Xi_{-\theta} - \iota_2 \Im\Xi_{-\theta}\}}\, d\mathbb{P}_{0,0} \tag{11.1}$$

is a probability measure (the statistics are bounded), and differentiation at $(0,0)$, justified by boundedness, gives $(\partial_{\iota_1} + i\partial_{\iota_2}) \operatorname{Cov}_{\iota_1,\iota_2}(\Xi_\alpha, \Xi_{\theta-\alpha})|_{(0,0)} = -\operatorname{cum}_3(\Xi_\alpha, \Xi_{\theta-\alpha}, \Xi_{-\theta})$, the third joint cumulant.

Theorem 11.1 (Connected kernel of the model)**.** *Let lattice frequencies tend to $\alpha, \theta-\alpha, -\theta$ with $0<\theta<1$. Then $R_0^{-1}\operatorname{cum}_3(\Xi_\alpha,\Xi_{\theta-\alpha},\Xi_{-\theta})\to\mathfrak{K}_\theta(\alpha)$ and $R_0^{-1}\operatorname{cum}_2(\Xi_\theta,\Xi_{-\theta})\to\theta$.*

Proof. For commuting multiplication operators $\mathsf{m}_1,\mathsf{m}_2,\mathsf{m}_3$, the third joint cumulant of the corresponding linear statistics of a determinantal process with projection kernel is [30]

$$\begin{aligned}\operatorname{cum}_3 = {}& \operatorname{Tr}(\mathsf{m}_1\mathsf{m}_2\mathsf{m}_3\Pi_{\mathfrak{n}}) - \operatorname{Tr}(\mathsf{m}_1\mathsf{m}_2\Pi_{\mathfrak{n}}\mathsf{m}_3\Pi_{\mathfrak{n}}) - \operatorname{Tr}(\mathsf{m}_1\mathsf{m}_3\Pi_{\mathfrak{n}}\mathsf{m}_2\Pi_{\mathfrak{n}}) - \operatorname{Tr}(\mathsf{m}_2\mathsf{m}_3\Pi_{\mathfrak{n}}\mathsf{m}_1\Pi_{\mathfrak{n}})\\ &+ \operatorname{Tr}(\mathsf{m}_1\Pi_{\mathfrak{n}}\mathsf{m}_2\Pi_{\mathfrak{n}}\mathsf{m}_3\Pi_{\mathfrak{n}}) + \operatorname{Tr}(\mathsf{m}_1\Pi_{\mathfrak{n}}\mathsf{m}_3\Pi_{\mathfrak{n}}\mathsf{m}_2\Pi_{\mathfrak{n}}),\end{aligned} \tag{11.2}$$

the third-order term of the expansion of $\log\det(\mathrm{Id}+\Pi_{\mathfrak{n}}(e^{\mathsf{m}}-1)\Pi_{\mathfrak{n}})$. Multiplication by $e^{2\pi iqy}$ shifts a mode by R_0q; a trace with two projections counts the overlap of two integer intervals of length $\mathfrak{n}$, and a trace with three their triple overlap. For the shifts $R_0\alpha$, $R_0(\theta-\alpha)$, $-R_0\theta$, whose sum is 0, the triple overlap has length $\max(\mathfrak{n}-R_0\max(|\alpha|,|\theta-\alpha|,\theta),0)$ for both cyclic orders, and the pair overlaps have lengths $R_0h(\cdot)+o(R_0)$. Dividing by R_0 gives the right side of (10.13). The same count for two operators gives $R_0^{-1}\operatorname{cum}_2(\Xi_\theta,\Xi_{-\theta})\to 1-h(\theta)=\theta$. □

Why Hypothesis W is plausible. A window $[T,2T]$ contains about $\mathcal{N}_T=TL_T/2\pi$ zeros of the Riemann zeta function at unit mean spacing after unfolding, and the one-particle prime amplitude in Hypothesis U is $-\mathcal{N}_T\theta/\sqrt{p}$. This equals the linear response of $\mathbb{E}\,\Xi_\theta$ to the tilt (11.1) with $\iota_1=2/\sqrt{p}$, $\iota_2=0$ and $R_0=\mathcal{N}_T$, namely $-\frac{\iota_1}{2}\operatorname{cum}_2(\Xi_\theta,\Xi_{-\theta})\sim-\mathcal{N}_T\theta/\sqrt{p}$, by the susceptibility θ of Theorem 11.1. If the zeros responded to the prime p like the model, the connected product in (10.11) would be $-\mathcal{N}_T\mathfrak{K}_\theta(\alpha)/\sqrt{p}$, which is the content of Hypothesis W. This is an analogy, not an argument: stationary sine-kernel statistics do not determine the shifted correlation (10.11), and convergence on compact sets would not control the singular weight and the tails. Note that $\mathfrak{K}_\theta$ vanishes where $|\alpha|+|\theta-\alpha|+\theta<2$ (Proposition 10.8), so the whole value $-2\Phi(\theta)/\theta$ comes from frequencies outside that region.

11.2. Numerical evidence

Conjecture 11.2. *For primes with $\theta=\log p/L_T$ in compact subsets of $(0,1)$, $\mathcal{R}_p^\gamma(\mathcal{I}_T)=1-\Phi(\theta)+o(1)$.*

For all 2262 primes $p\le 20\,000$ we compute $\mathcal{R}_p^\gamma$ and $\mathcal{R}_p^\tau$ on the five windows of ordinals above $k=1000$, with $\theta=\log p/\operatorname{median}_k\log(\tau_k/2\pi)$, and compare the primes with $0<\theta<1$ with $1-\Phi(\theta)$. Table 7 separates individual errors from bin means over θ-bins of width 0.05 containing at least five primes; Table 8 lists the bins of the two top windows, and Figure 7 all bins.

Table 7: Errors of $\mathcal{R}_p^\gamma-(1-\Phi(\theta))$ over the primes $p\le 20\,000$ with $\theta<1$.

window of k	primes	max individual error	RMS error	max |bin-mean error|
$(10^3,10^4]$	159	0.34557	0.09425	0.02637
$(10^4,10^5]$	901	0.44085	0.08511	0.02371
$(10^5,5\cdot10^5]$	2262	0.27857	0.06429	0.02972
$(5\cdot10^5,10^6]$	2262	0.31682	0.06086	0.03531
$(10^6,K]$	2262	0.16425	0.03309	0.01951

Table 8: Bin means of $\mathcal{R}_p^\gamma$ and $\mathcal{R}_p^\tau$ in the two top windows; "law" is the mean of $1-\Phi(\theta)$ over the same primes, s.e. the standard error of the mean difference.

window of k	θ-bin	primes	$\mathcal{R}_p^\gamma$	law	difference	s.e.	$\mathcal{R}_p^\tau$
$(5\cdot 10^5, 10^6]$	$[0.30, 0.35)$	6	0.8985	0.8933	+0.0052	0.0003	0.8845
	$[0.35, 0.40)$	8	0.8592	0.8549	+0.0043	0.0001	0.8515
	$[0.40, 0.45)$	13	0.8174	0.8127	+0.0047	0.0002	0.8180
	$[0.45, 0.50)$	22	0.7671	0.7619	+0.0052	0.0004	0.7829
	$[0.50, 0.55)$	33	0.7114	0.7059	+0.0054	0.0004	0.7496
	$[0.55, 0.60)$	54	0.6513	0.6443	+0.0070	0.0007	0.7205
	$[0.60, 0.65)$	86	0.5849	0.5753	+0.0096	0.0008	0.6960
	$[0.65, 0.70)$	142	0.5090	0.4979	+0.0111	0.0009	0.6771
	$[0.70, 0.75)$	234	0.4239	0.4103	+0.0136	0.0012	0.6689
	$[0.75, 0.80)$	380	0.3305	0.3123	+0.0181	0.0017	0.6905
	$[0.80, 0.85)$	630	0.2225	0.2018	+0.0207	0.0021	0.7548
	$[0.85, 0.90)$	645	0.1355	0.1002	+0.0353	0.0034	0.8411
$(10^6, K]$	$[0.30, 0.35)$	7	0.8952	0.8909	+0.0043	0.0001	0.8779
	$[0.35, 0.40)$	12	0.8556	0.8518	+0.0038	0.0002	0.8426
	$[0.40, 0.45)$	16	0.8126	0.8085	+0.0041	0.0003	0.8076
	$[0.45, 0.50)$	27	0.7669	0.7629	+0.0040	0.0002	0.7749
	$[0.50, 0.55)$	48	0.7126	0.7076	+0.0050	0.0003	0.7406
	$[0.55, 0.60)$	77	0.6518	0.6453	+0.0065	0.0004	0.7085
	$[0.60, 0.65)$	129	0.5845	0.5761	+0.0084	0.0006	0.6815
	$[0.65, 0.70)$	222	0.5074	0.4974	+0.0100	0.0007	0.6640
	$[0.70, 0.75)$	368	0.4218	0.4099	+0.0119	0.0009	0.6524
	$[0.75, 0.80)$	627	0.3271	0.3120	+0.0151	0.0011	0.6642
	$[0.80, 0.85)$	718	0.2393	0.2198	+0.0195	0.0016	0.7015

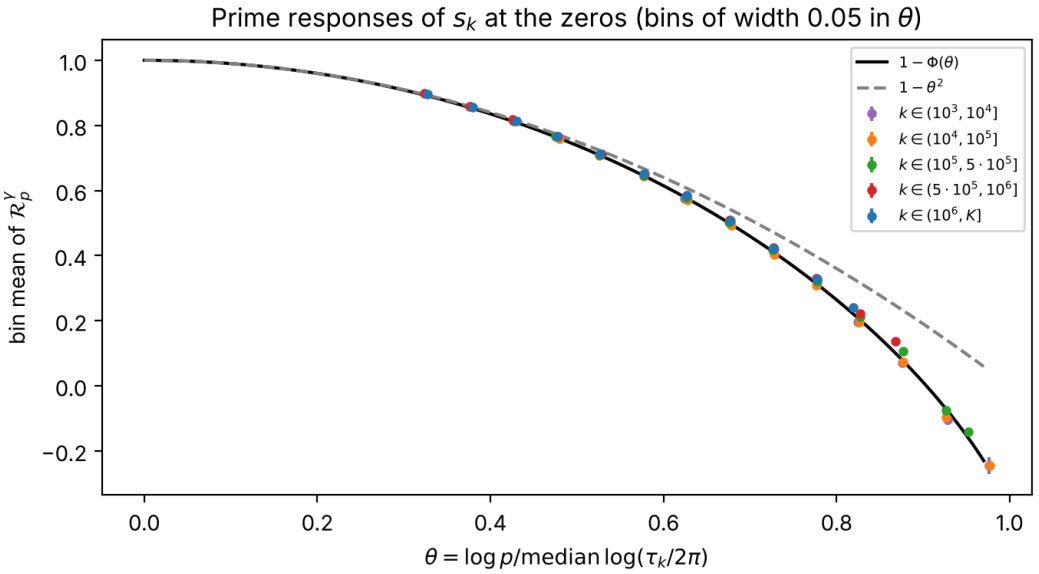


Figure 7: Bin means of $\mathcal{R}_p^\gamma$ (bars: descriptive standard error, see the text) against $1-\Phi(\theta)$ (solid) and $1-\theta^2$ (dashed), for five windows of ordinals.

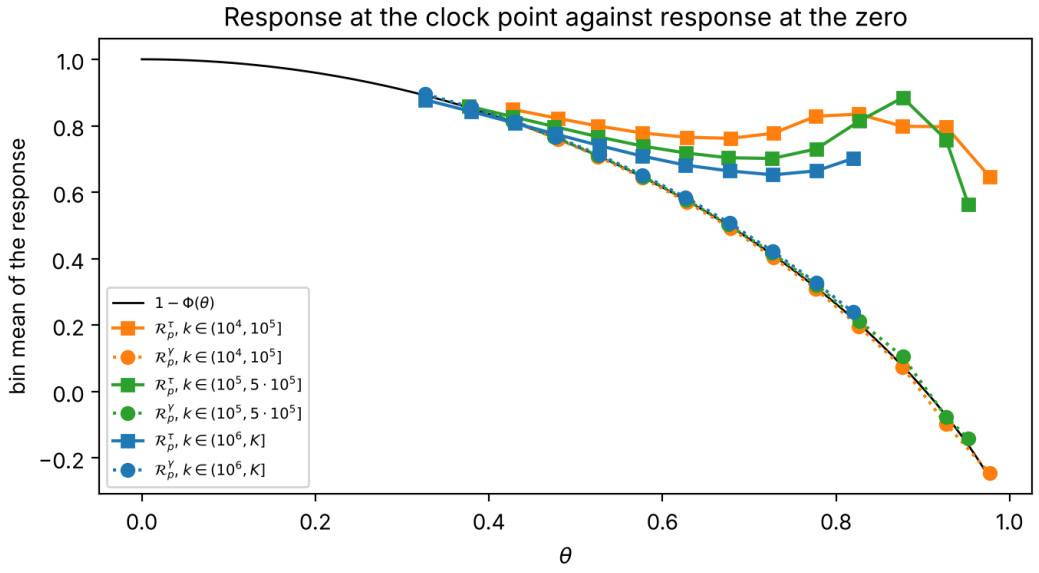


Figure 8: Bin means of the response at the clock points, $\mathcal{R}_p^\tau$ (squares), and at the zeros, $\mathcal{R}_p^\gamma$ (circles), for three windows of ordinals; solid: $1-\Phi(\theta)$.

The bin means follow $1-\Phi(\theta)$ and separate from $1-\theta^2$ for $\theta \geq 0.6$. The residuals are not noise: in the two top windows every bin mean exceeds the candidate, by about 0.004 at $\theta \approx 0.3$ and 0.02–0.035 at $\theta \approx 0.8$–0.9, many standard errors. In every bin the excess is smaller in the top window than in the preceding one, by 6–23%; since θ is computed with the median height of each window, a bin contains different primes in the two windows, and two windows do not determine a rate of decay. The data are consistent with Conjecture 11.2 only together with a finite-height correction of a few hundredths. The standard errors treat the primes of a bin as independent; since all responses are computed on the same ordinates, they are descriptive only. The response at the clock points follows a different curve, with a plateau near 0.65–0.70 for $\theta \in [0.65, 0.80]$ in the top window (Figure 8), as Remark 6.4 allows.

12. Conclusions

12.1. What was known

Before this work, the relation between the RvM clock and the ordinates of the zeros of the Riemann zeta function was known at two levels. Pointwise, $\delta_k = \tau_k - \gamma_k$ is bounded, with $\limsup |\delta_k| \leq 0.6331$ as an immediate consequence of the bound of Bellotti and Wong [2]. On average, s_k is asymptotically Gaussian with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae for the Gram displacement proved by Korolev [20, 19], so $\langle \delta_k^2 \rangle \sim 2\log\log T/\log^2 T$: the clock is close to the zeros in height, but it misplaces almost every zero by an unbounded number of positions. Selberg's approximation [28, 12] replaces $S(t)$ by short prime sums in t-mean, and França–LeClair and LeClair [10, 22, 11] write the equation of an individual zero with an Euler phase and a growing number of primes, with computations at a fixed number of primes.

12.2. What this paper adds

1. *An exact dictionary between clock-type models and the zeros* (Theorems 4.1 and 4.2). For any C^1 model F and any continuous $\mathfrak{g}$, sums of $\mathfrak{g}$-increments of $k - \frac{1}{2} - F(\gamma_k)$ over the ordinates are exactly dF-integrals of $\mathfrak{g}(N-F)$ up to two boundary terms. For powers this gives the even moments with the constant $\frac{1}{12}$ per zero; for exponentials it transfers the signed law of $S(t)$ to s_k (Corollary 4.4).

2. *The prime information needed by the clock* (Theorems 5.1 and 5.11). The corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$ holds at the true ordinates with mean-square ordinal defect $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1+\sqrt{\log(\log T/\log X)})$, also with a cutoff varying at every index. In the main term each factor e in $\log X$ removes $\frac{1}{2\pi^2}$ from the mean square; the error term is not resolved at this scale. The defect is bounded in mean square if and only if the primes reach a fixed power of the height.

3. *A computable clock that is closer to the zeros by an order of magnitude* (Theorem 7.3, Corollaries 7.4 and 7.5). The one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$ uses only τ_k and the primes. With $X = T^{1/\sqrt{\log\log T}}$, or with the cutoff $\tau_k^{1/\sqrt{\log\log \tau_k}}$, its mean-square error in height is $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$, against $(2+o(1))\log\log T/\log^2 T$ for the bare clock; in units of zeros the mean square drops from $\frac{1}{2\pi^2}\log\log T$ to $\frac{1}{4\pi^2}\log\log\log T$. This is the main finding: the information in the primes up to a slowly decreasing power of the height cancels all but a $\log\log\log T/(2\log\log T)$ part of the mean square of the

error of the clock. The restriction on c comes from the transport between τ_k and γ_k (Lemma 7.2); a root of the corrected equation gains nothing where it can be selected (Proposition 7.6).

4. *The pointwise envelope* (Proposition 8.1, Theorem 8.3, Proposition 8.5). $\limsup|\tau_k-\gamma_k| = 2\pi\limsup|S(t)|/\log t$ exactly, and no slowly growing prime correction changes it. Backlund's method with the reflected exponent profile gives $\limsup|S(t)|/\log t \le \frac{1}{2\pi\log 2}\int_0^1 \mathfrak{e}(\sigma)\,d\sigma/\sqrt{\sigma(2-\sigma)}$; Bourgain's exponent pair gives $0.0968160\ldots$, hence $\limsup|\tau_k-\gamma_k| < 0.60832$, and the explicit inputs alone give $|S(T)| \le 0.09783\log T + 0.722\log\log T + 13$ for all $T \ge e$.

5. *An exact finite-height account and the prime responses* (Theorem 6.1, Section 10). The gain of τ_k^X is the diagonal response minus a cross term, with the exact nonnegative term χ_k. The responses tend to 1 for primes up to $(\log T)^{2-\varepsilon}$ unconditionally, the cross term vanishes for $X \le T^{1/4-\varepsilon}$, and on the data it switches on at $X \approx (\tau/2\pi)^{1/2}$ through resonant pairs $pq \approx \tau_k/2\pi$. For $p = T^\theta$ the response at the zeros is reduced exactly to a pair correlation shifted by a prime frequency; under two named hypotheses it is $1-\Phi(\theta)$, which the data follow up to a systematic excess of a few hundredths.

12.3. Why a finer knowledge of $\tau_k - \gamma_k$ matters

The clock is the deterministic part of the ordinates, and δ_k carries all the arithmetic. Statistics of δ_k and s_k are statistics of the zeros of the Riemann zeta function that do not depend on an unfolding of the ordinates or on Gram intervals. Theorem A turns questions about the ordinates into questions about $N-F$ in t-mean, so any improvement of the t-mean description of $S(t)$ (sharper forms of Selberg's approximation, longer prime polynomials, mixed moments with prime powers) passes to the zeros. Theorem D shows that the size of $\tau_k^X - \gamma_k$ is governed by two separate quantities: the prime information missing below X, which Theorem B measures, and the transport $S_X(\gamma_k) - S_X(\tau_k)$, whose observed anticorrelation with the defect (Table 5) is not explained by any proved estimate. Pointwise, Proposition 8.1 identifies the extreme displacement of the clock with the extreme size of $S(t)$, so that every bound for one is a bound for the other. The measured responses $\mathcal{R}_n^\tau$, $\mathcal{R}_n^\gamma$ and the cross terms Cr_X are exact finite-height invariants of the zeros, which any model of the zeros must reproduce; the response $1-\Phi(\theta)$ is a function that the stationary pair correlation does not determine.

12.4. Future work

Question 12.1. For fixed $c \in (0, \frac{1}{2}]$, do the bounded means of $u_k(T^c)^2$ converge, and to $(2\pi^2)^{-1}\log(1/c)$ plus which constant?

Question 12.2. Does $\langle r_k(T^c)^2\rangle_{\mathcal{I}_T}$ stay bounded for fixed c, that is, does the transport cancel against the defect as in Table 5? Can a second Newton step improve (7.7)? Can Theorems 5.1 and 7.3 be extended to $c > \frac{1}{2}$?

Question 12.3. Can Hypotheses U and W be proved for $p = T^\theta$, that is, can the terms of Proposition 10.9 that contain Selberg's remainder be evaluated to $o(p^{-1/2})$? What is the finite-height correction to $1-\Phi(\theta)$ seen in Table 8?

Theorem 4.1 applies to any counting staircase and any C^1 model, in particular to the zeros of a primitive Dirichlet L-function; an analogue of Theorem 5.1 would additionally need Selberg's approximation for $L(s,\chi)$ [29, 12] and uniformity in the conductor.

Declaration of competing interest

The author declares no competing interests.

Data availability

All programs, the input-generation script with its certificates, the numerical outputs (CSV and JSON) and the figures are in the public GitHub repository `ordinal_prime_cutoff_v10` (link to be inserted). The ordinates are Odlyzko's table [27]; the two input files (arrays of 2 001 052 little-endian double-precision numbers) are regenerated from that table by `make_inputs_v10.py`. Their SHA-256 identifiers are
`fafb9436ce6a506de58c96c9bc9765d3942de3f15c921e99e0015b3e945fe0ac` (ordinates) and
`b4c6758f0aef3724d3345f894ff82d5bbcb0f6139025b8b266091f713faa9504` (clock). The computations used Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, Matplotlib 3.11.2 and mpmath 1.3.0.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used an AI assistant (Claude, Anthropic) to edit the text, check proofs and write analysis programs. The author reviewed and edited the content and takes full responsibility for it.

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